

RACER TURBO SYSTEM 2020

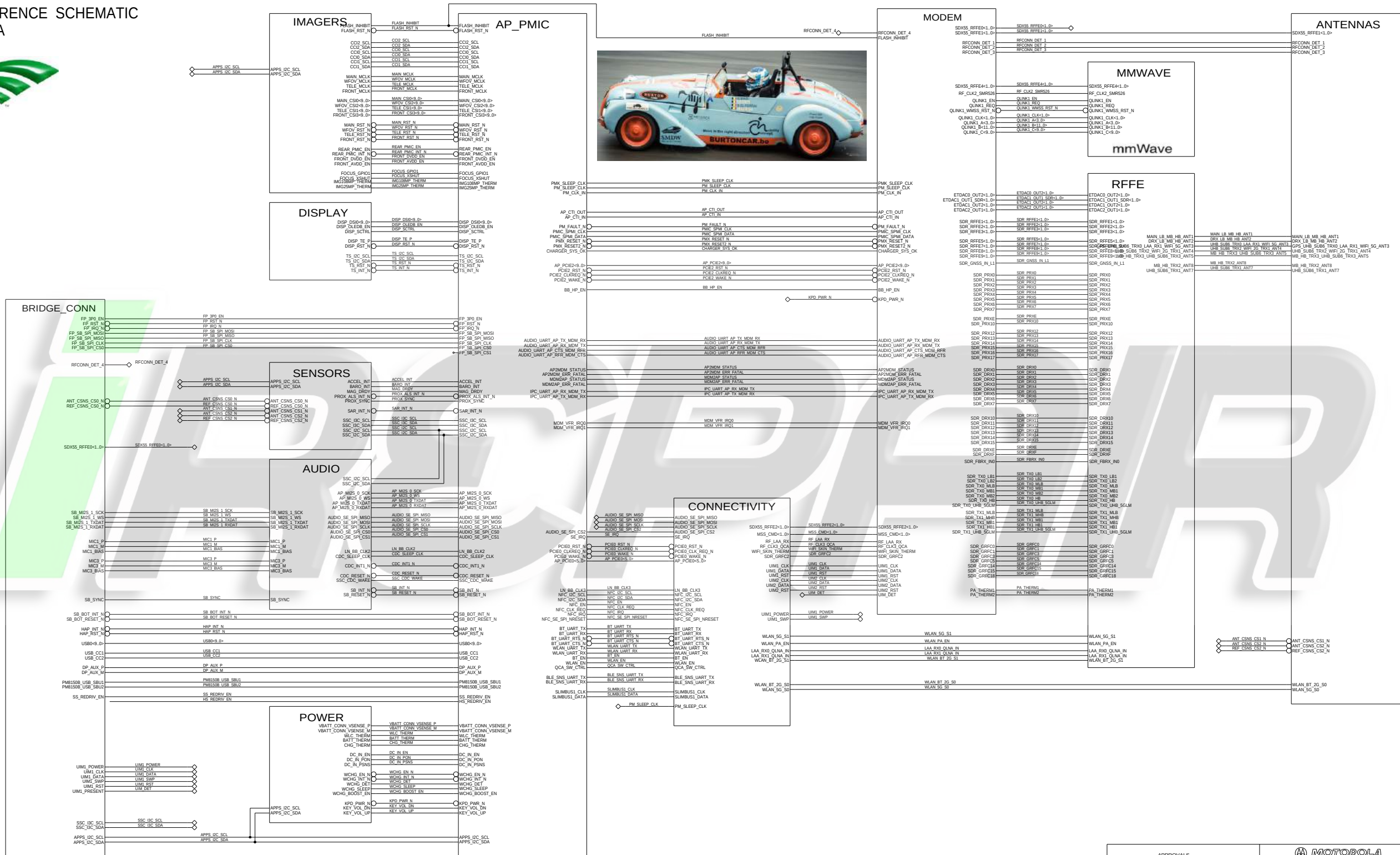
BASED ON REFERENCE SCHEMATIC
LD20-PM222-11_A



ANIBAL GARCIA VENEZUELA

REVISIONS

ZONE	LTR	DESCRIPTION	DATE	APPROVED



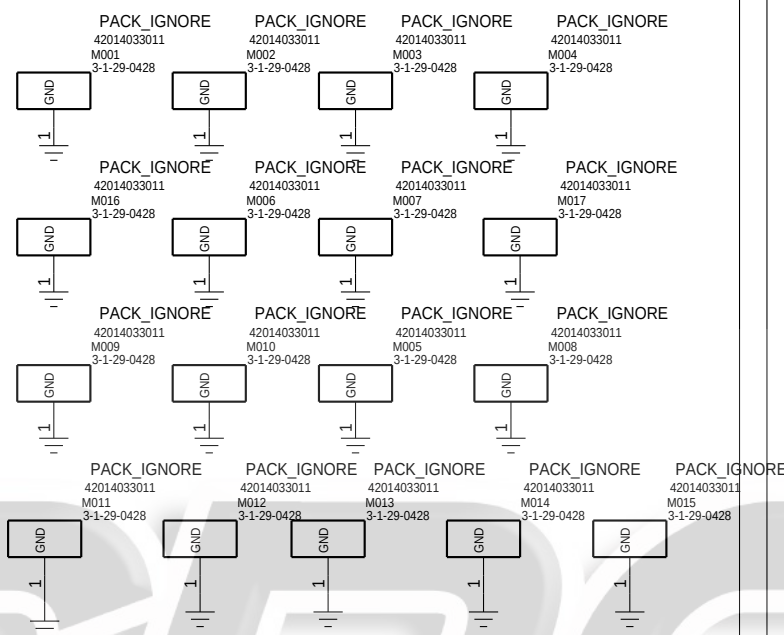
APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Mon Dec 02 17:23:21 2019		sr28c51351_b	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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CADD FILE NAME:		PAGE 1 OF 97	

ME/SHIELDS

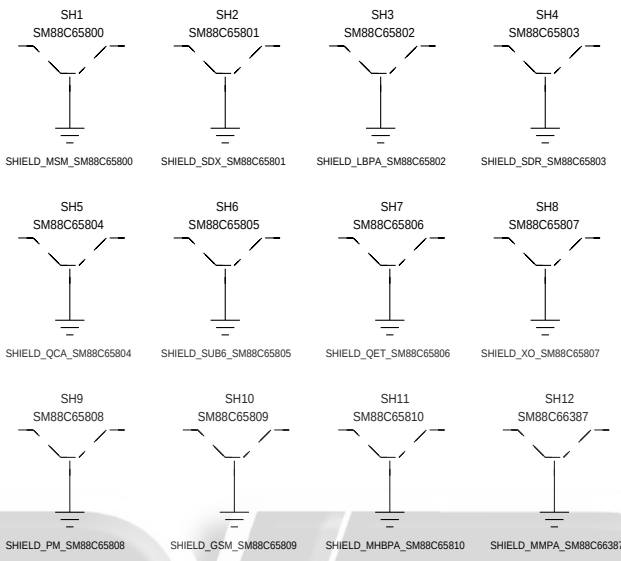
REFERENCE DESIGNATOR RANGES BY BLOCK

- 000 - 499: ANTENNA
- 500 - 799: POWER
- 1000-2999: SM8250 + PM8250 + PM8150A + PM8150B
- 3000-3799: SDX55M + PMK8002 + PMX55 + SDR865
- 3800-3999: SMR526 + QTM525
- 4000-4499: CONNECTIVITY
- 4500-4999: AUDIO
- 5000-5199: DISPLAY
- 5200-5999: IMAGERS
- 6000-6499: SENSORS
- 7000-8999: RFFE

GROUND CONTACTS FROM PCB TO REAR



SOLDERED SHIELDS



SMT BOSS ABOVE 108MP CONNECTOR

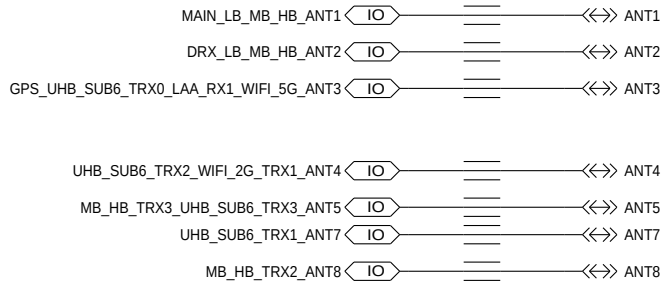


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:07:05 2020		sr28c51351_b	
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		CAD FILE NAME: PAGE 2 OF 97	

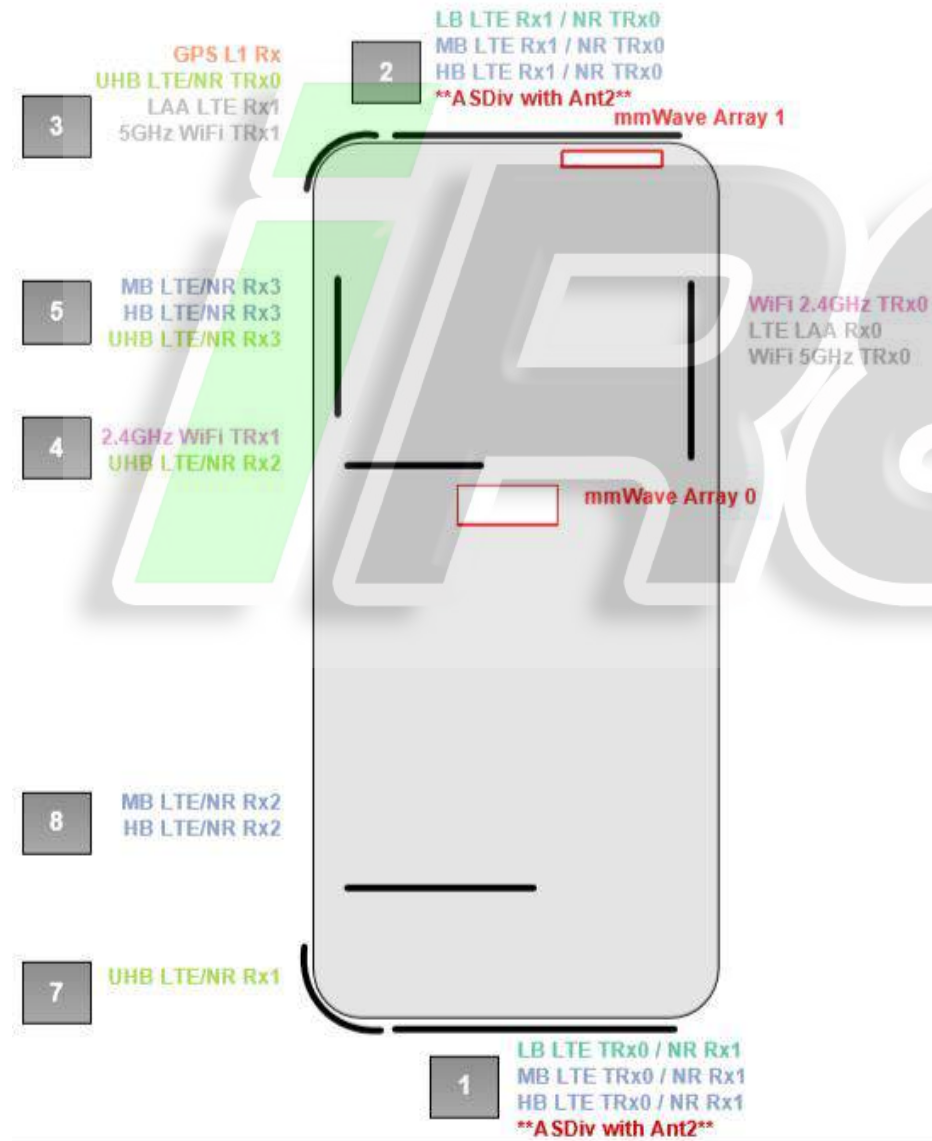
ANT: PORTS

REFDES: 000 - 499

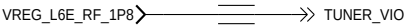
ANTENNA



Viewed from Front (Display) Side

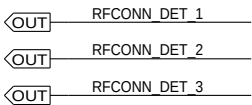


SUPPLIES

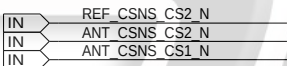


VERIFY POWER SUPPLIES TP PMIC

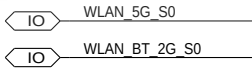
INPUTS /OUTPUTS - RF_CON_DET



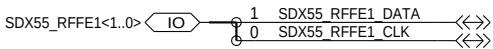
CAP_SENSE



BT WIFI ANTENNA



INTERFACE MIPI



APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
Tue Feb 18 16:06:56 2020

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

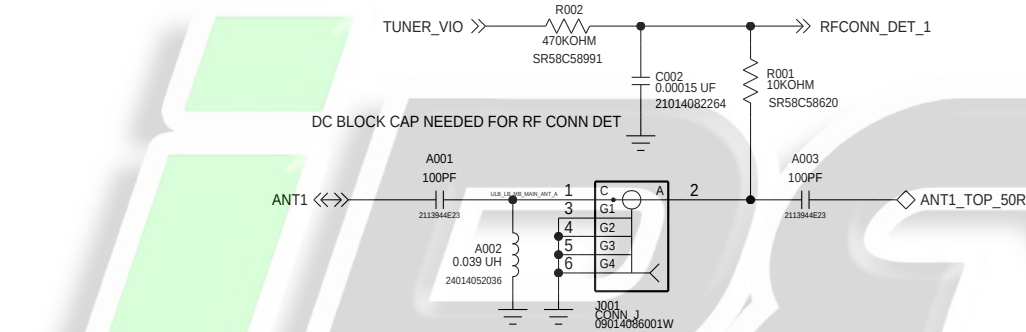
antennas

PROJECT NAME
B sr28c51351_b.cpm

REV
03

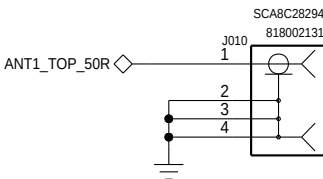
CAD FILE NAME:

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ANT1 COAX

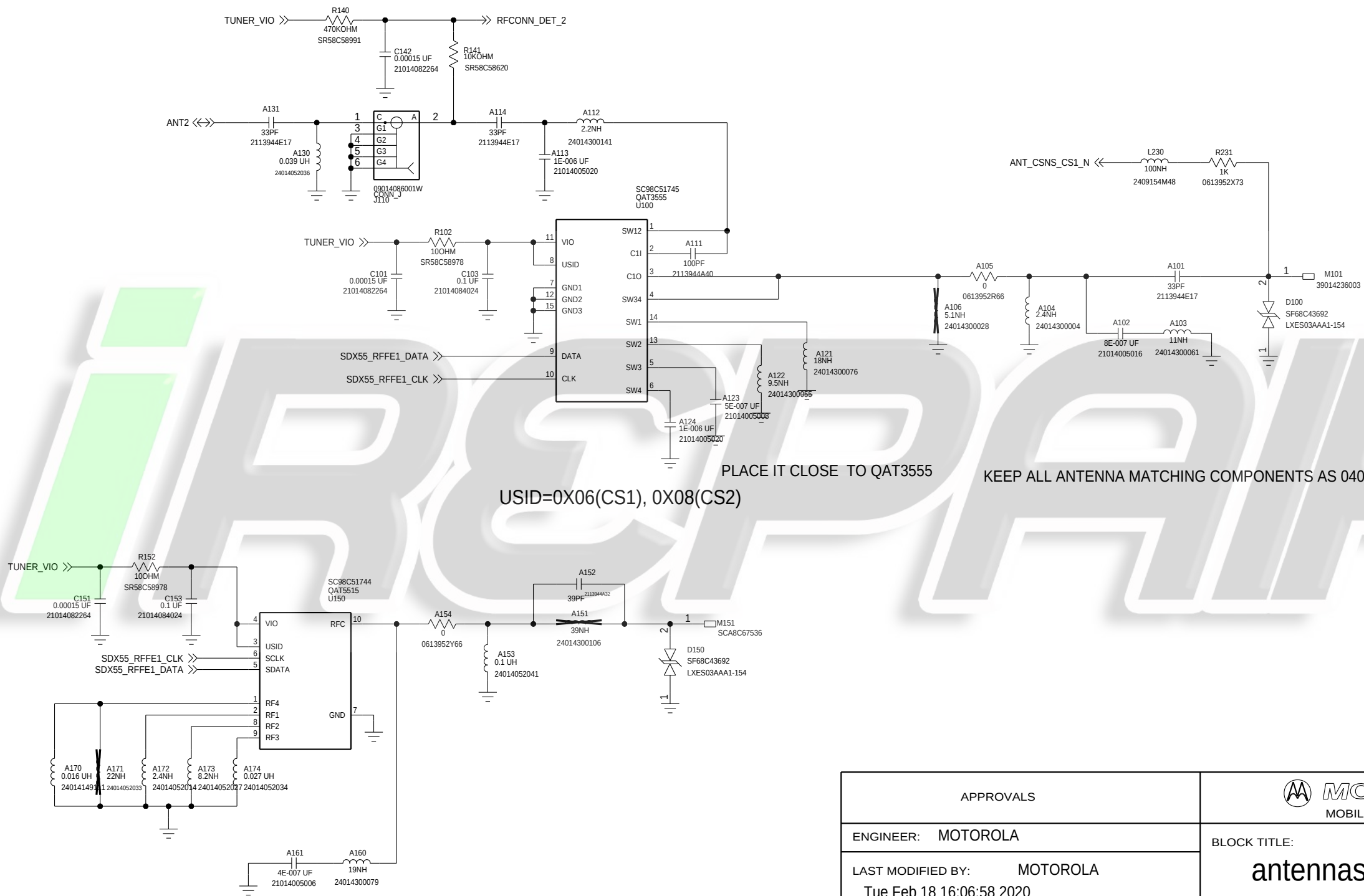
TOP PCB



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: antennas	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:57 2020			
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		CAD FILE NAME: PAGE 4 OF 97	

ANT2: LB+MB+HB TRX1 DIV

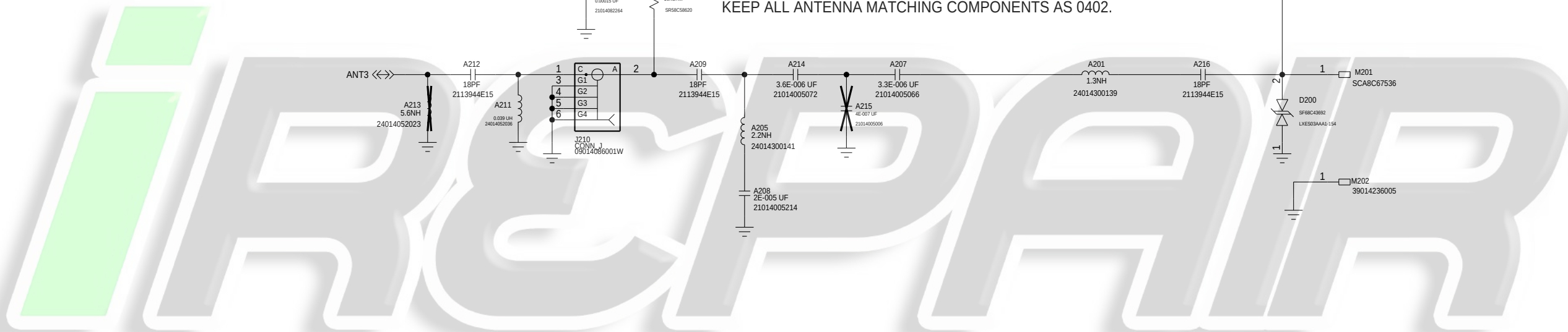
REFDES: 100-199



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:58 2020		antennas	
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		B sr28c51351_b.cpm	
		REV 03	
		PAGE 5 OF 97	

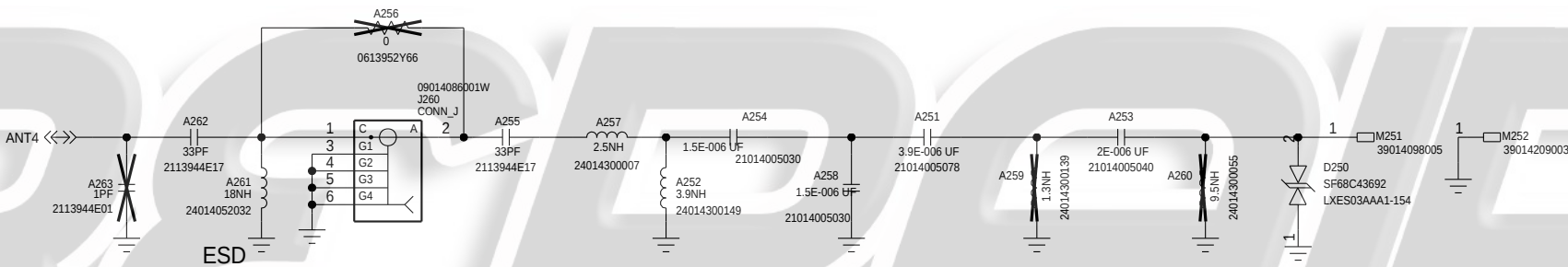
ANT3: GPS+UHB/SUB6 TRX0+LAA RX1+WIFI 5GHZ TRX1

REFDES : 200-249



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:59 2020		antennas	
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		CAD FILE NAME: PAGE 6 OF 97	

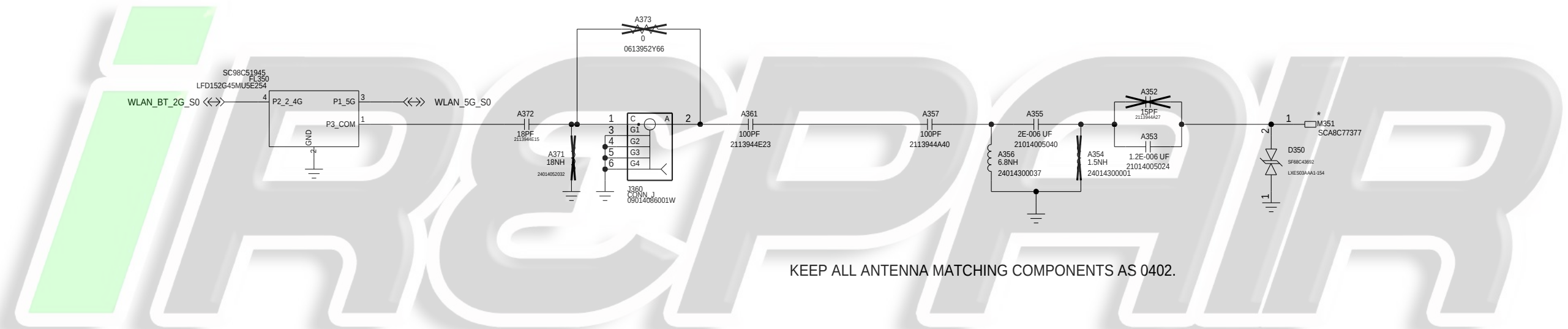
ANT4: UHB/SUB6 TRX2+WIFI 2.4G TRX1
REFDES: 250-299



KEEP ALL ANTENNA MATCHING COMPONENTS AS 0402.

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:07:00 2020		antennas	
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		REV 03	
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		PAGE 7 OF 97	

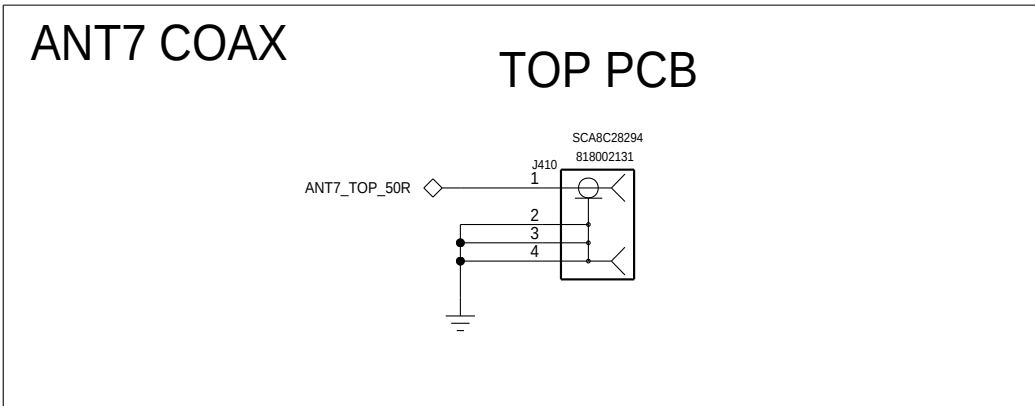
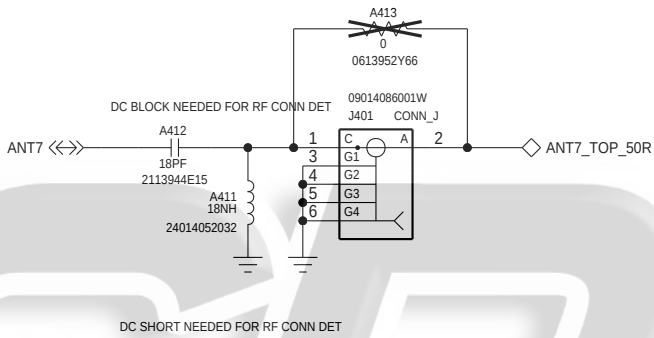
ANT6: BT/WIFI S0+LAA RX0
REFDES: 350-399



KEEP ALL ANTENNA MATCHING COMPONENTS AS 0402.

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: antennas	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:07:01 2020			
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		CAD FILE NAME: PAGE 9 OF 97	

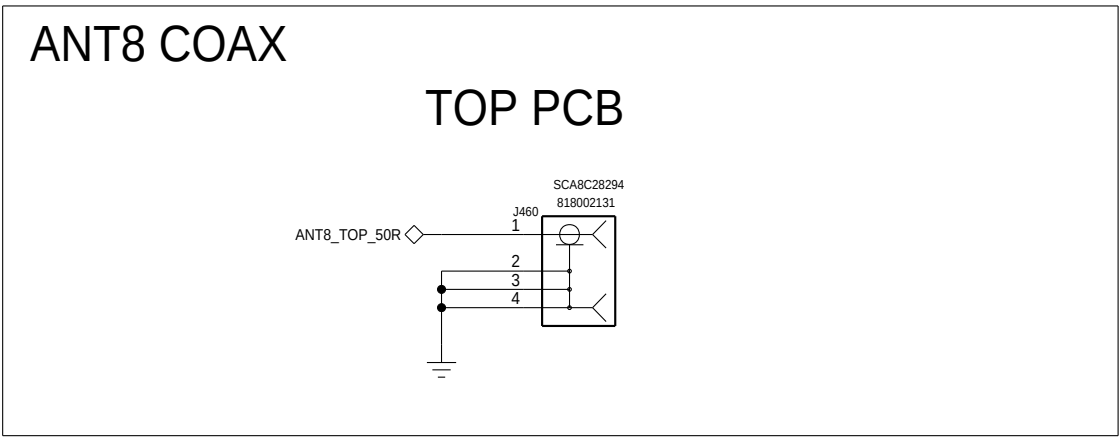
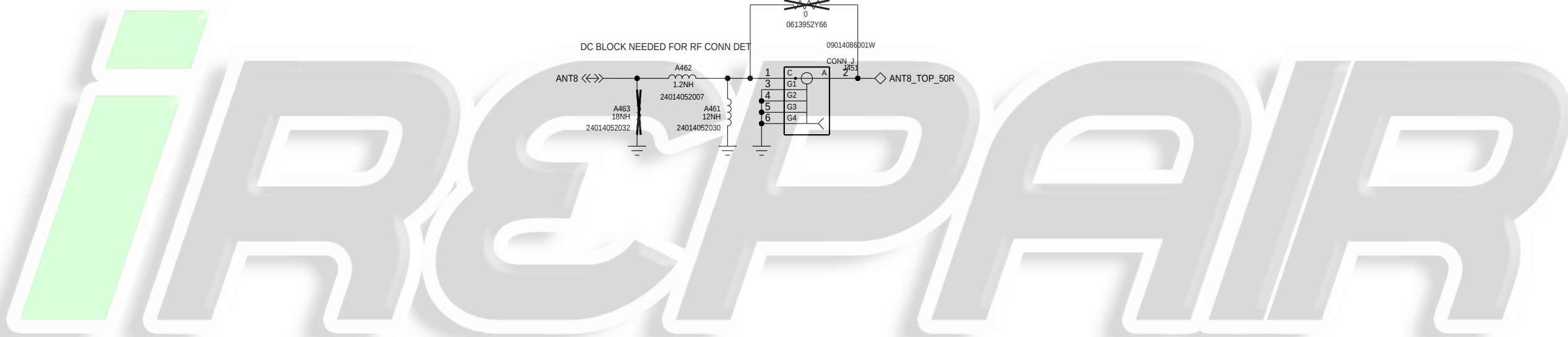
ANT7: UHB/SUB6 TRX1
REFDES: 400-449



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: antennas	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:07:02 2020			
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ANT8: MB/HB TRX2

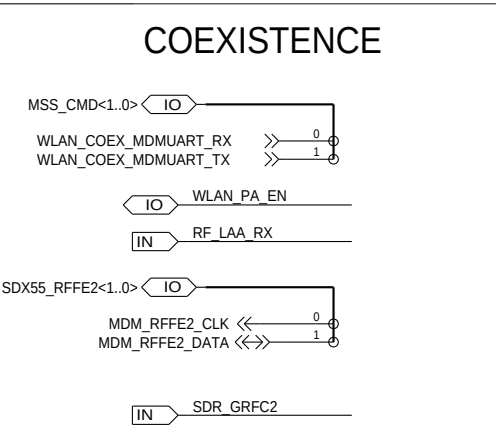
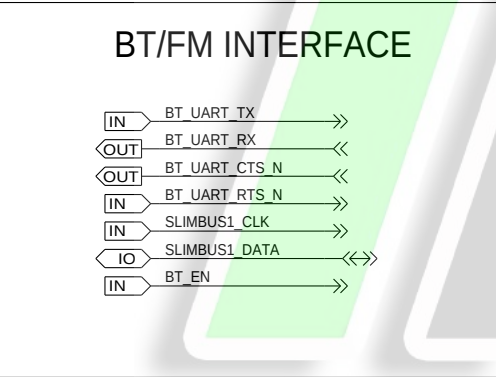
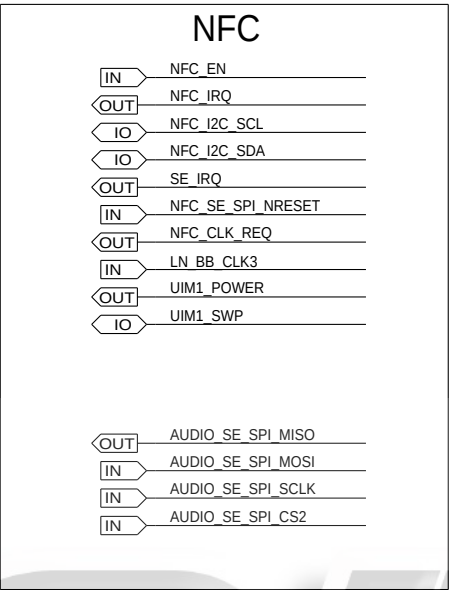
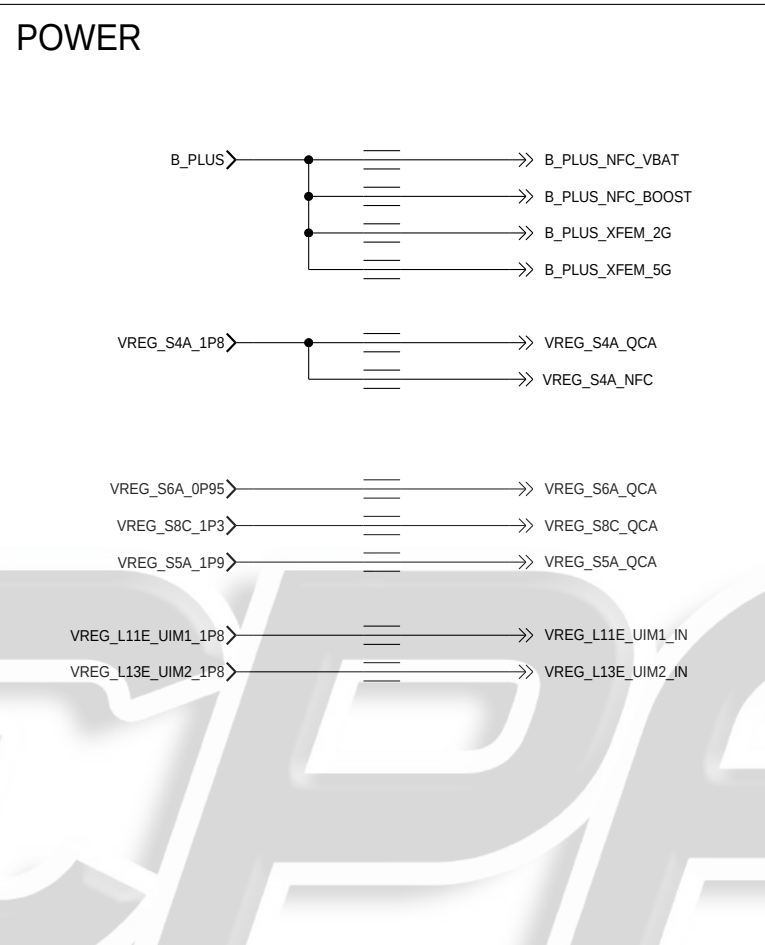
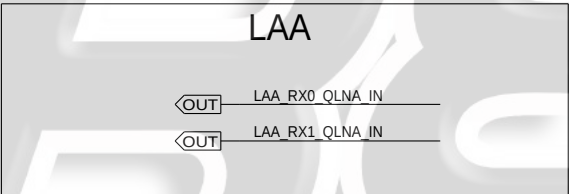
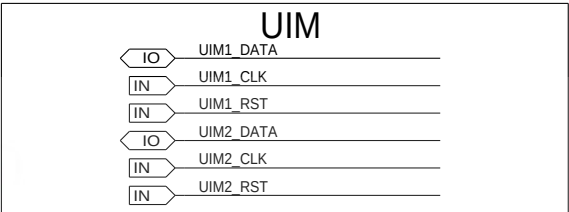
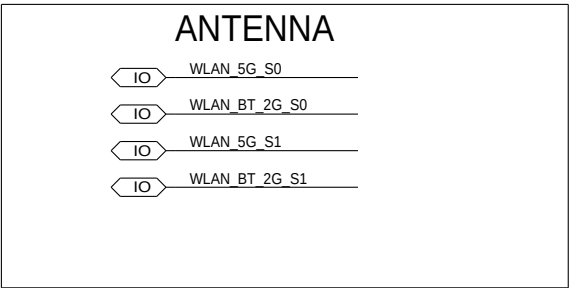
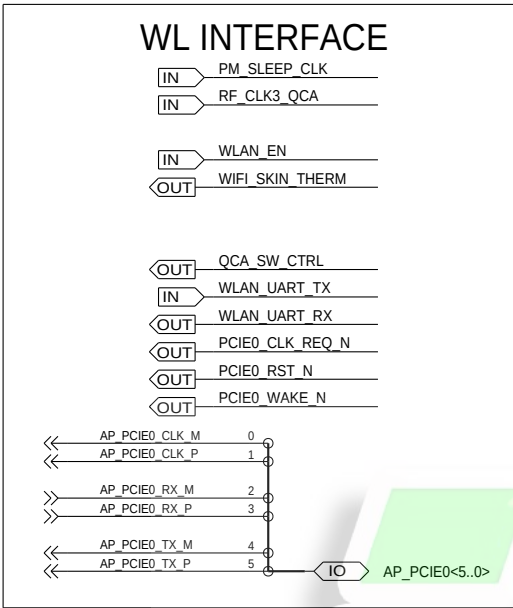
REFDES: 450-499



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:07:03 2020			
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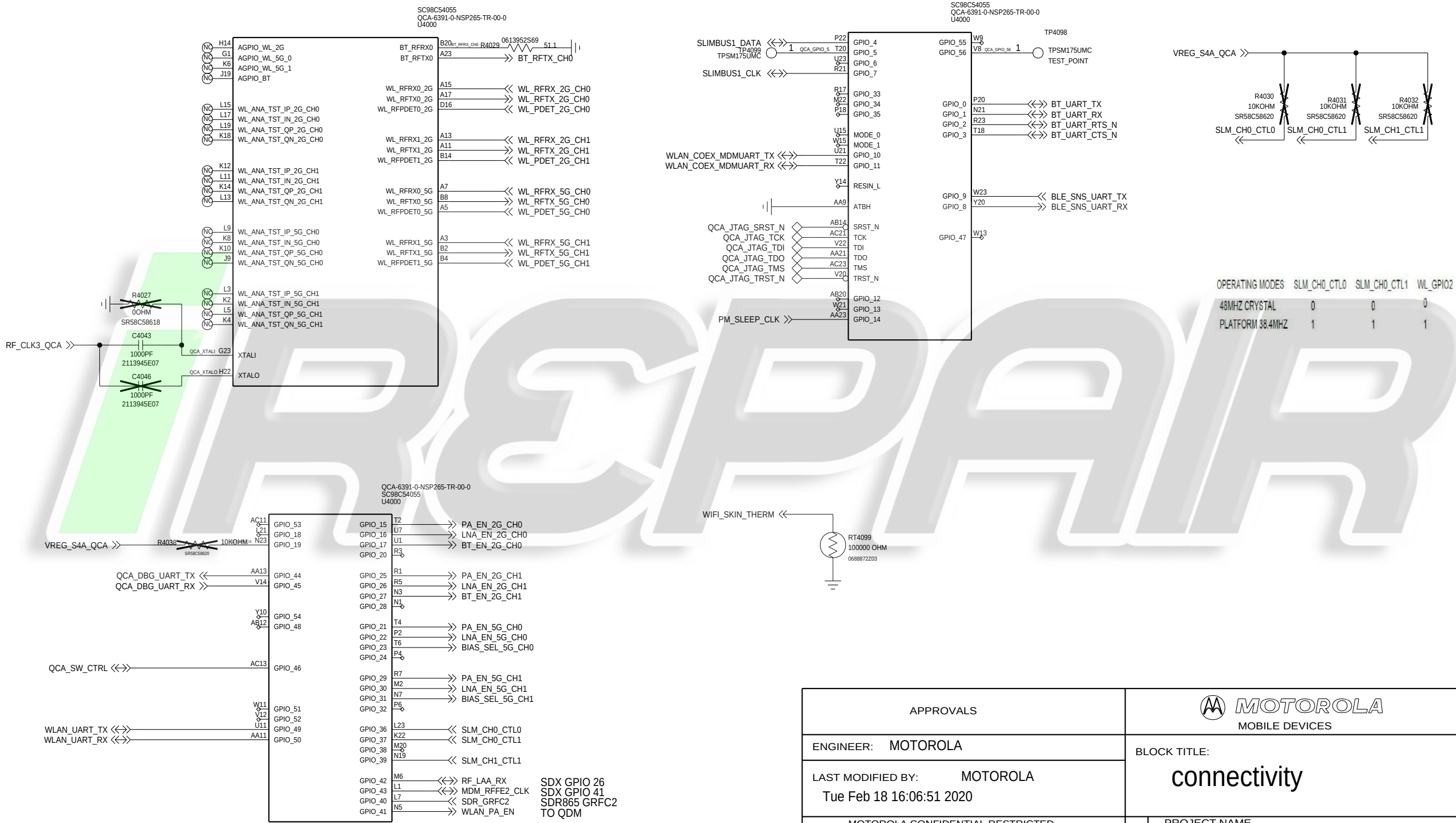
CONN: PORTS

REF 4000-4499



APPROVALS		 MOTOROLA MOBILE DEVICES		
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LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:49 2020				
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		B sr28c51351_b.cpm		03
		CAD FILE NAME:		PAGE 12 OF 97

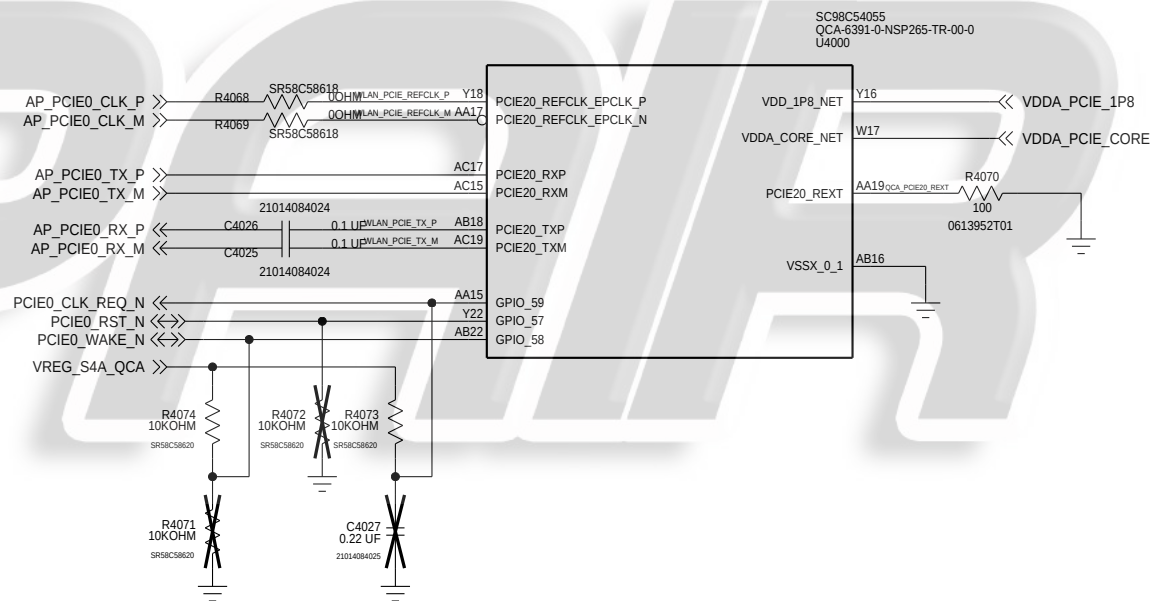
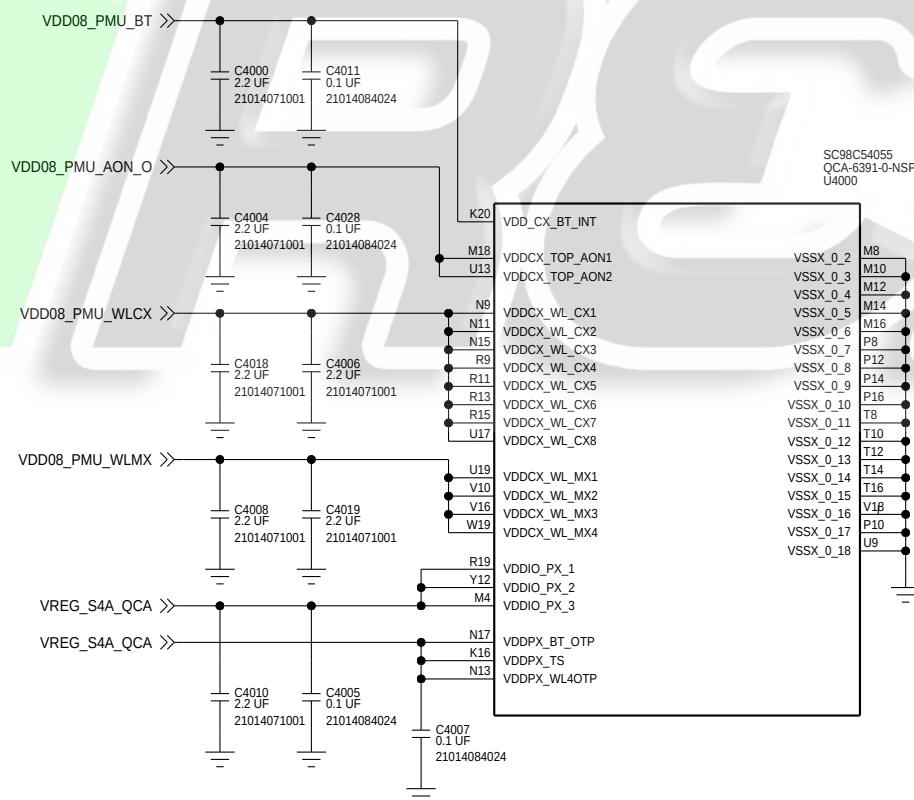
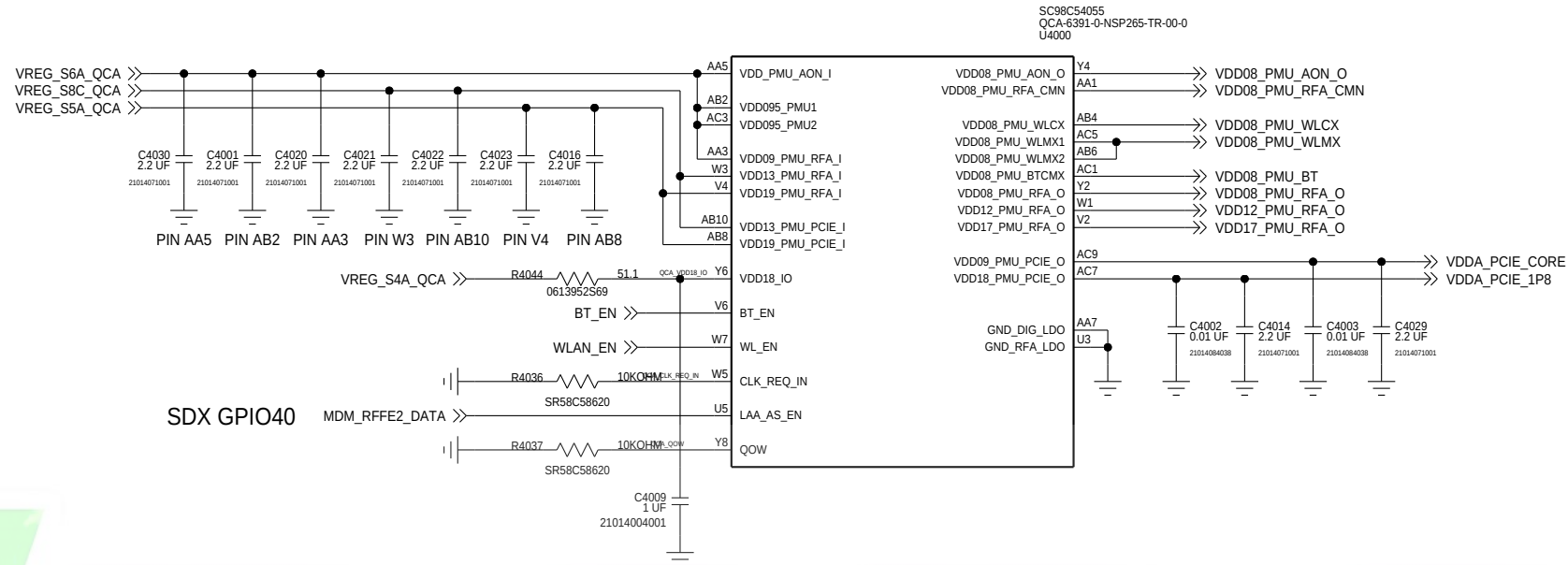
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REF 4000-4099



BASED ON 80-WL020-41 REV. B

APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:51 2020		connectivity	
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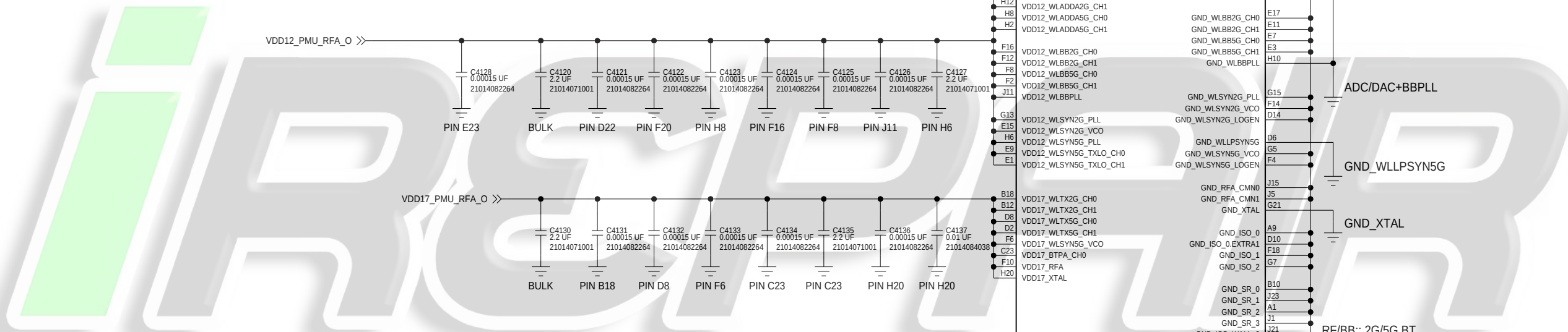
CONN: QCA6391 PWR/GND/PCIE
REF 4000-4099



APPROVALS		BLOCK TITLE:	
ENGINEER: MOTOROLA		connectivity	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:52 2020		PROJECT NAME B sr28c51351_b.cpm	
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		PAGE 14 OF 97	

BASED ON 80-WL020-41 REV. B

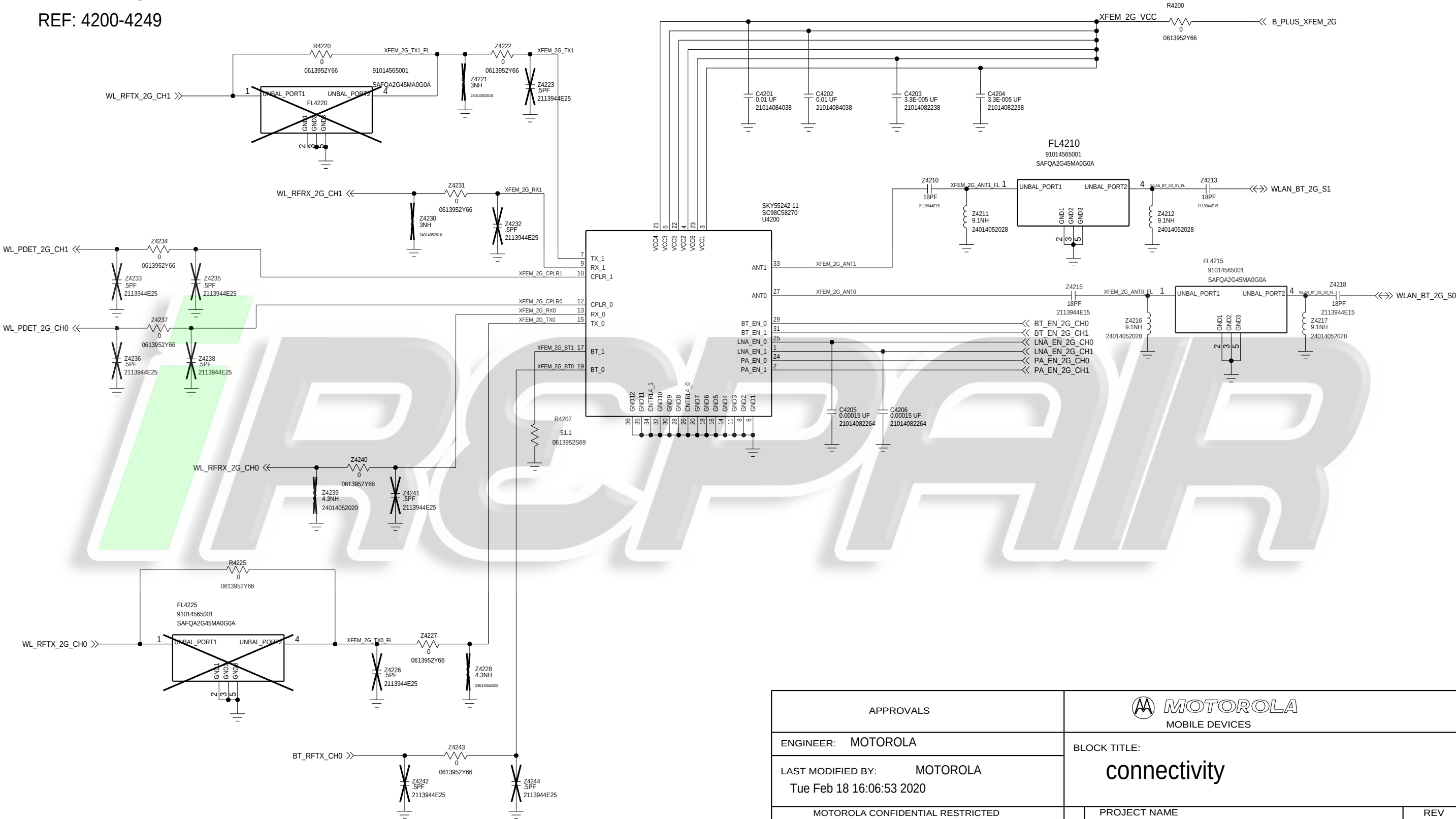
ANIBAL GARCIA VENEZUELA



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: connectivity	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:52 2020			
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		CAD FILE NAME:	
		PAGE 15 OF 97	

CONN: QCA6391 2.4GHZ XFEM

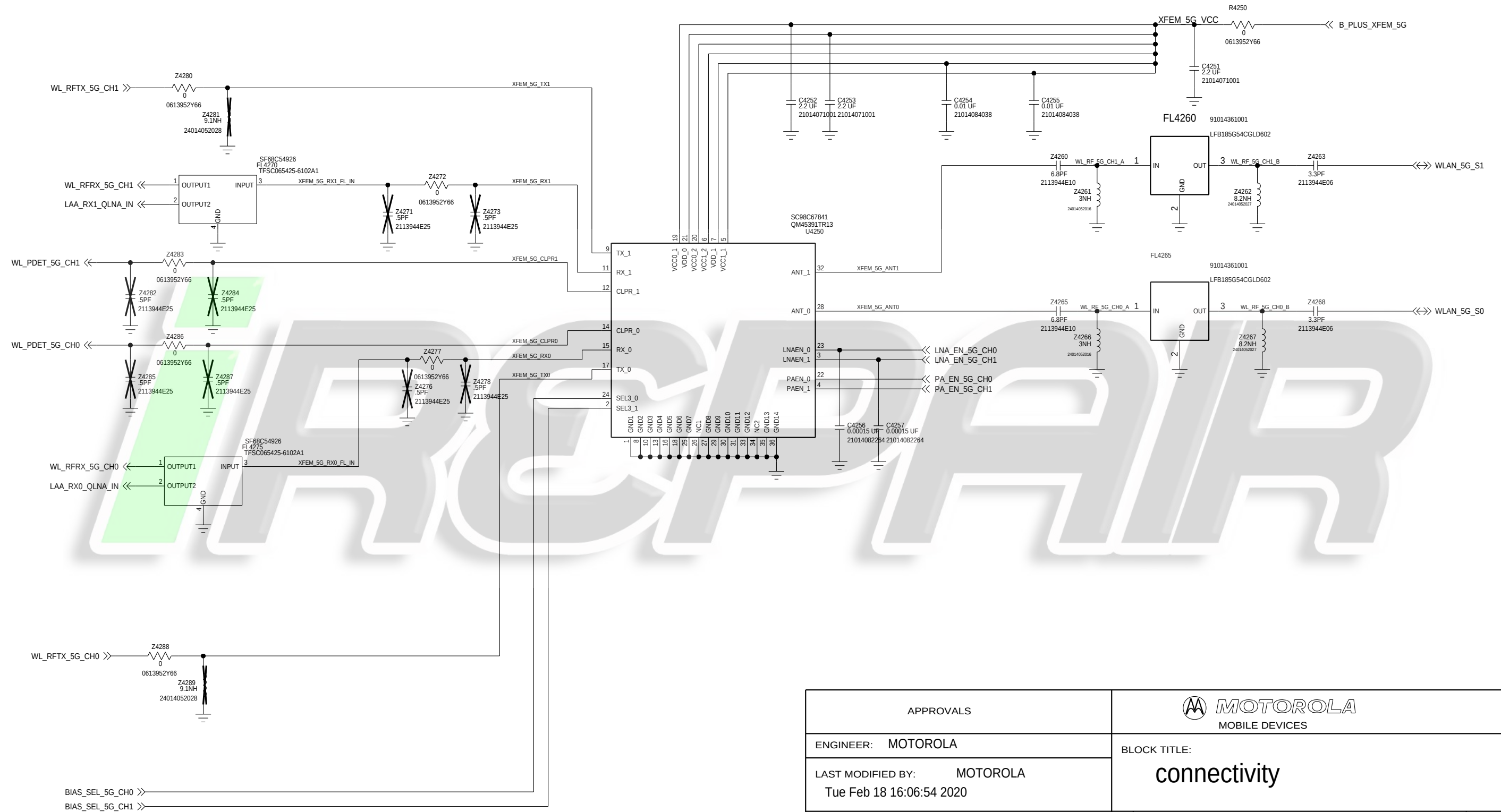
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BASED ON 80-WL020-41 REV. B

APPROVALS		 MOTOROLA MOBILE DEVICES	
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		REV 03	
		PAGE 16 OF 97	

CONN: QCA6391 5GHZ XFEM
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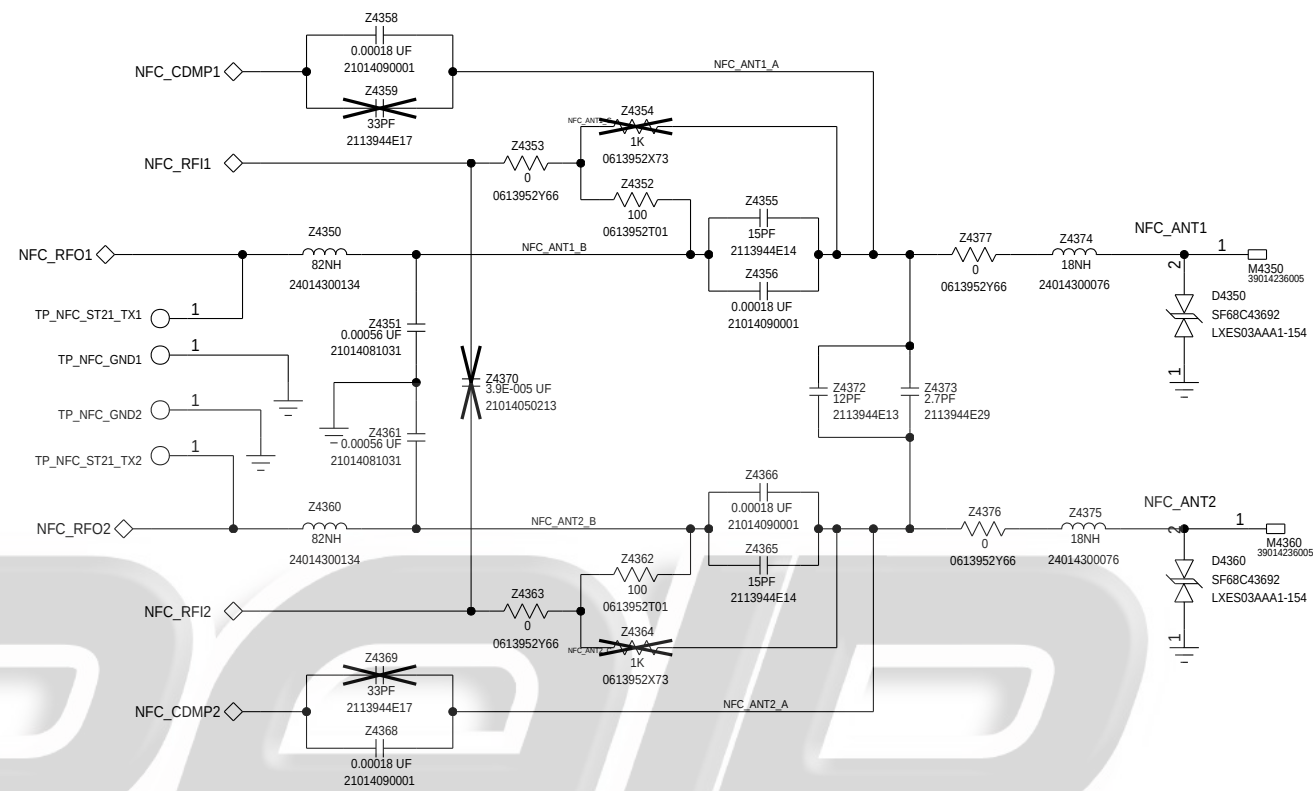


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ENGINEER: MOTOROLA	BLOCK TITLE: connectivity			
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:54 2020				
MOTOROLA CONFIDENTIAL RESTRICTED		B	PROJECT NAME sr28c51351_b.cpm	REV 03
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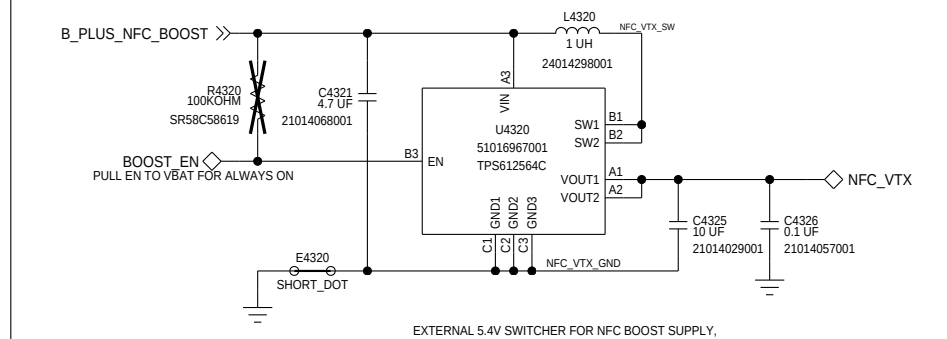
CONN: ST54J NFC

REF: 4300-4399

ANTENNA MATCHING



NFC BOOST



APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
Tue Feb 18 16:06:55 2020

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BLOCK TITLE:

connectivity

PROJECT NAME

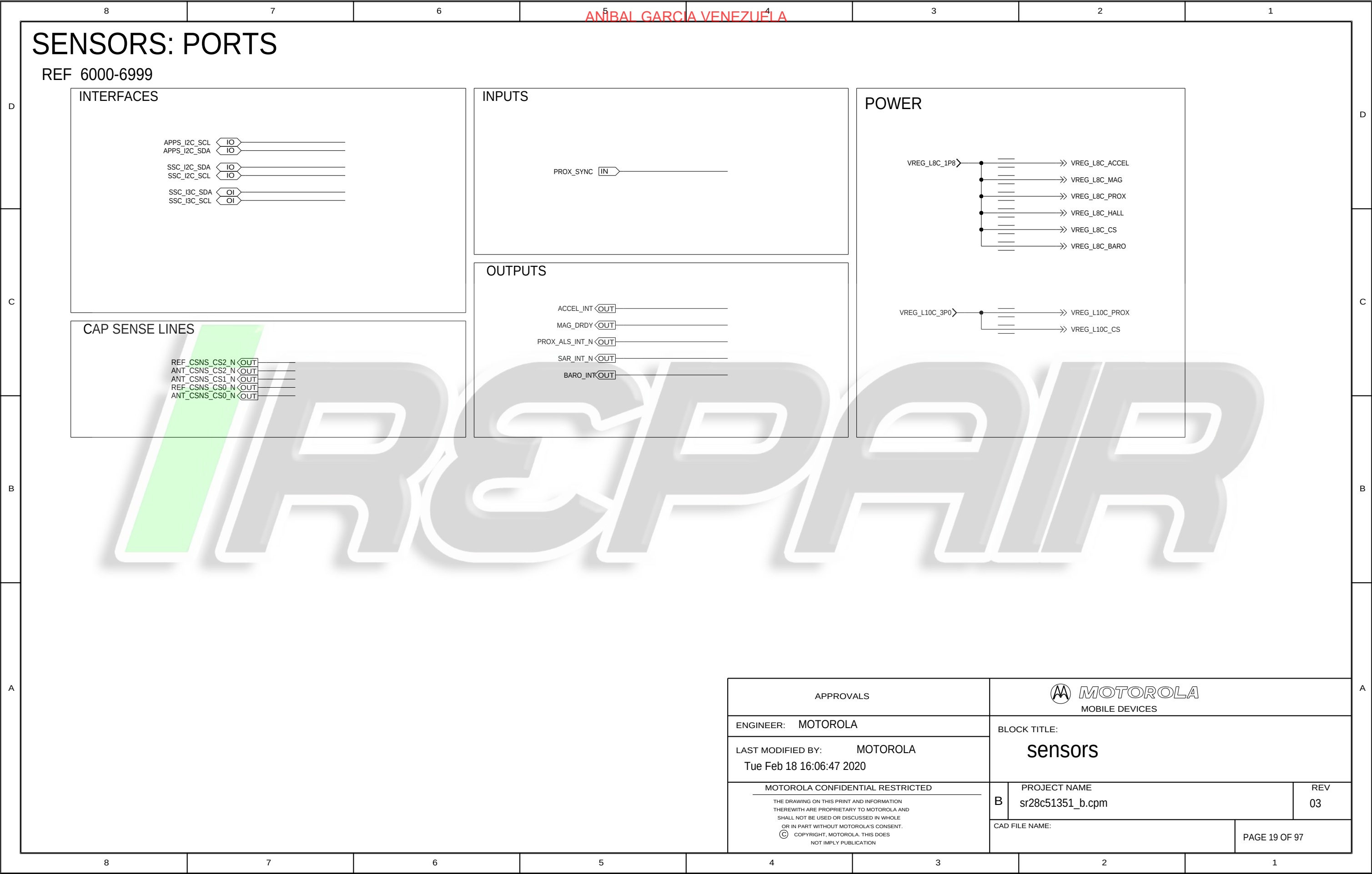
B sr28c51351_b.cpm

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03

CAD FILE NAME:

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APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: sensors	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:47 2020			
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		CAD FILE NAME:	PAGE 19 OF 97

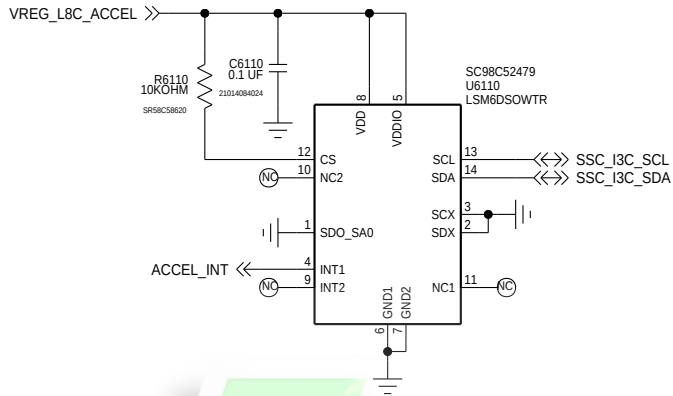
SENSORS: MOTION

REF: 6100-6159

ACCELEROMETER/GYROSCOPE

REF:6110-6119

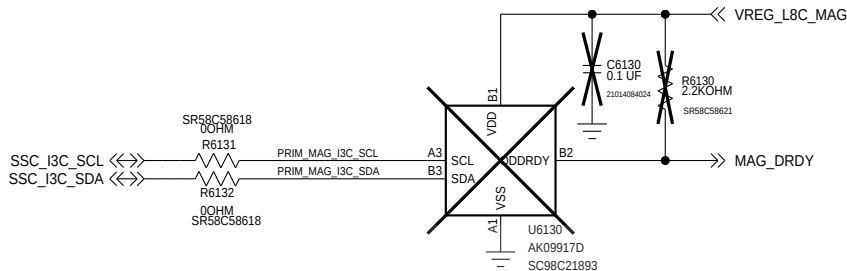
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MAGNETOMETER

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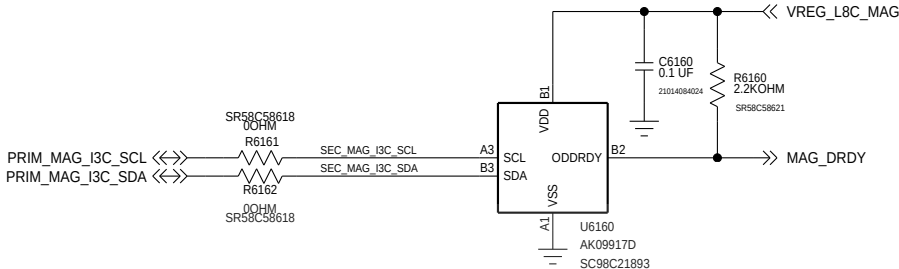
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MAGNETOMETER (SECONDARY)

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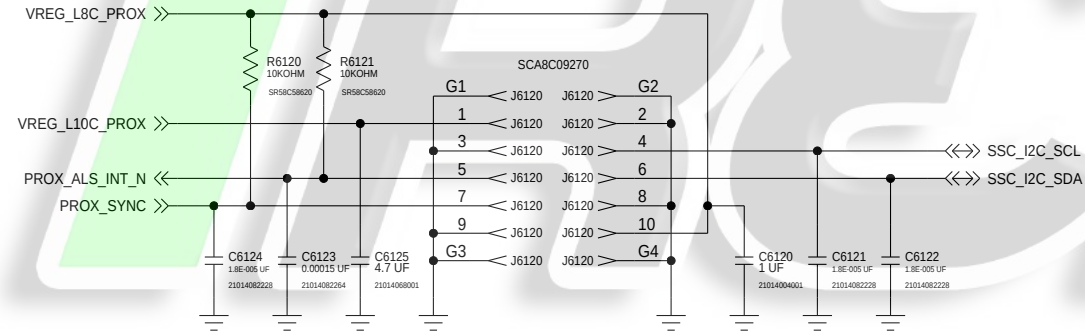
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PROXIMITY / AMBIENT LIGHT

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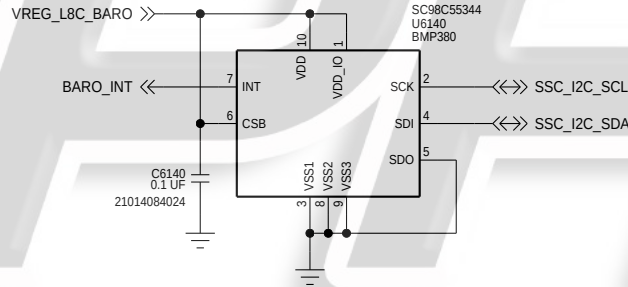
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BAROMETER

REF:6140-6149

I2C ADDR:0X76



APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
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BLOCK TITLE:

sensors

PROJECT NAME

B sr28c51351_b.cpm

CAD FILE NAME:

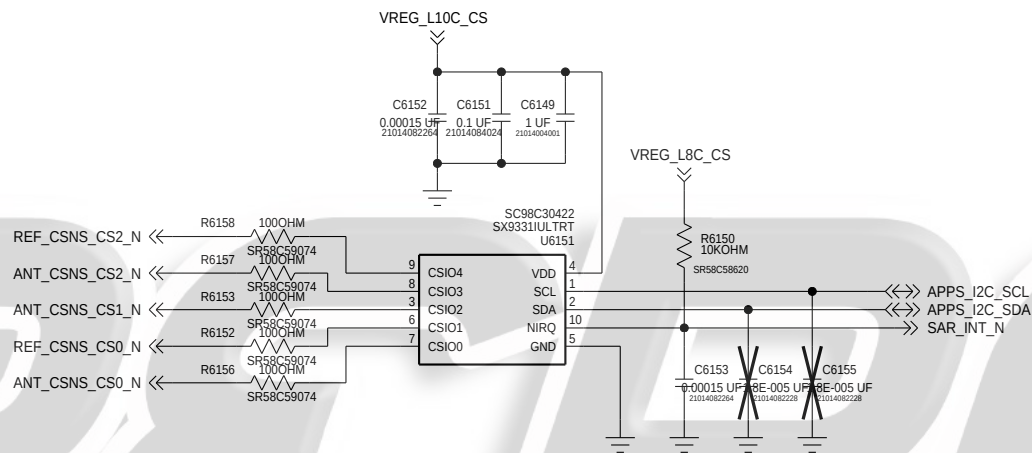
REV

03

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SENSORS: CAP SENSE

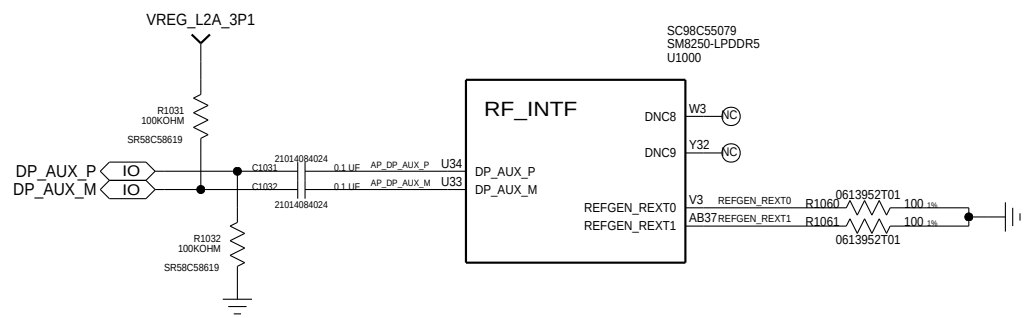
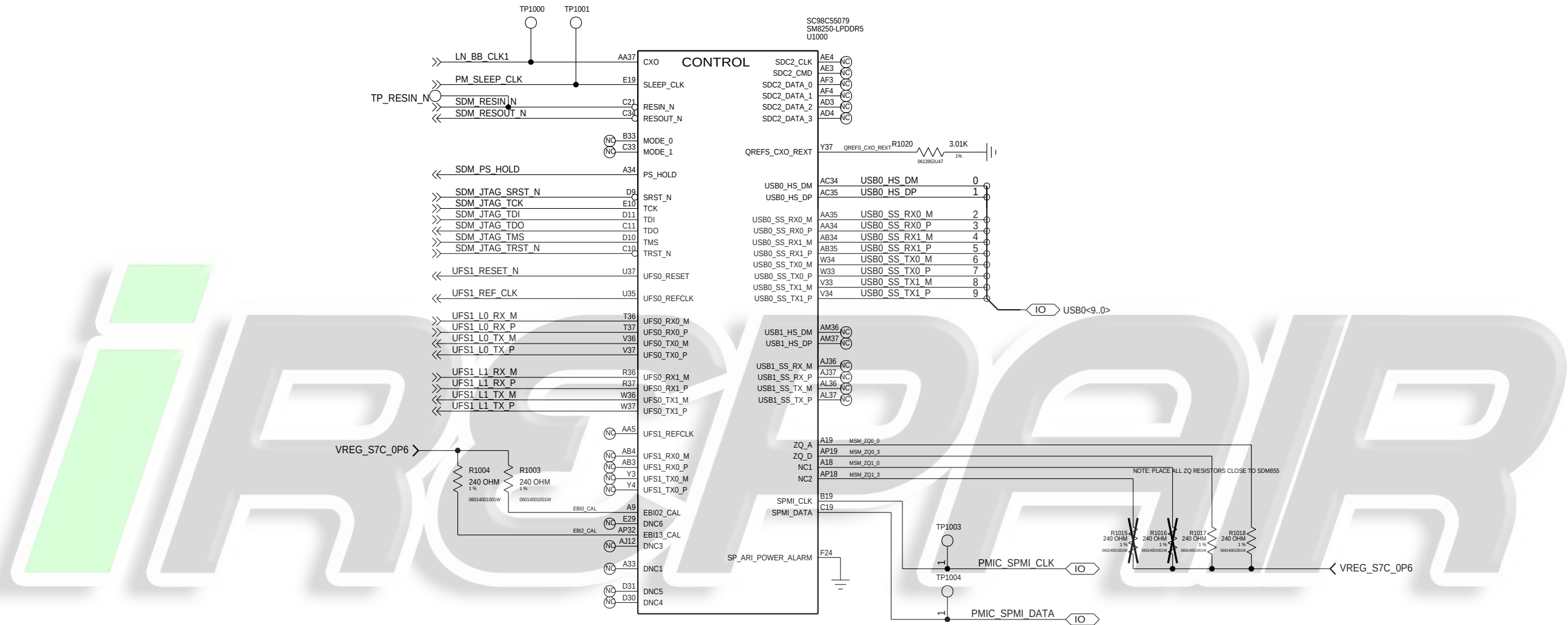
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PRIMARY I2C ADDR:0X28
SECONDARY I2C ADDR:0X2C



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: MMI		BLOCK TITLE: sensors		
LAST MODIFIED BY: MMI Tue Feb 18 16:06:48 2020				
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		B sr28c51351_b.cpm		03
		CAD FILE NAME:		PAGE 21 OF 97

SM8250: CONTROL

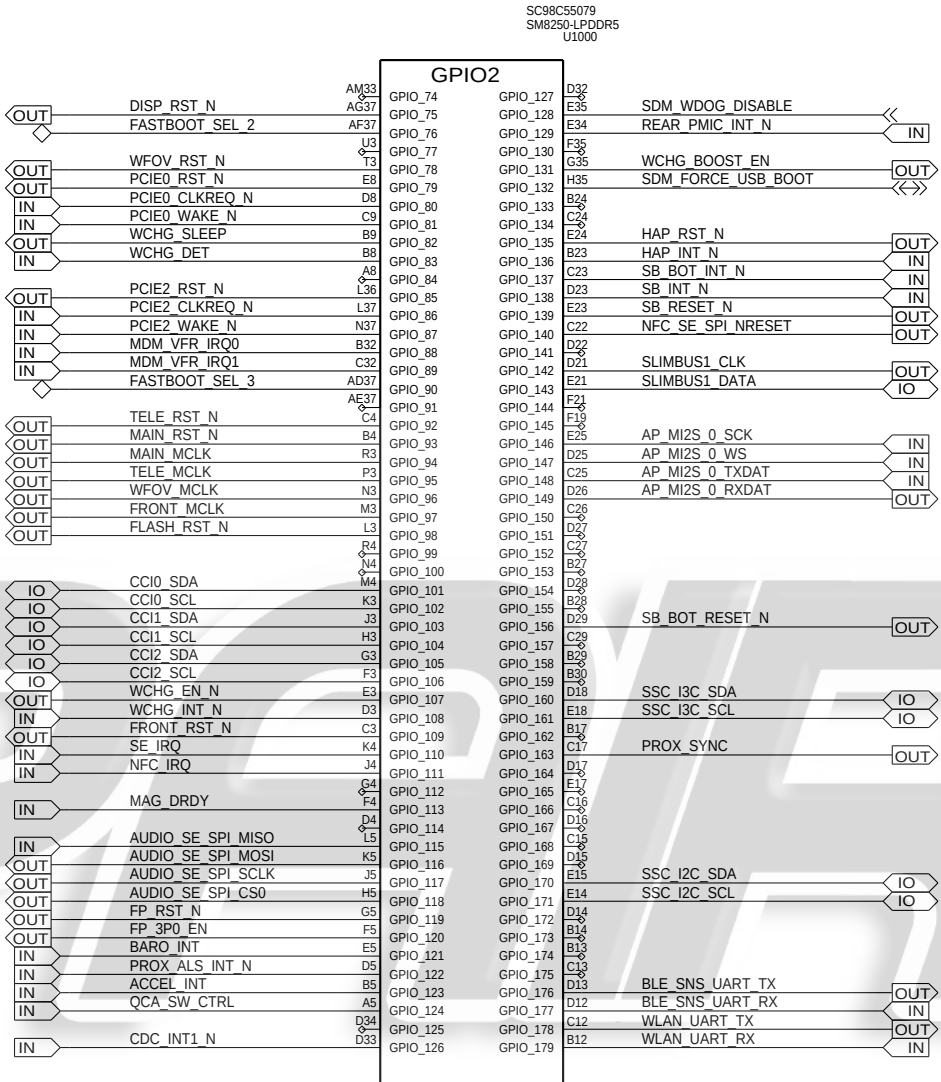
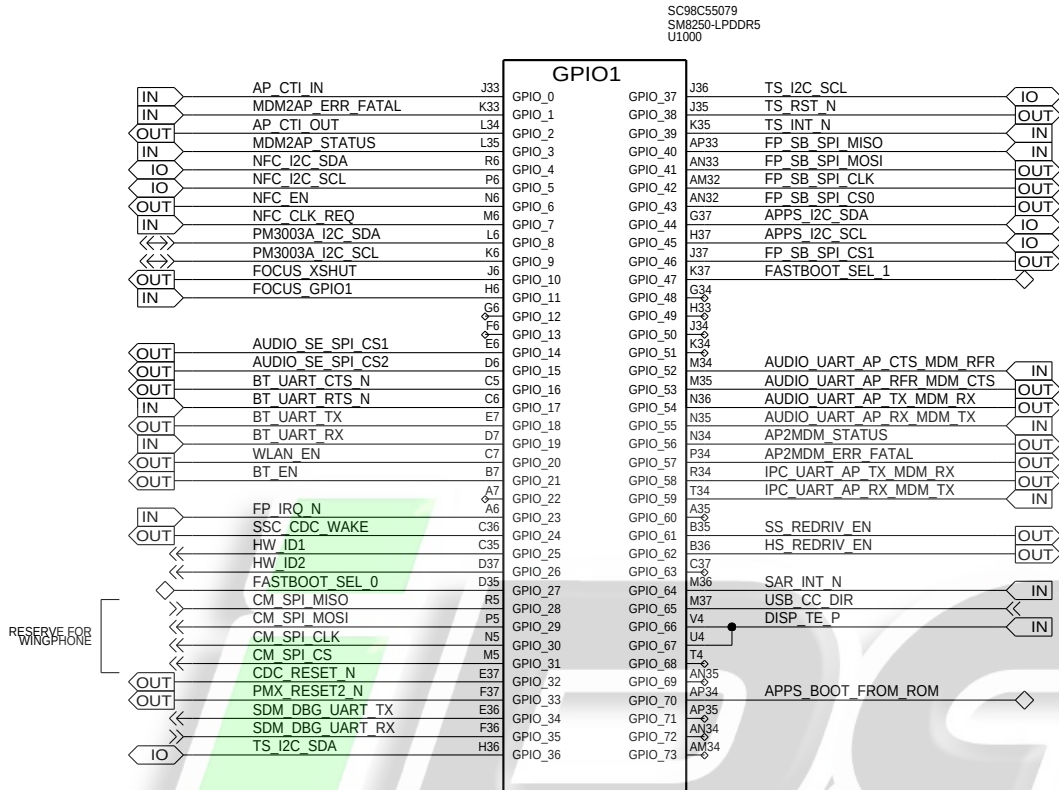
REFDES: 1000-1099



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: ap_pmic	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:28 2020		PROJECT NAME B sr28c51351_b.cpm	
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SM8250: GPIO 1/2

REFDES: 1100-1199



BOOT CONFIG PINS
MUST BE FLOATING OR PULLED DOWN AT BOOT TIME
ADD A SYNONYM HERE INSTEAD OF RENAMING THE GPIO

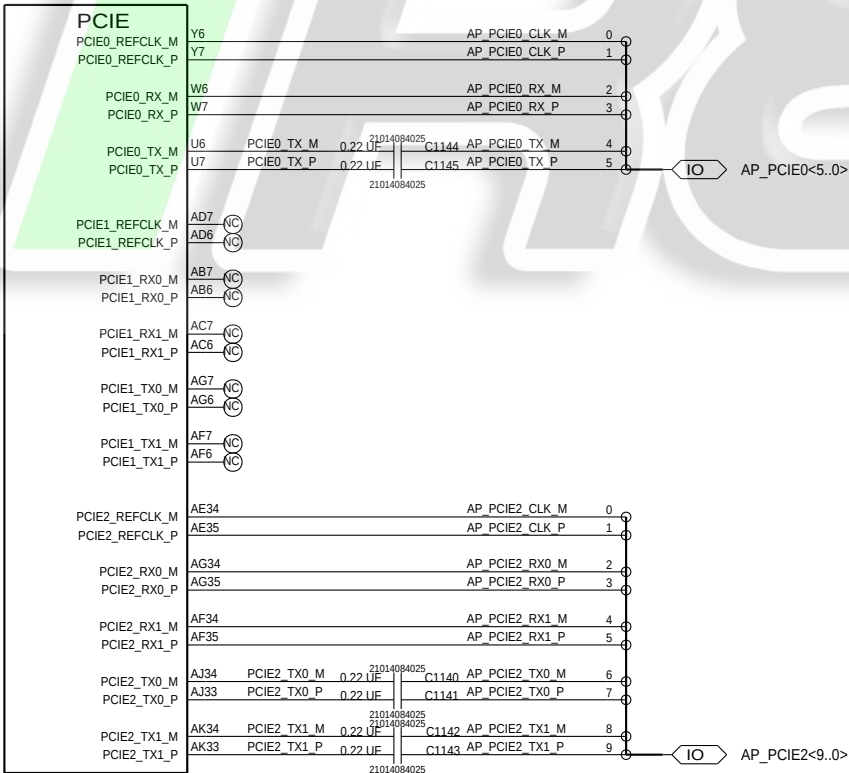
- SDM_WDOG_DISABLE
- FASTBOOT_SEL_0
- FASTBOOT_SEL_1
- FASTBOOT_SEL_2
- FASTBOOT_SEL_3
- APPS_BOOT_FROM_ROM
- SDM_FORCE_USB_BOOT

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: ap_pmic	
LAST MODIFIED BY: MOTOROLA Fri Dec 27 14:21:43 2019		PROJECT NAME B sr28c51351_b.cpm	
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		CAD FILE NAME: PAGE 23 OF 97	

SM8250: MIPI/PCIE

REFDES: 1100-1199

SC98C55079
SM8250-LPDDR5
U1000



REAR MAIN CAMERA

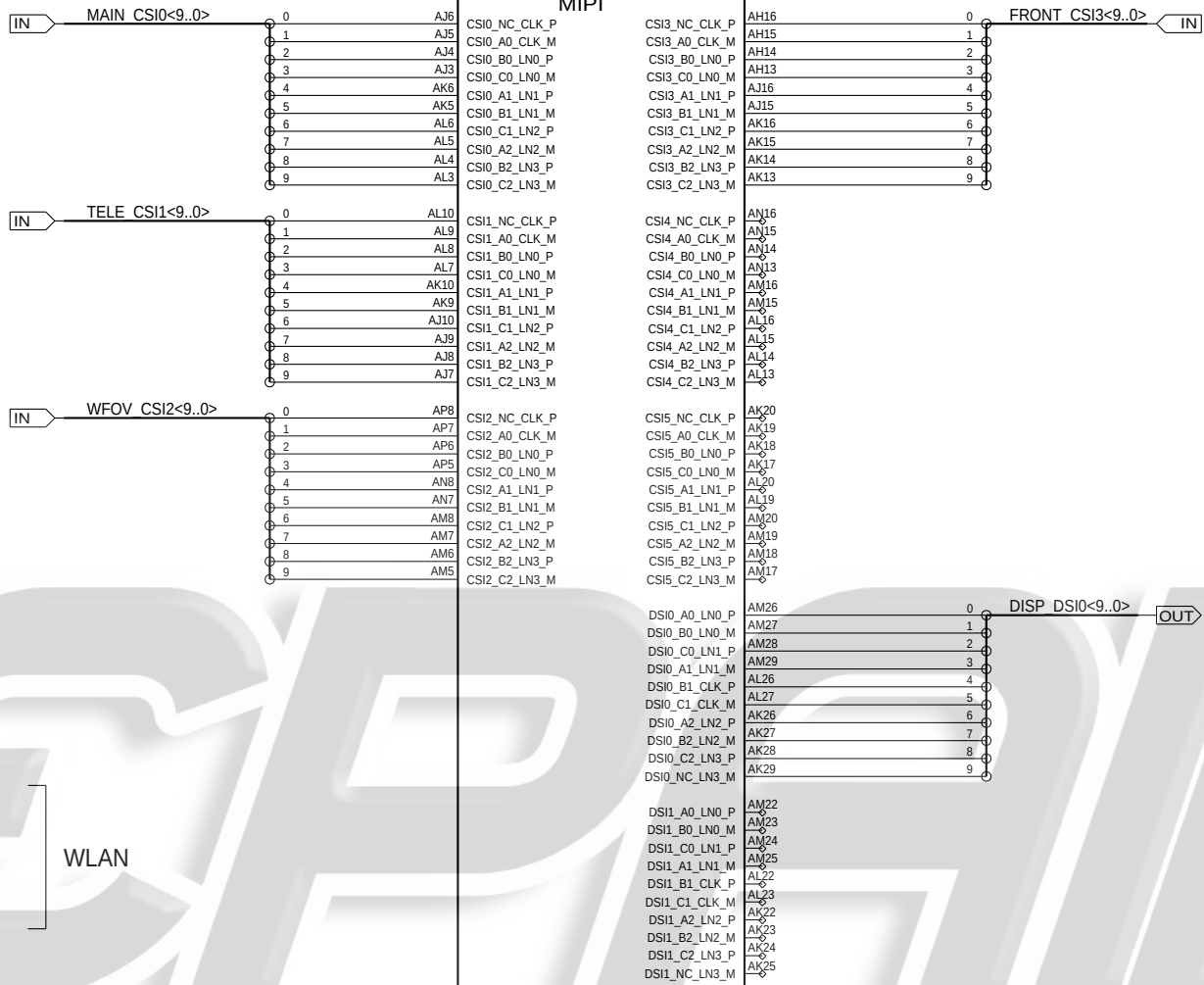
REAR ZOOM CAMERA

REAR WFOV CAMERA

WLAN

MODEM

MIPI



FRONT CAMERA

DISPLAY

APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
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BLOCK TITLE:

ap_pmic

B

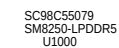
PROJECT NAME
sr28c51351_b.cpm

REV
03

CAD FILE NAME:

PAGE 24 OF 97

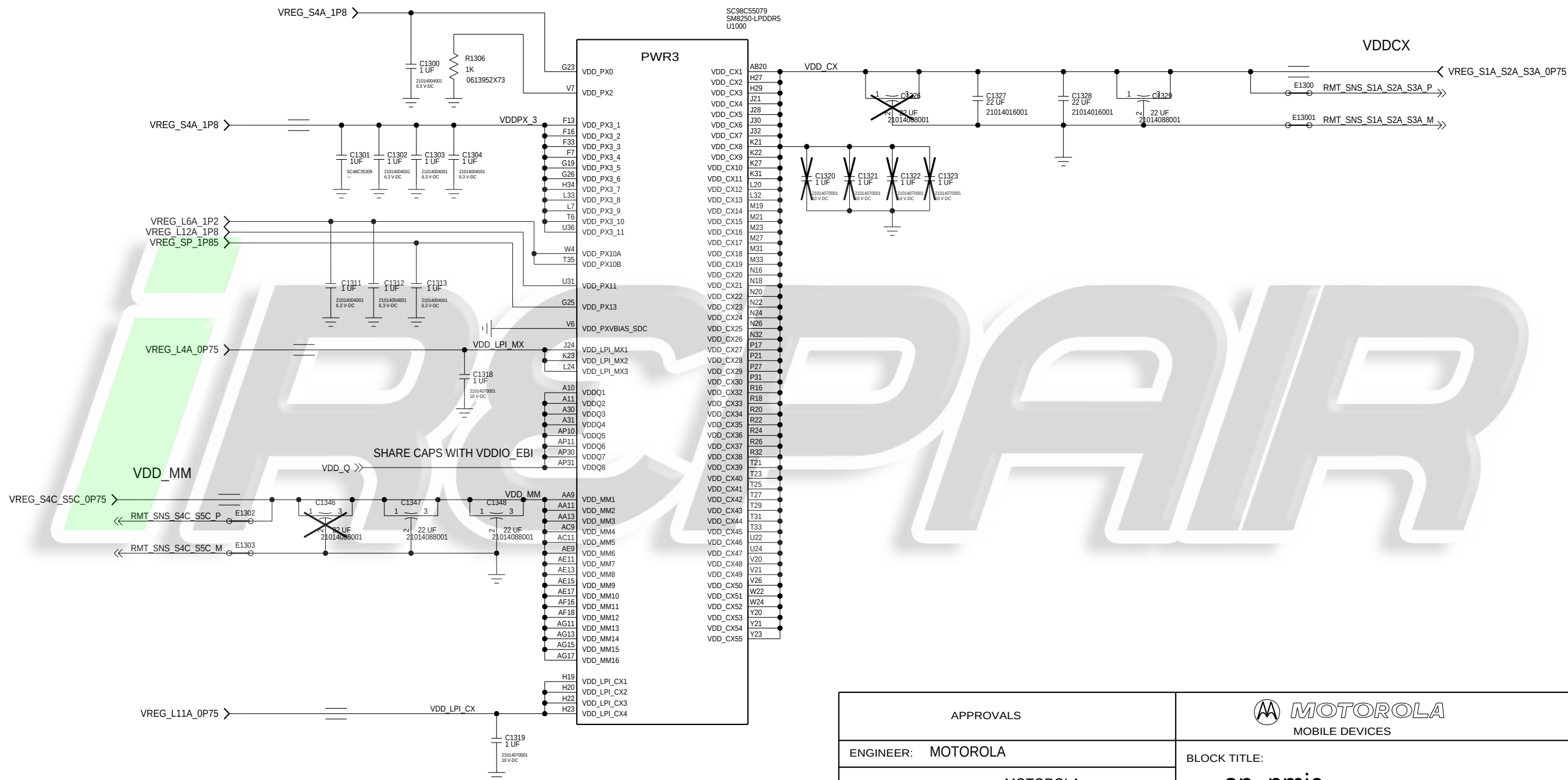
REFDES: 1200-1299



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA	BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:32 2020			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		sr28c51351_b.cpm	03
		CAD FILE NAME:	PAGE 26 OF 97

SM8250: PWR 3

REFDES: 1300-1399



APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
Tue Feb 18 16:06:33 2020

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MOTOROLA

MOBILE DEVICES

BLOCK TITLE:

ap_pmic

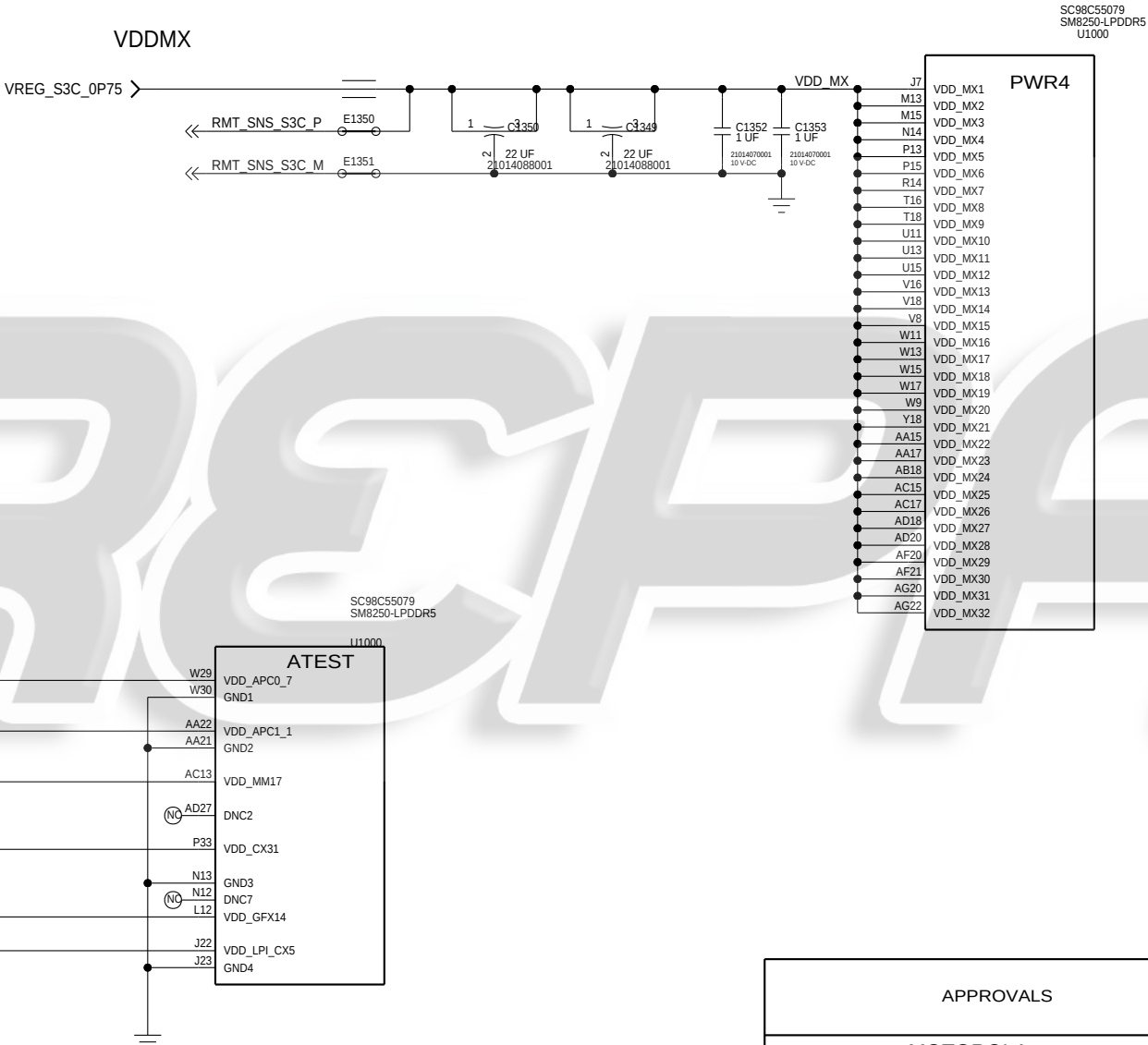
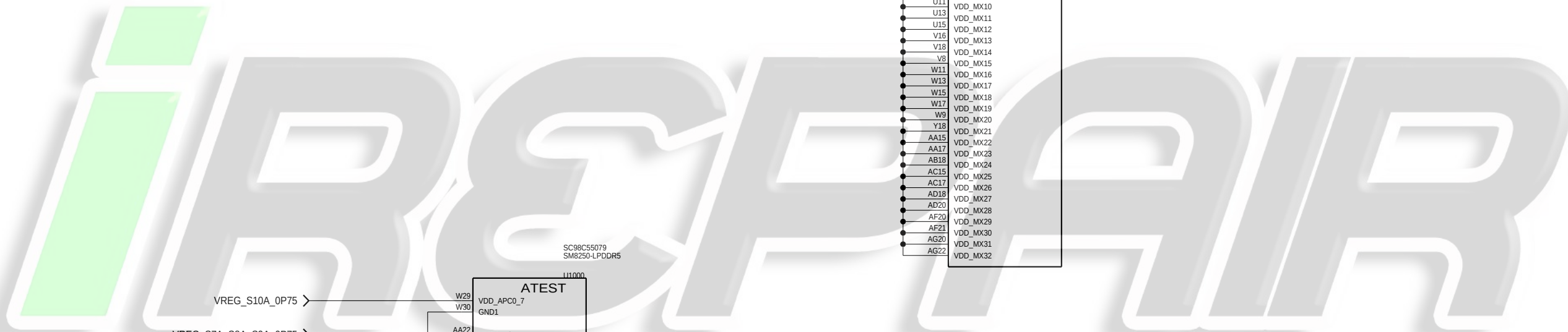
B

PROJECT NAME
sr28c51351_b.cpmREV
03

CAD FILE NAME:

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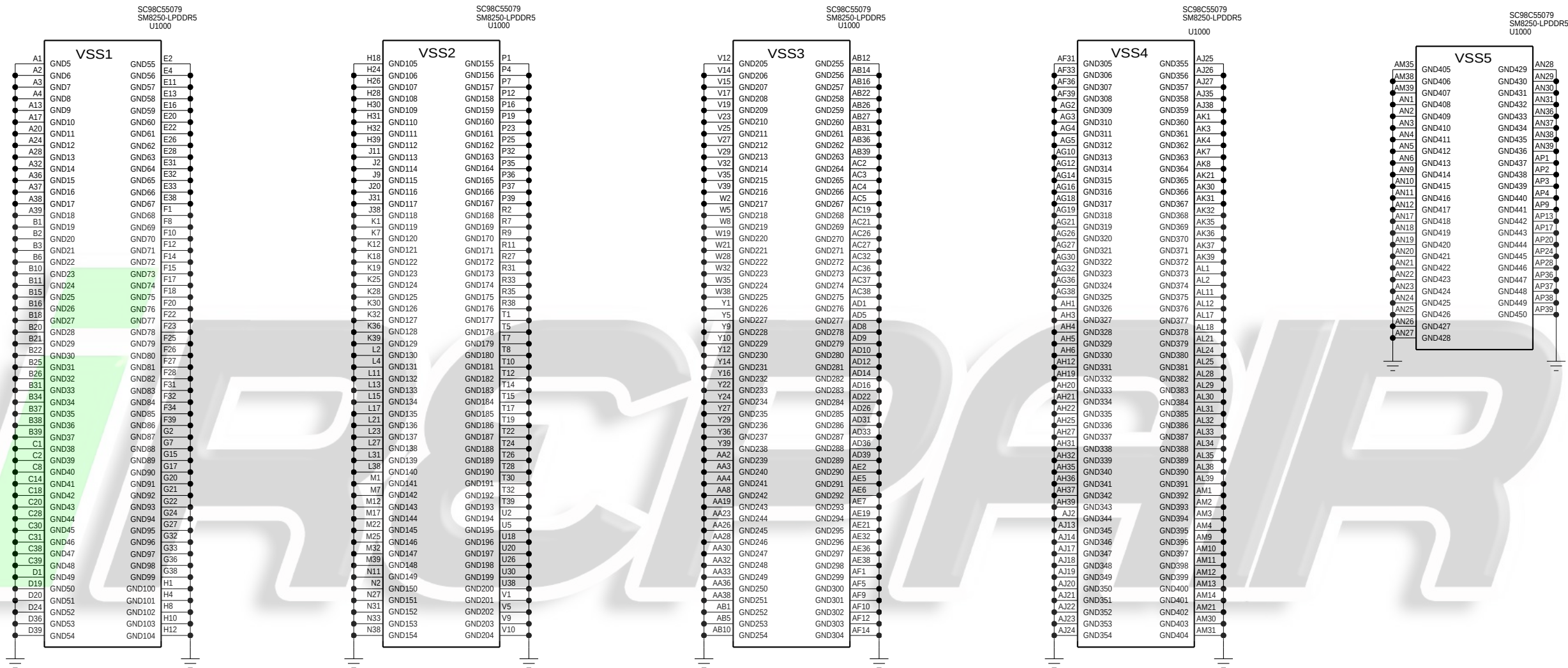
SM8250: PWR 4
REFDES: 1300-1399



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:34 2020		ap_pmic	
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		CAD FILE NAME: PAGE 28 OF 97	

SM8250: GND

REFDES: 1100-1199

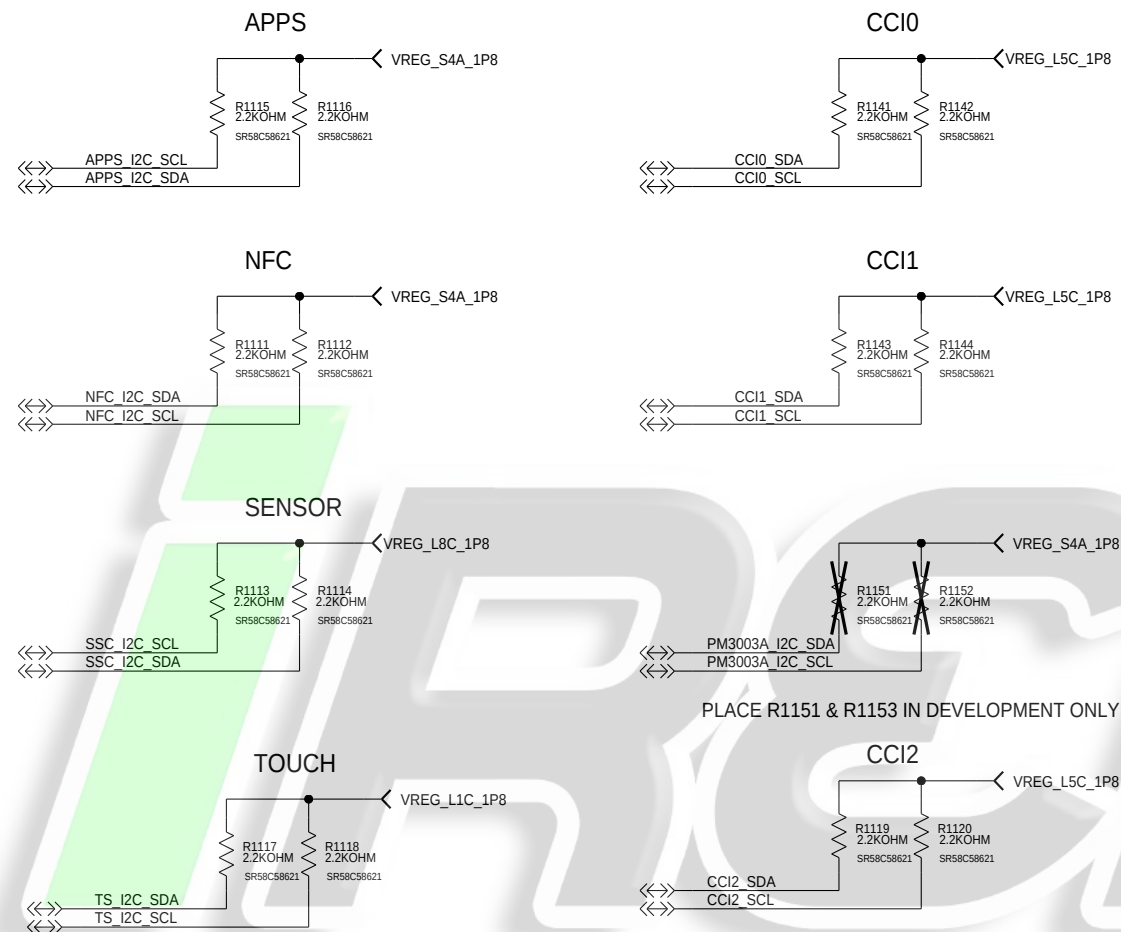


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: ap_pmic	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:35 2020			
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		CAD FILE NAME:	PAGE 29 OF 97

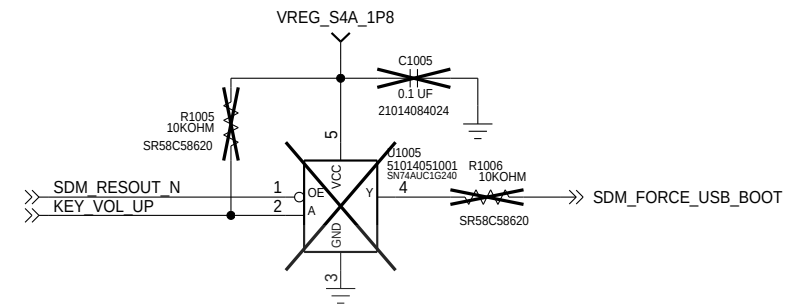
SM8250: SUPPORT

REFDES: 1000-1099

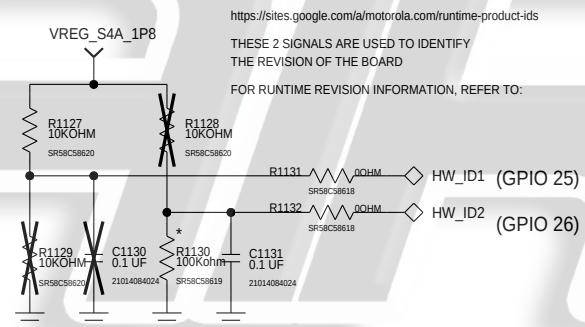
I2C PULL-UP RESISTORS



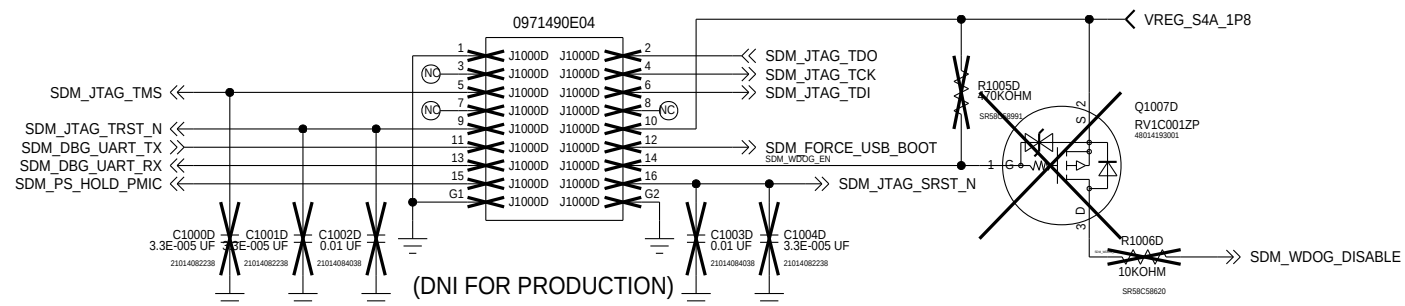
(DNI FOR PRODUCTION AND LAB ENTRY)

FORCE FLASH SUPPORT
WITH VOL UP KEY

PRODUCT HW ID



JTAG:MINI



APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
Tue Feb 18 16:06:36 2020

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

ap_pmic

B

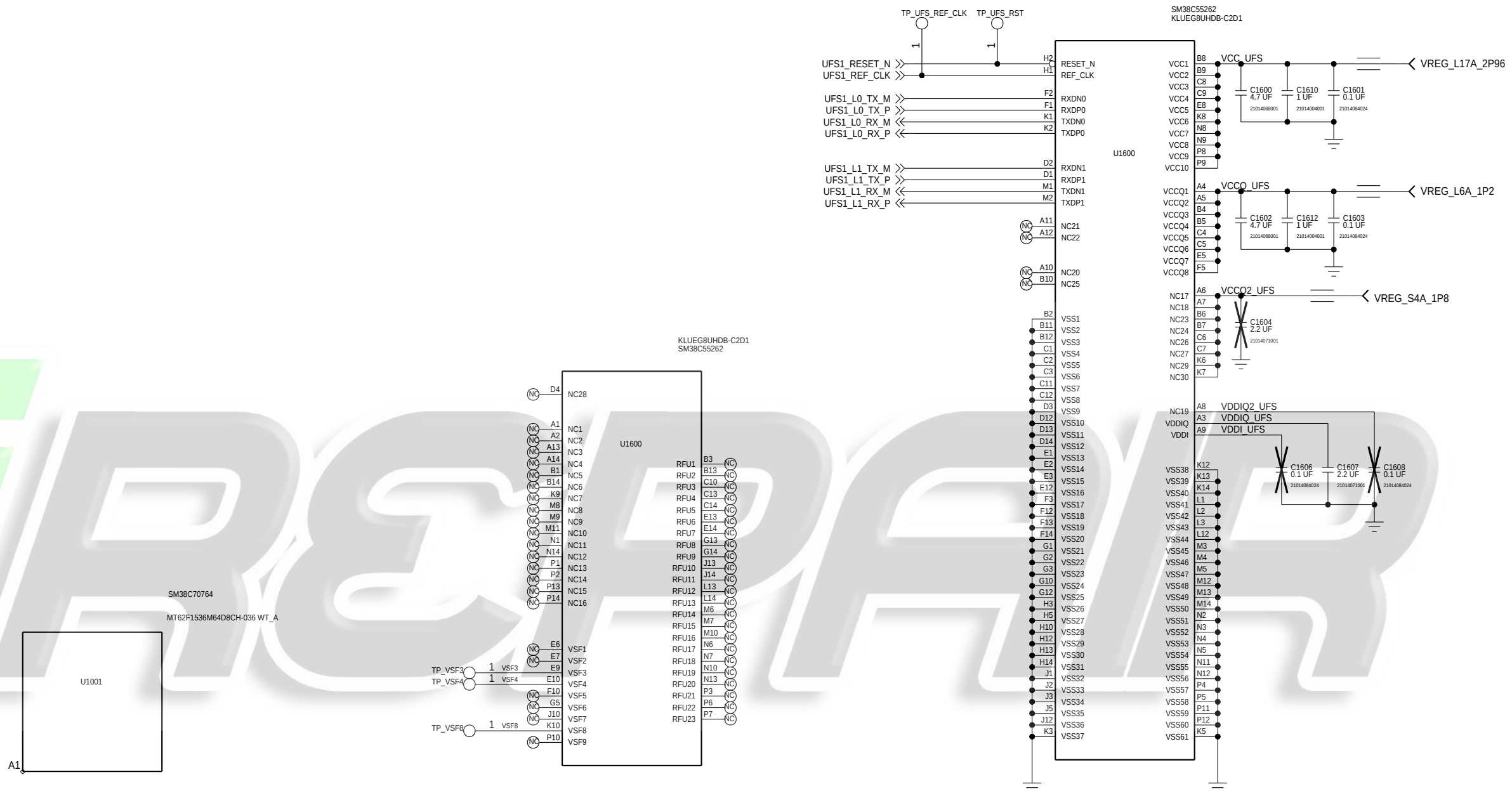
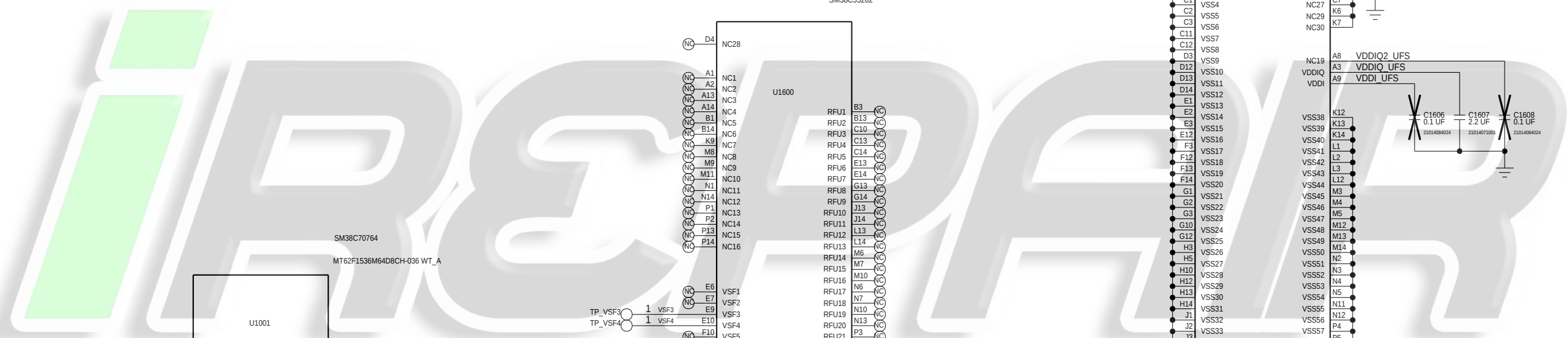
PROJECT NAME
sr28c51351_b.cpmREV
03

CAD FILE NAME:

PAGE 30 OF 97

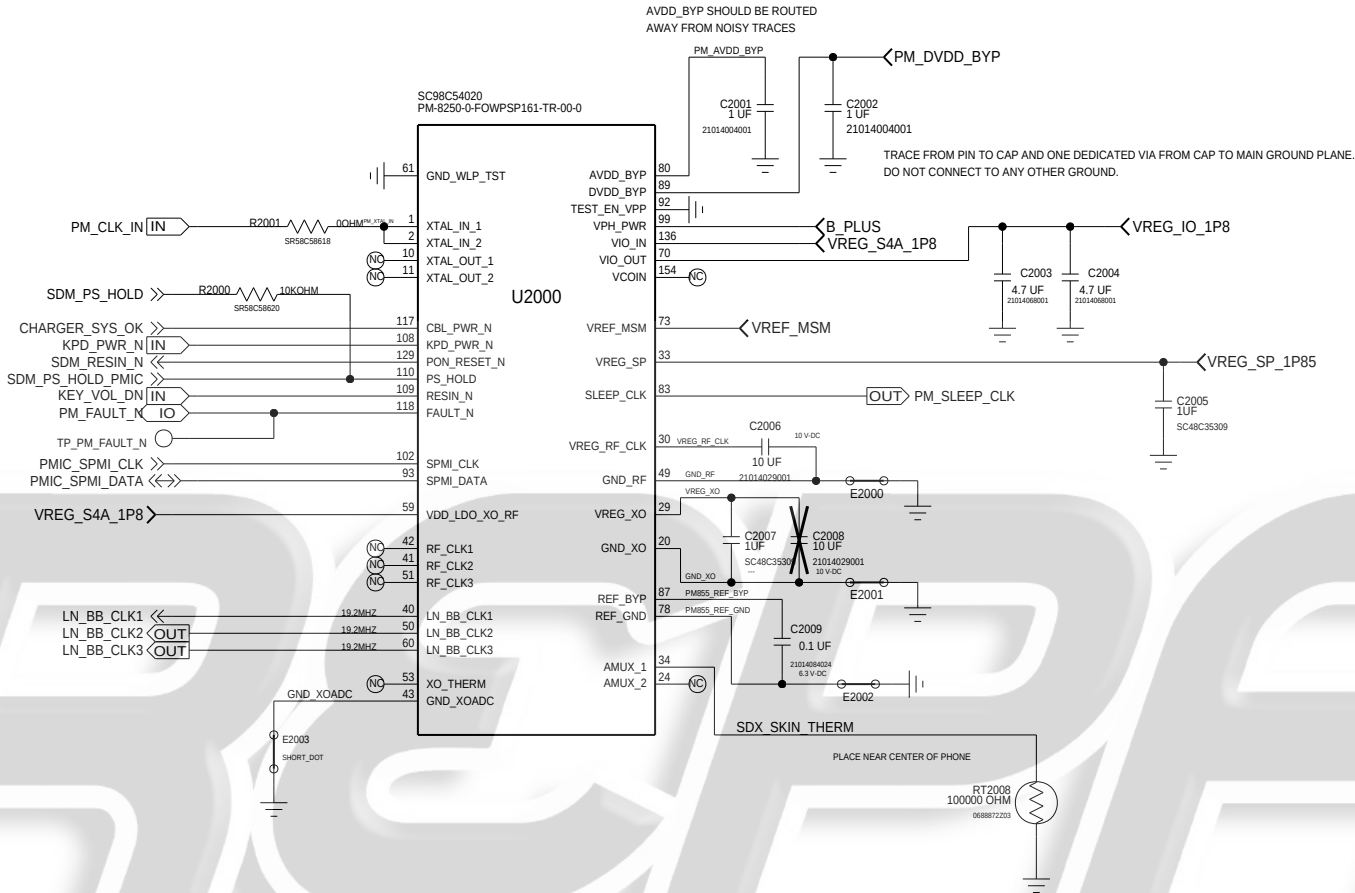
MEMORY: UFS/POP LPDDR5X

REFDES: 1600-1699



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:36 2020		ap_pmic	
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		CAD FILE NAME: PAGE 31 OF 97	

PM8250: CTRL/GPIO/AMUX
REFDES: 2000-2099



DEDICATED TRACE FROM REF_BYP TO CAP.
DEDICATED TRACE FROM REF_GND TO CAP.
DEDICATED VIA TO MAIN GROUND PLANE RIGHT UNDER REF_GND PIN
OR AS CLOSE TO THE PIN AS POSSIBLE.
DO NOT ADD VIA AT C117

REF_BYP AND REF_GND SHOULD BE
ROUTED AWAY FROM NOISY TRACES

CRITICAL LAYOUT GUIDELINES FOR CLOCKS
COUPLING TO ANY RFCLK OR LNBCLK OUTPUT FROM EACH AGGRESSOR SHOULD BE < 50 FF.
1. SMPS COMPONENTS (BUCK, BOOST, CHARGER)
2. VSW_SX TRACES (BUCK, BOOST, CHARGER)
3. ALL OTHER RFCLK, LNBCLK
4. ANY HIGH SPEED DIGITAL SIGNAL
5. ANY OTHER NOISY SIGNALS

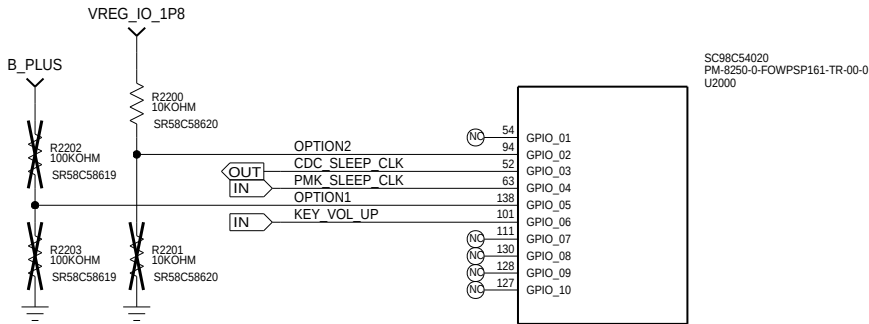
CRITICAL LAYOUT GUIDELINES FOR VREF_RF (GO/PMICLAYOUTCHECKLIST)
PREFERRED: SHORT, DEDICATED ROUTE TO VREG_RF CAP,
AWAY FROM NOISY SIGNALS AND DEDICATED VIA TO MAIN GROUND PLANE AT CAP OR PMIC PIN.
ACCEPTABLE: DEDICATED VIA DIRECTLY TO MAIN GROUND PLANE AT THE GND_CLKSXO PIN
AND AT VREG_RF CAP (WITHOUT ROUTING TO CAP).

DO NOT CONNECT TO ANY OTHER GROUND.
LOOP DCR MUST BE <1000 M?
LOOP INDUCTANCE MUST BE <3 NH

GND_XOADC
SKINNY TRACE TO XTAL COMPONENTS GROUND.
ONE DEDICATED VIA TO MAIN GROUND PLANE AS CLOSE TO PMIC PIN AS POSSIBLE

CRITICAL LAYOUT GUIDELINES FOR VREG_XO
SHORT, DEDICATED ROUTE TO VREG_XO CAP,
AWAY FROM NOISY SIGNALS AND DEDICATED VIA TO MAIN GROUND PLANE AT CAP.

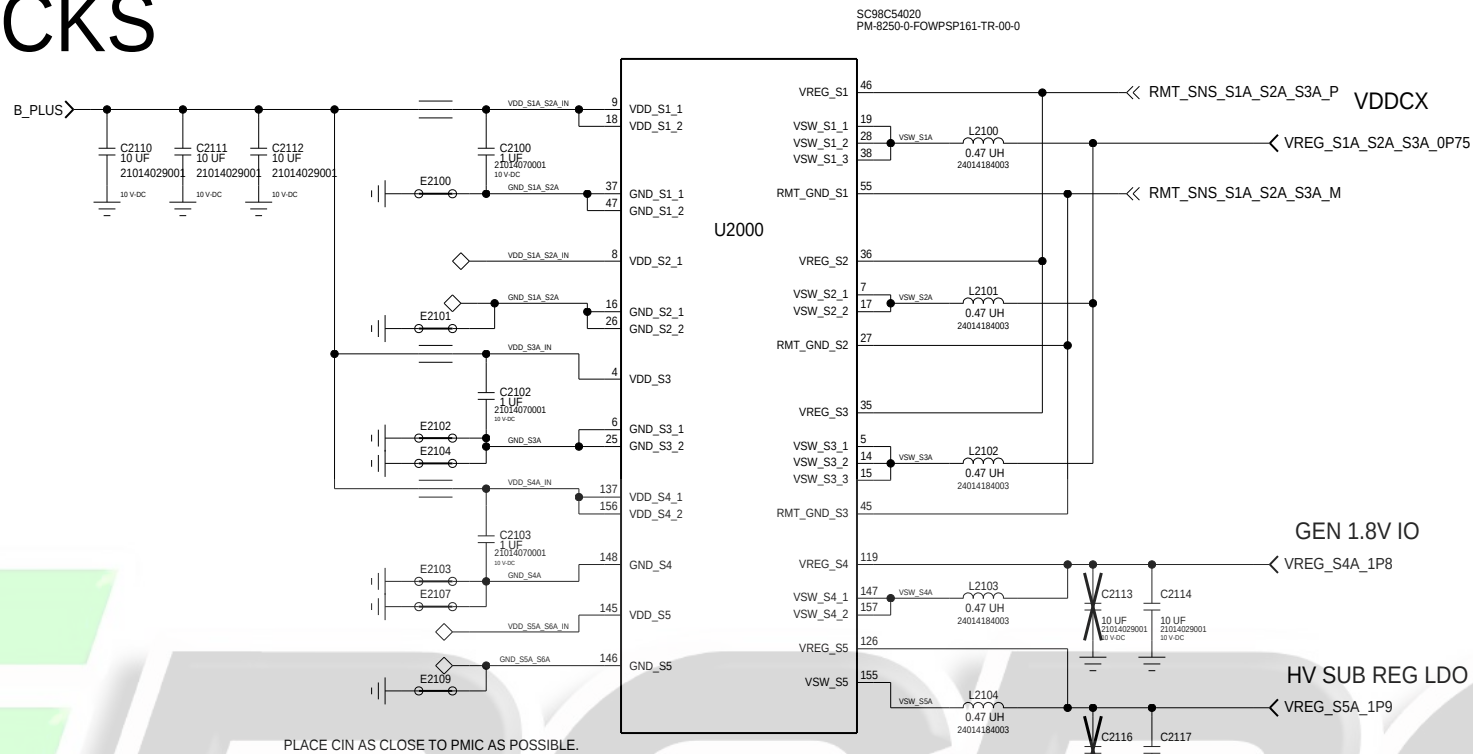
DO NOT CONNECT TO ANY OTHER GROUND EXCEPT THE MAIN GND PLANE.



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: ap_pmic	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:37 2020		PROJECT NAME B sr28c51351_b.cpm REV 03	
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PM8250: BUCKS

REFDES: 2100-2199



VREG_S1A IS REMOTE DIFF SENSE,
VREG_SX & RMT_GND_SX SHOULD BE ROUTED AS A DIFFERENTIAL PAIR
DO NOT ROUTE VREG_SX & RMT_GND_SX NEAR VIN_SX
DO NOT ROUTE VREG_SX & RMT_GND_SX NEAR VSW_SX UNLESS SEPARATED BY GROUND SHIELD

VREG_S2A IS REMOTE DIFF SENSE,
VREG_SX & RMT_GND_SX SHOULD BE ROUTED AS A DIFFERENTIAL PAIR
DO NOT ROUTE VREG_SX & RMT_GND_SX NEAR VIN_SX
DO NOT ROUTE VREG_SX & RMT_GND_SX NEAR VSW_SX UNLESS SEPARATED BY GROUND SHIELD

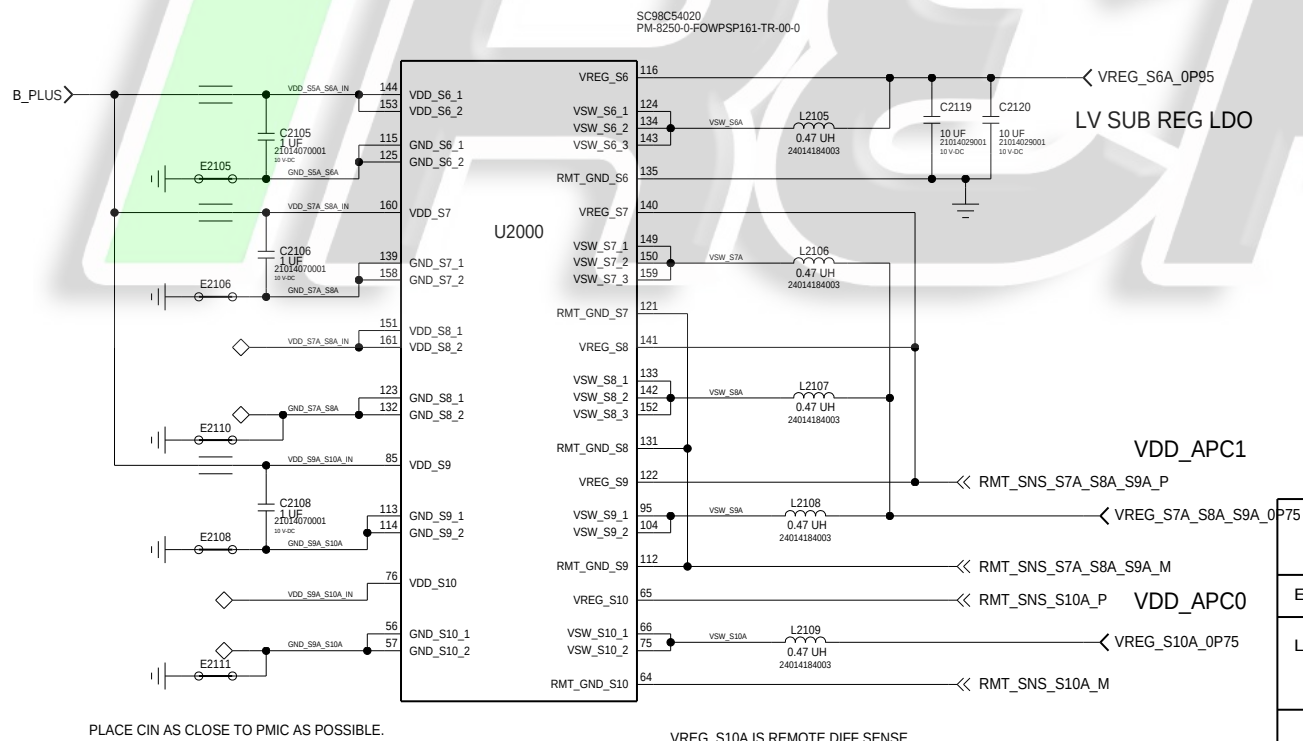
VREG_S3A IS REMOTE DIFF SENSE,
VREG_SX & RMT_GND_SX SHOULD BE ROUTED AS A DIFFERENTIAL PAIR
DO NOT ROUTE VREG_SX & RMT_GND_SX NEAR VIN_SX
DO NOT ROUTE VREG_SX & RMT_GND_SX NEAR VSW_SX UNLESS SEPARATED BY GROUND SHIELD

VREG_S4A IS LOCAL SENSE,
CONNECT VREG TO THE CAPACITOR

VREG_S5A IS LOCAL SENSE,
CONNECT VREG TO THE CAPACITOR

VREG_S6A IS LOCAL DIFF SENSE,
VREG_SX & RMT_GND_SX SHOULD BE ROUTED AS A DIFFERENTIAL PAIR
DO NOT ROUTE VREG_SX & RMT_GND_SX NEAR VIN_SX
DO NOT ROUTE VREG_SX & RMT_GND_SX NEAR VSW_SX UNLESS SEPARATED BY GROUND SHIELD

VREG_S9A_S8A_S7A IS REMOTE DIFF SENSE,
VREG_SX & RMT_GND_SX SHOULD BE ROUTED AS A DIFFERENTIAL PAIR
DO NOT ROUTE VREG_SX & RMT_GND_SX NEAR VIN_SX
DO NOT ROUTE VREG_SX & RMT_GND_SX NEAR VSW_SX UNLESS SEPARATED BY GROUND SHIELD



VREG_S10A IS REMOTE DIFF SENSE,
VREG_SX & RMT_GND_SX SHOULD BE ROUTED AS A DIFFERENTIAL PAIR
DO NOT ROUTE VREG_SX & RMT_GND_SX NEAR VIN_SX
DO NOT ROUTE VREG_SX & RMT_GND_SX NEAR VSW_SX UNLESS SEPARATED BY GROUND SHIELD

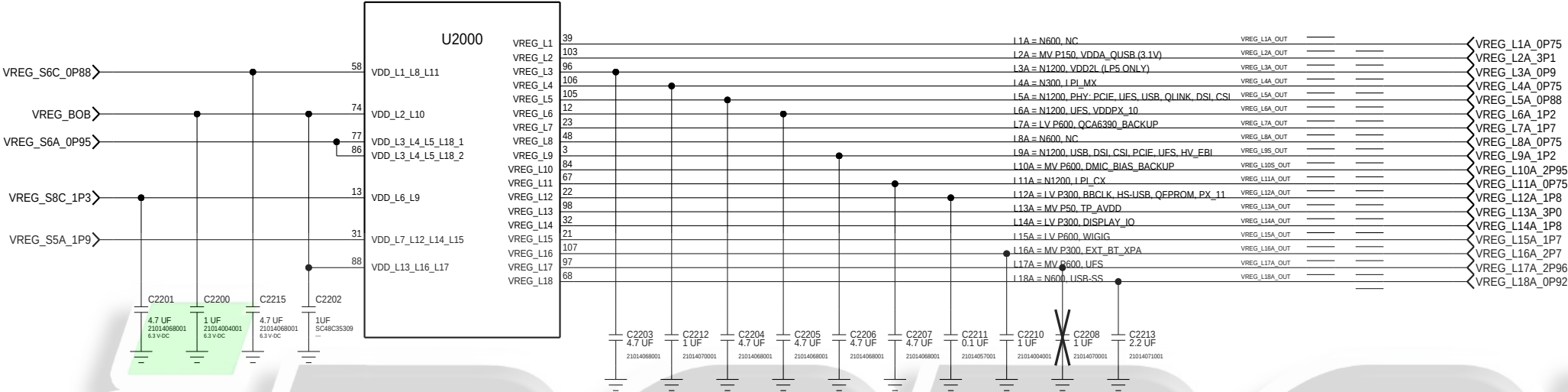
0P75	APPROVALS		 MOTOROLA MOBILE DEVICES	
	ENGINEER: MOTOROLA		BLOCK TITLE: ap_pmic	
	LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:38 2020			
SHIELD	MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT.  COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B PROJECT NAME sr28c51351_b.cpm REV 03	
			CAD FILE NAME: PAGE 33 OF 97	

PM8250: LDOS/GND

REFDES: 2200-2299

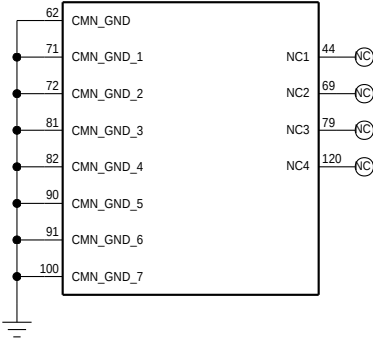
P-TYPE LDOS ARE PSUEDO-CAPLESS. CAPACITORS CAN BE PLACED NEAR LOAD.
REGULATORS POWERING PMIC INTERNAL CIRCUITS NEED CAPACITORS NEAR PMIC.
N-TYPE LDOS WITH >300MA RATING NEED LOCAL CAPACITORS NEAR PMIC.
SC98C54020
PM-8250-0-FOWPSP161-TR-00-0

L2A, L4A,L7A,L10A,L12A,L13A,L14A,L15A,L16A AND L17A ARE PSEUDO-CAPLESS



SOME LDO OUTPUT CAPACITORS USE 10UF FOR BOM CONSOLIDATION PURPOSES,
EVEN THOUGH THE LDO TYPE ONLY REQUIRES 4.7UF.
IF PREFERRED, THE 10UF CAPACITORS CAN BE REDUCED TO 4.7UF. REQUIRED CAPACITANCE
FOR EACH LDO TYPE CAN BE FOUND IN APP NOTE: UNDERSTANDING LDO REGULATORS (80-VT310-125).

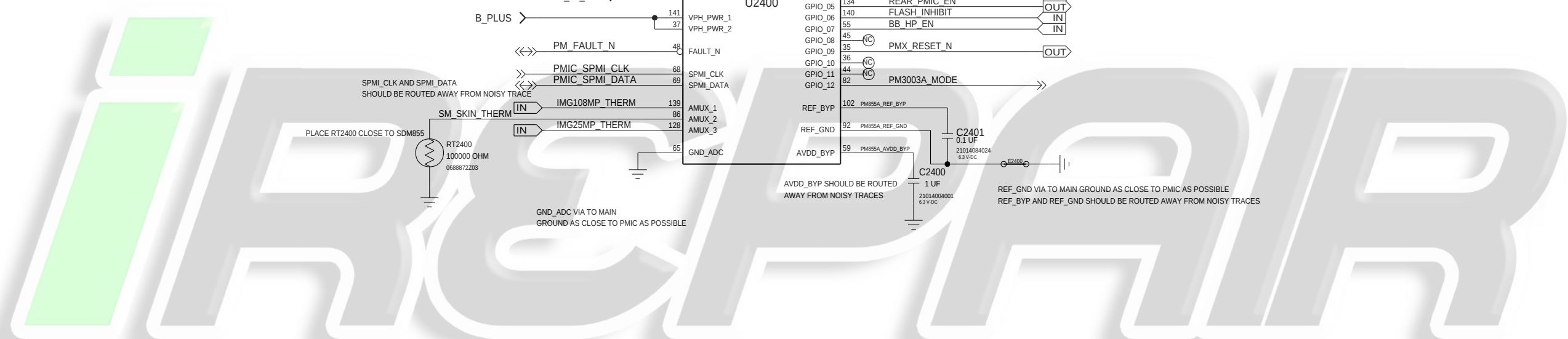
SC98C54020
PM-8250-0-FOWPSP161-TR-00-0
U2000



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: ap_pmic	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:39 2020			
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		CAD FILE NAME: PAGE 34 OF 97	

PM8150A: CTRL/GPIO

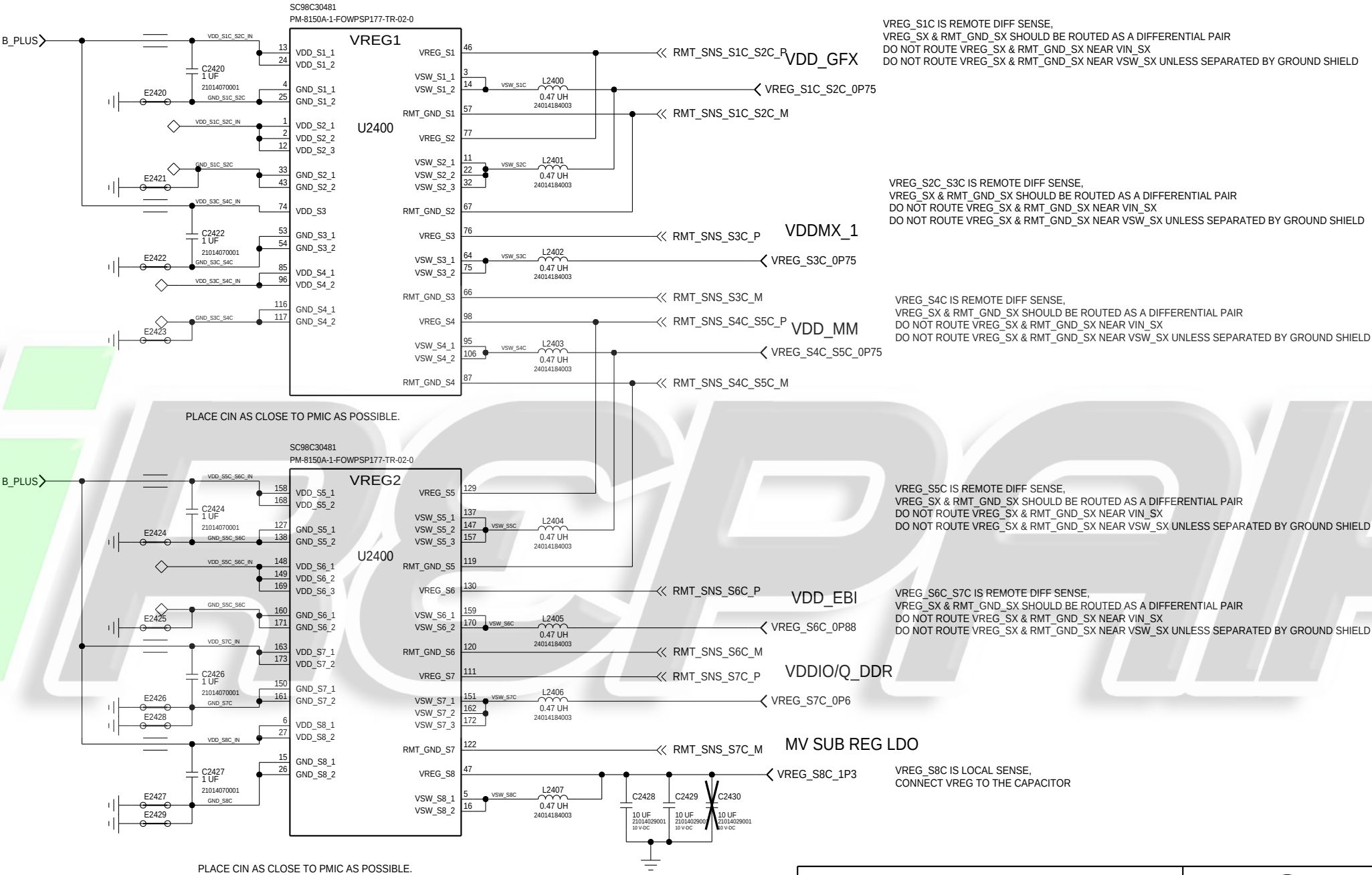
REFDES: 2400-2499



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:40 2020		ap_pmic	
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		CAD FILE NAME: PAGE 35 OF 97	

PM8150A: BUCKS

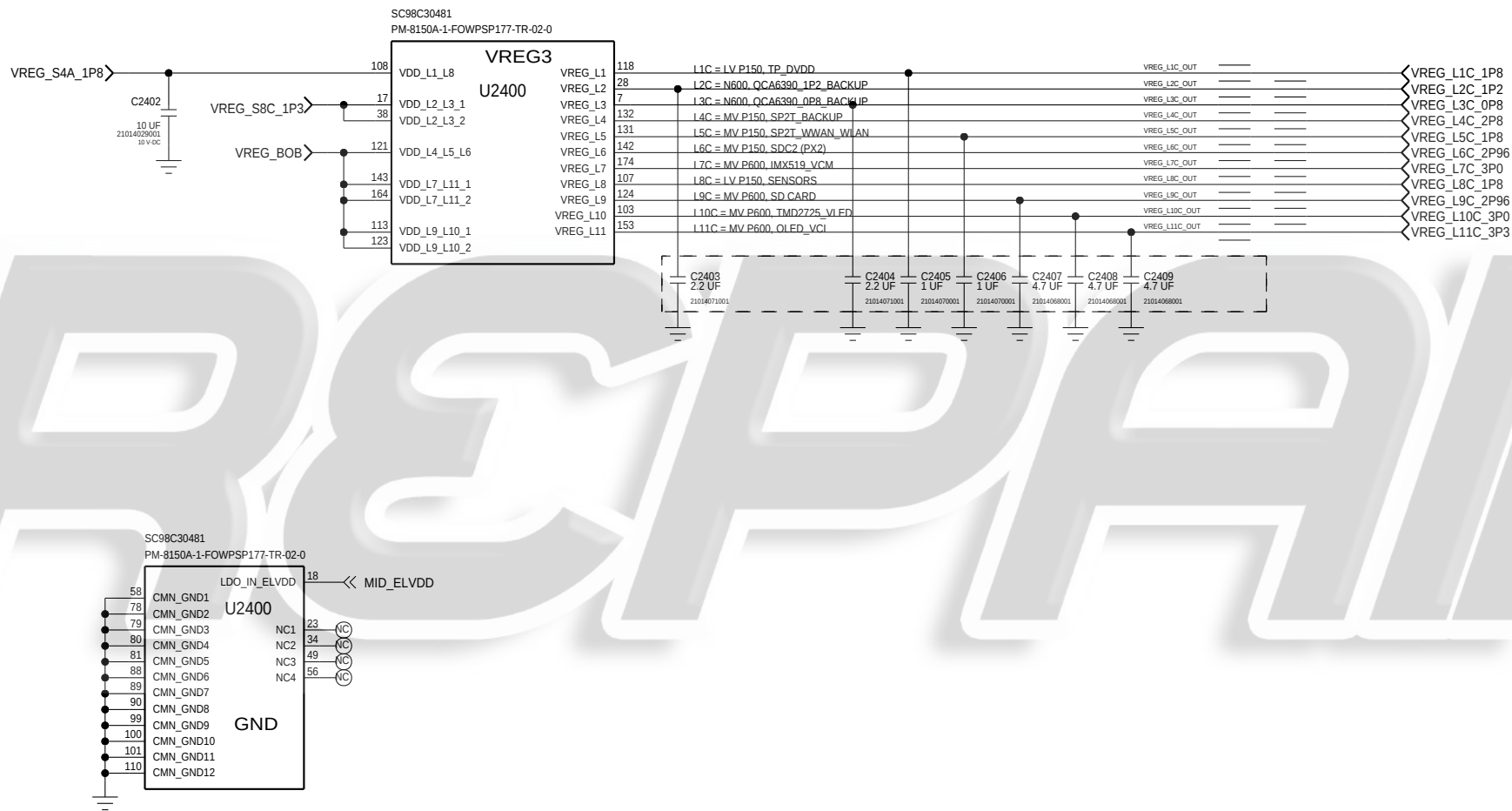
REFDES: 2400-2499



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: ap_pmic	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:40 2020			
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		B sr28c51351_b.cpm	03
		CAD FILE NAME:	PAGE 36 OF 97

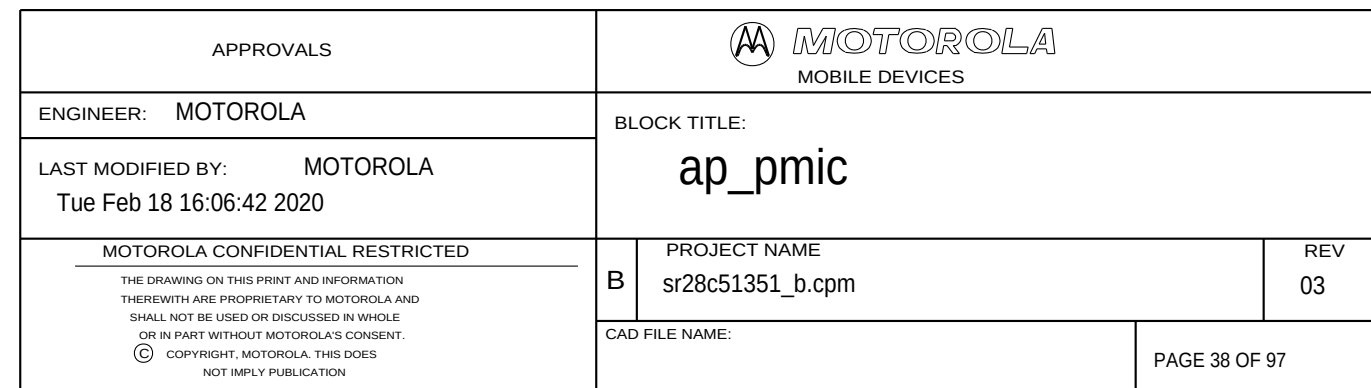
PM8150A: LDOS/GND

REFDES: 2400-2499



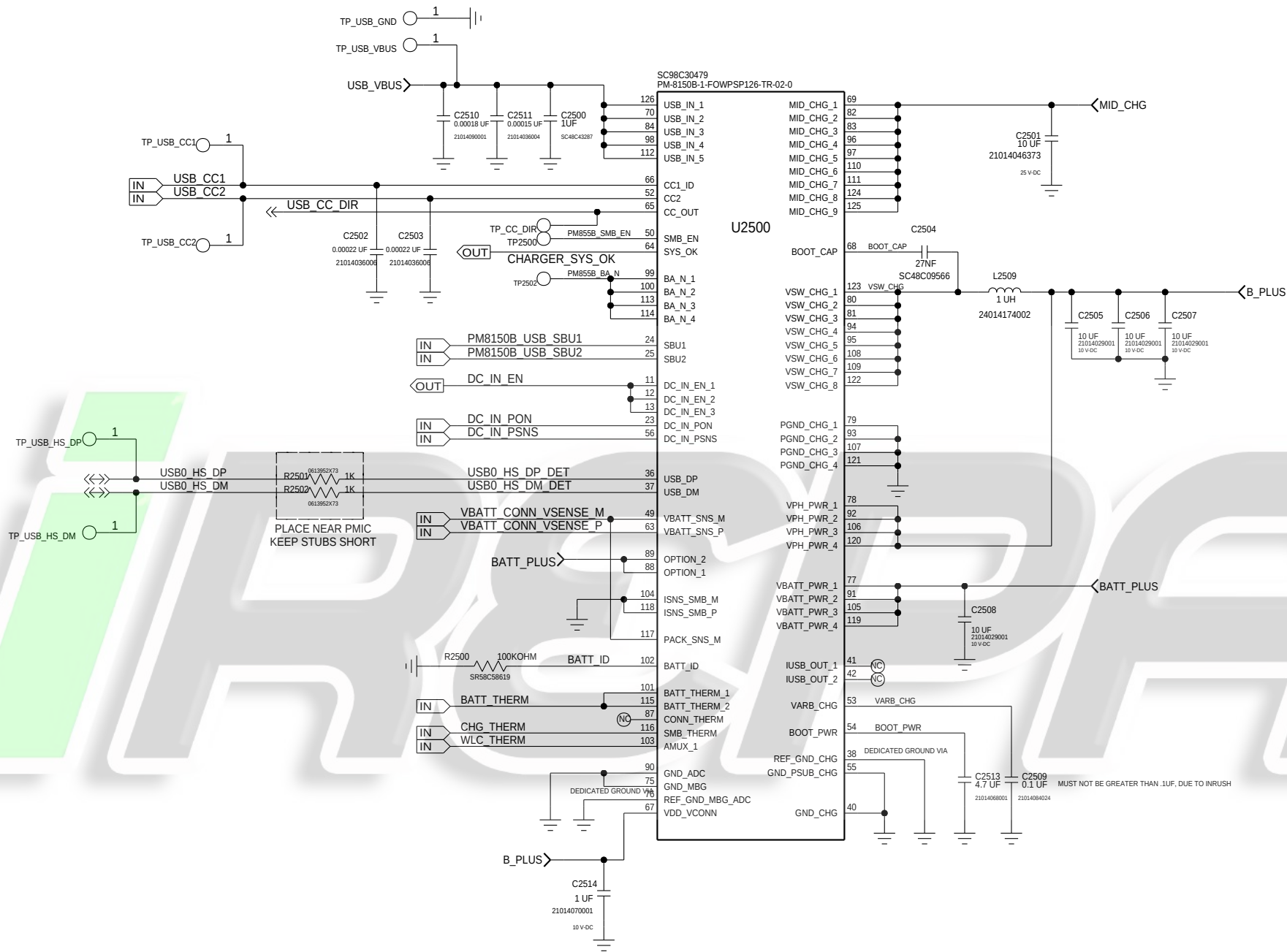
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:41 2020		ap_pmic	
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		B	sr28c51351_b.cpm
		REV 03	
		CAD FILE NAME:	
		PAGE 37 OF 97	

REFDES:2400-2499



PM8150B: CHARGE/FG

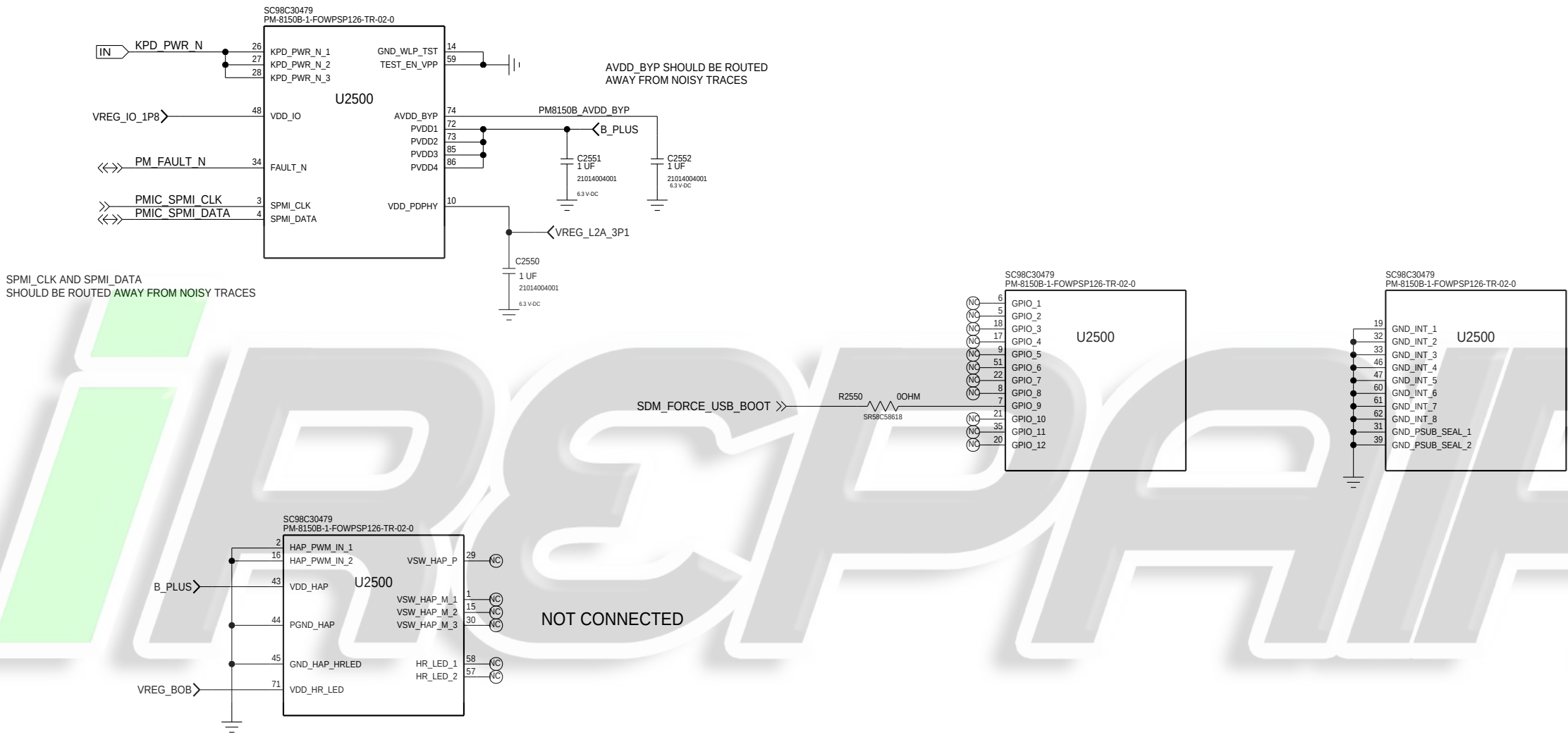
REFDES: 2500-2599



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:43 2020		ap_pmic	
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		REV 03	
		PAGE 39 OF 97	

PM8150B: CTRL/HAPTICS

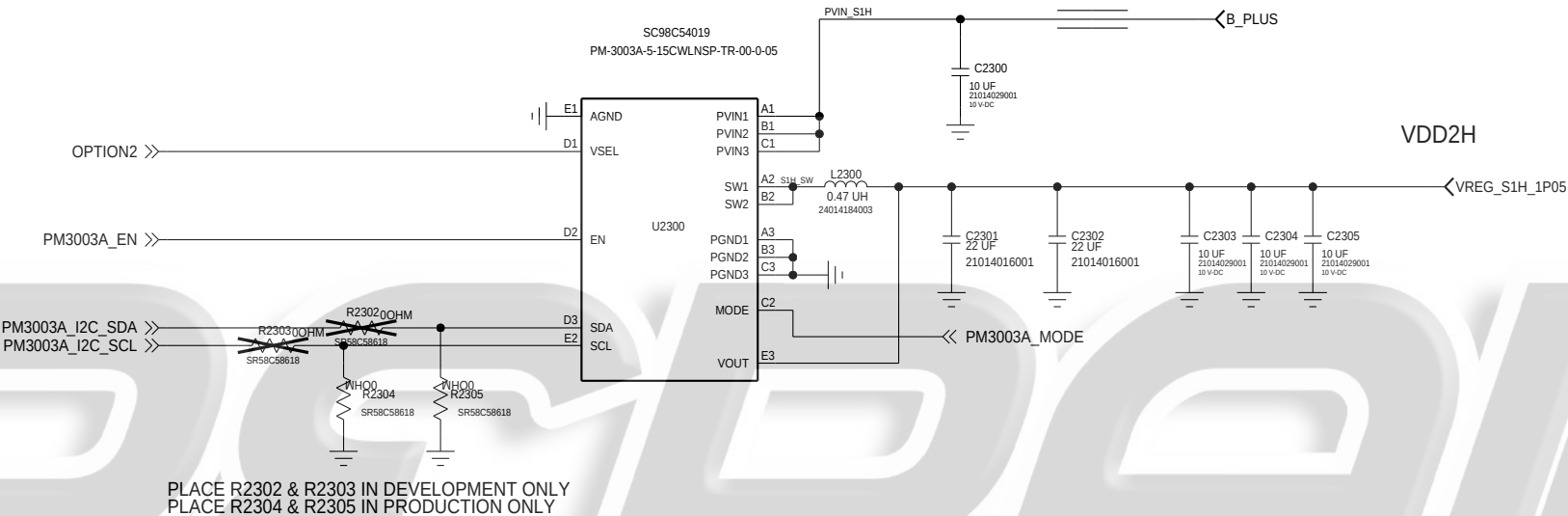
REFDES: 2500-2599



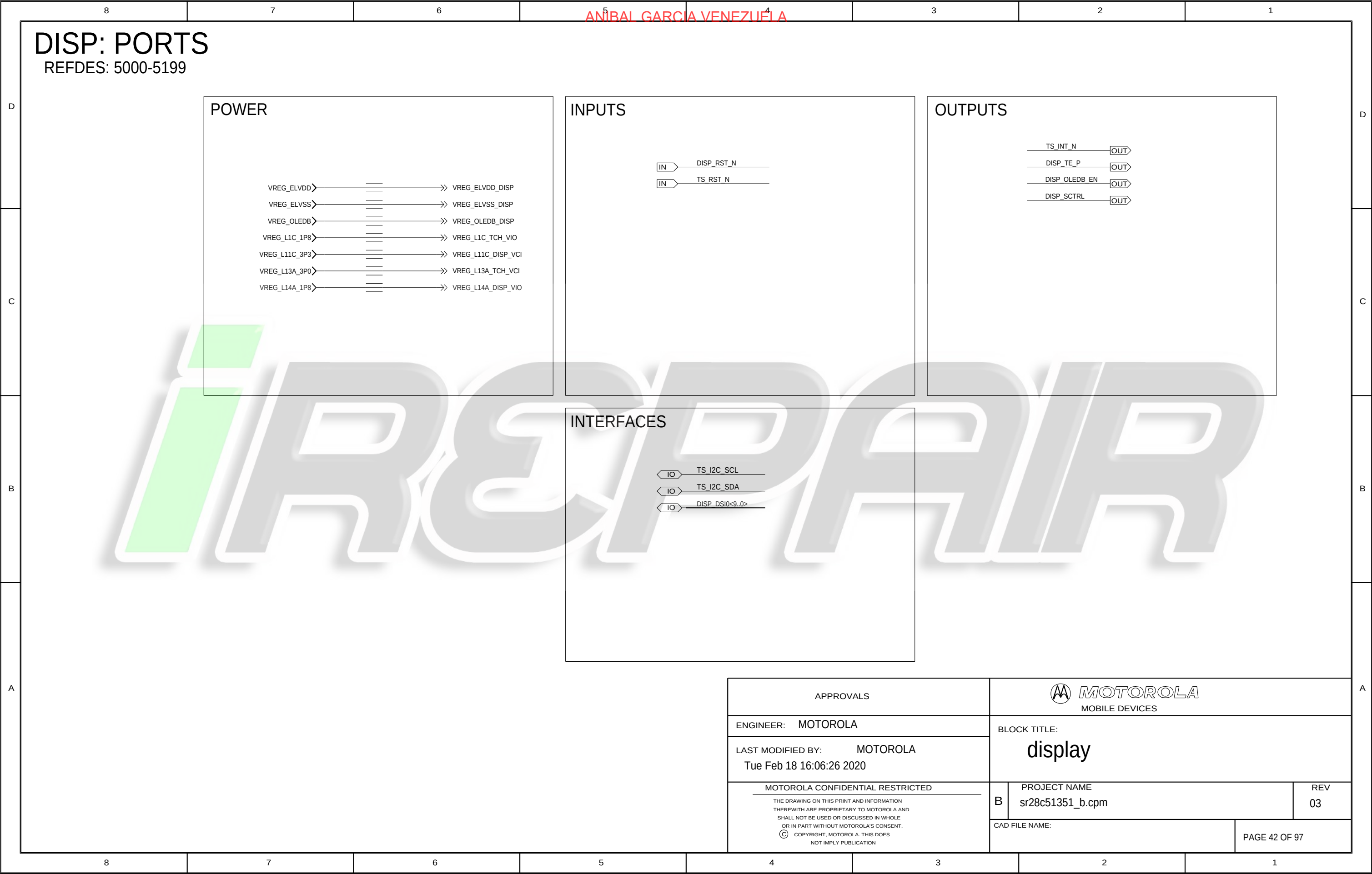
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:44 2020		ap_pmic	
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		CAD FILE NAME: PAGE 40 OF 97	

PM3003A: BUCK

REFDES:2300-2350

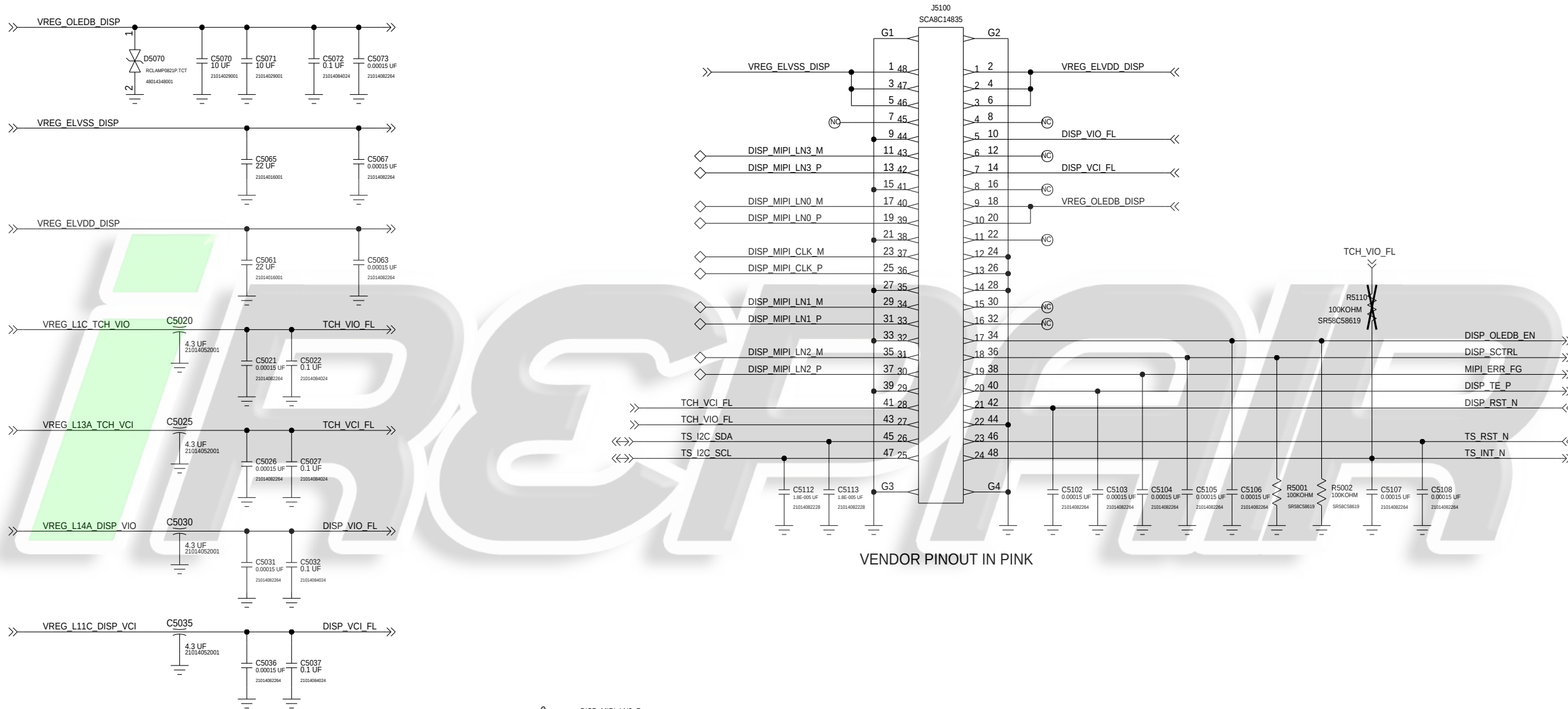


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:45 2020		ap_pmic	
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		CAD FILE NAME: PAGE 41 OF 97	

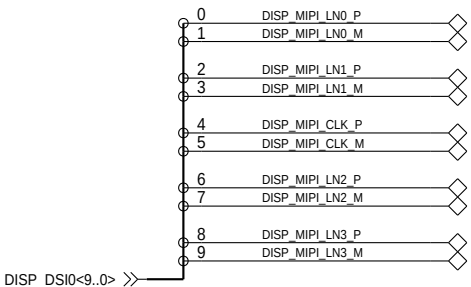


DISP: CONNECTOR

REFDES: 5000-5199



VENDOR PINOUT IN PINK

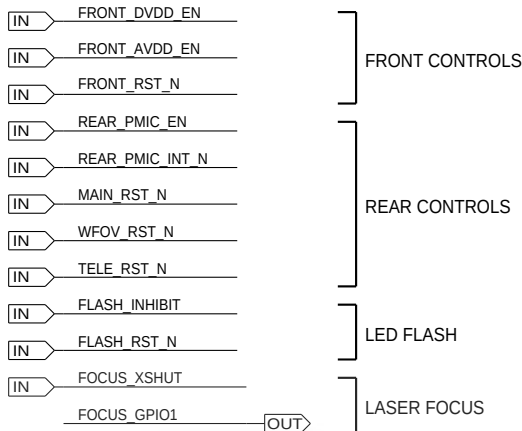


APPROVALS		BLOCK TITLE:	
ENGINEER: MOTOROLA		display	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:27 2020		PROJECT NAME B sr28c51351_b.cpm	
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CAD FILE NAME:		PAGE 43 OF 97	

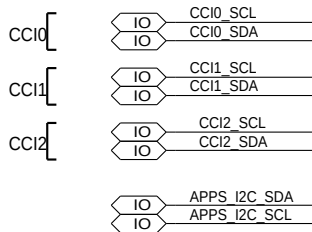
IMG: PORTS

REFDES: 5200-5999

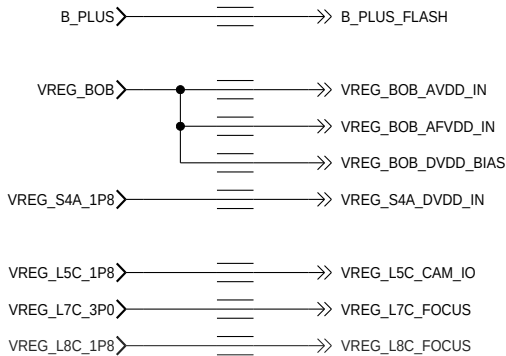
GPIO



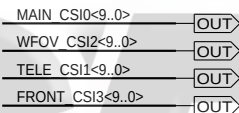
INTERFACES



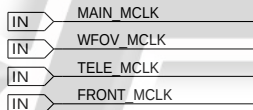
SUPPLIES



CSI OUTPUTS



CLOCKS



APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
Tue Feb 18 16:06:19 2020

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

imagers

B

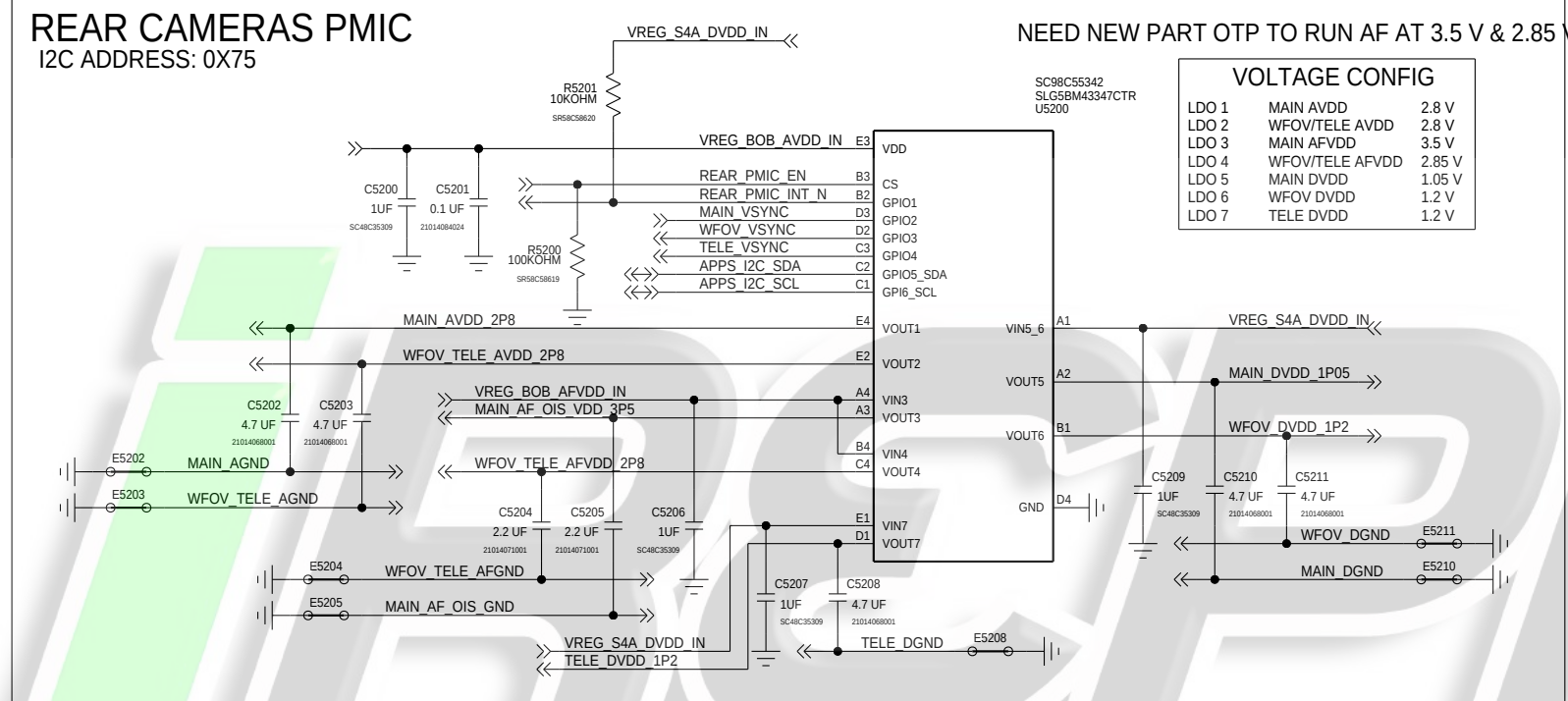
PROJECT NAME
sr28c51351_b.cpm

REV
03

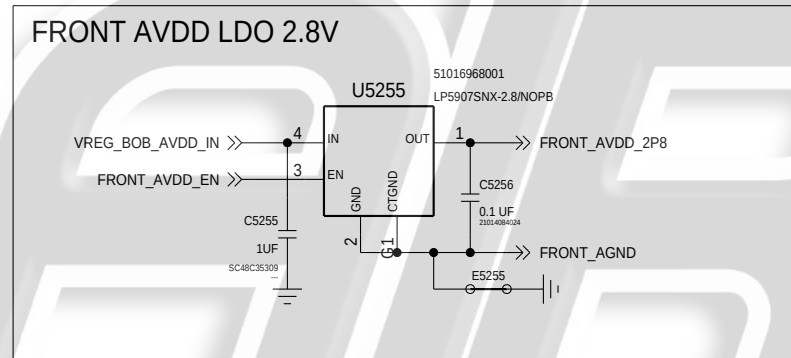
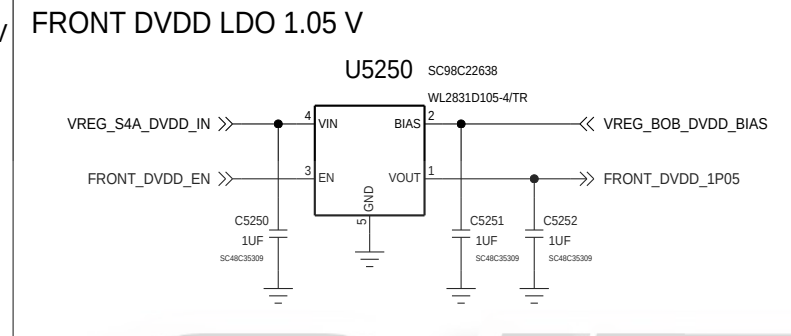
CAD FILE NAME:

PAGE 44 OF 97

REFDES: 5200-5299



VOLTAGE CONFIG		
LDO 1	MAIN AVDD	2.8 V
LDO 2	WFOV/TELE AVDD	2.8 V
LDO 3	MAIN AFVDD	3.5 V
LDO 4	WFOV/TELE AFVDD	2.85 V
LDO 5	MAIN DVDD	1.05 V
LDO 6	WFOV DVDD	1.2 V
LDO 7	TELE DVDD	1.2 V

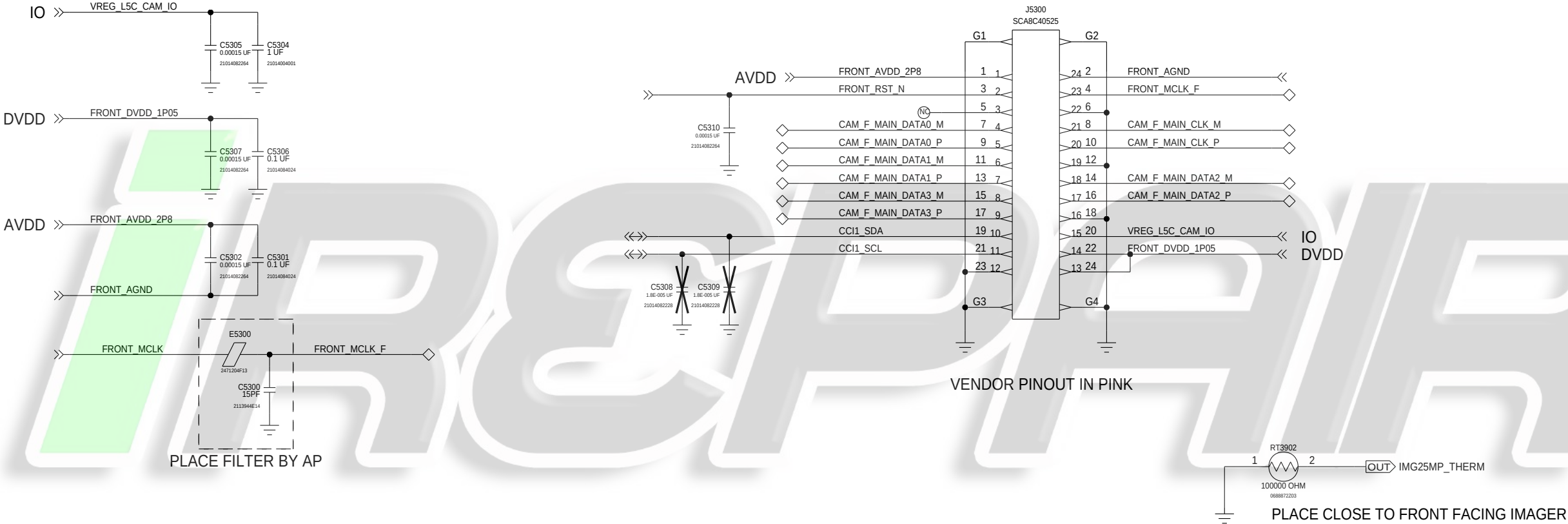


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: imagers	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:21 2020			
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		CAD FILE NAME:	PAGE 45 OF 97

IMG: FRONT CONNECTOR

REFDES: 5300-5399

FRONT 25MP CONNECTOR



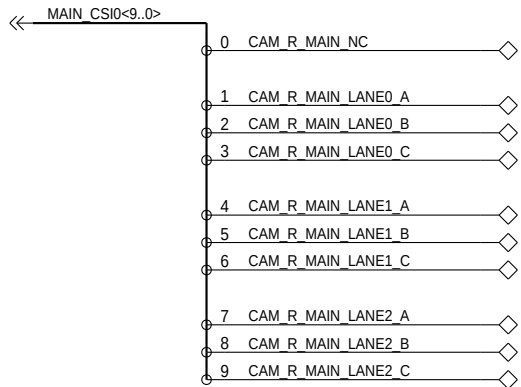
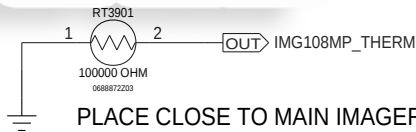
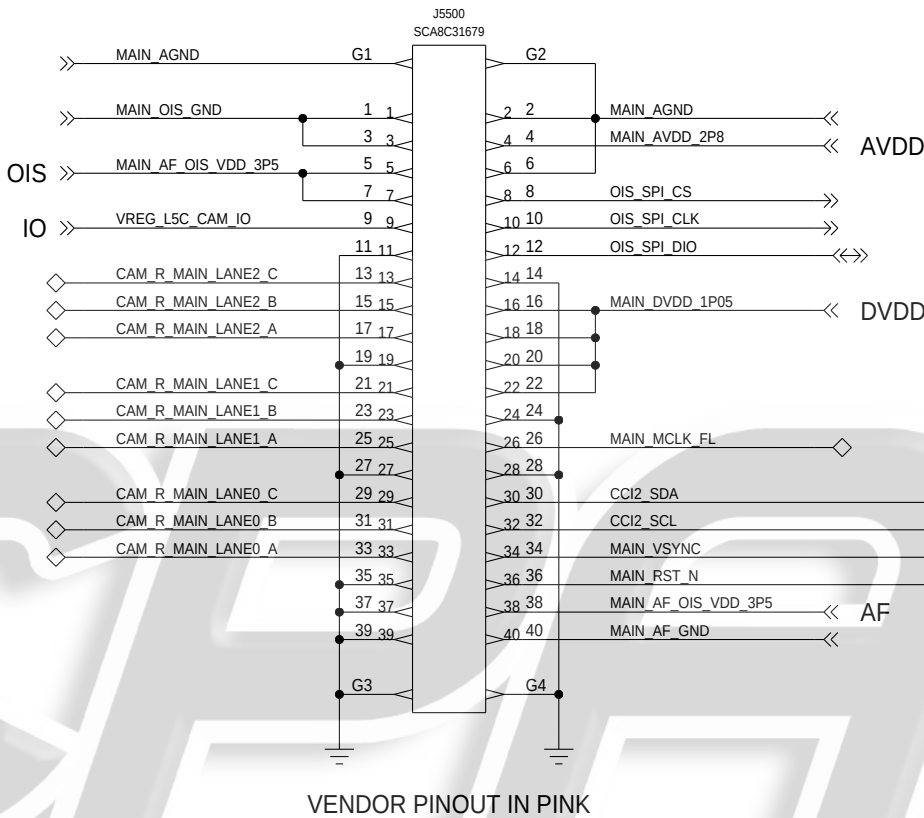
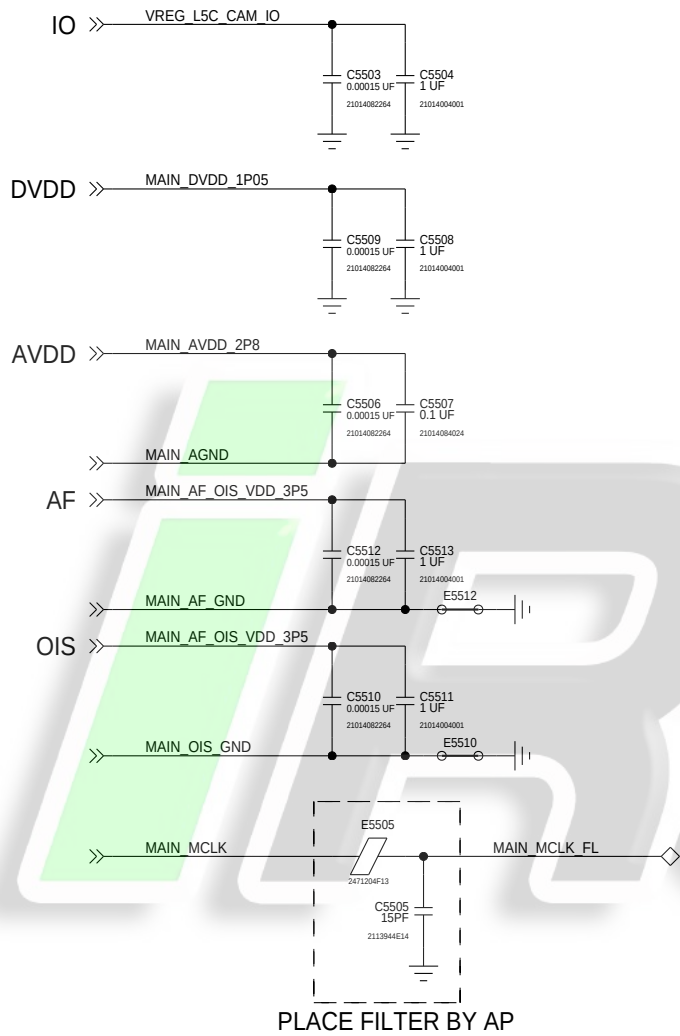
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	1	CAM_F_MAIN_CLK_M	
	2	CAM_F_MAIN_DATA0_P	
	3	CAM_F_MAIN_DATA0_M	
	4	CAM_F_MAIN_DATA1_P	
	5	CAM_F_MAIN_DATA1_M	
	6	CAM_F_MAIN_DATA2_P	
	7	CAM_F_MAIN_DATA2_M	
	8	CAM_F_MAIN_DATA3_P	
	9	CAM_F_MAIN_DATA3_M	

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: imagers	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:22 2020		PROJECT NAME B sr28c51351_b.cpm	
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IMG: REAR MAIN CONNECTOR

REFDES: 5500-5599

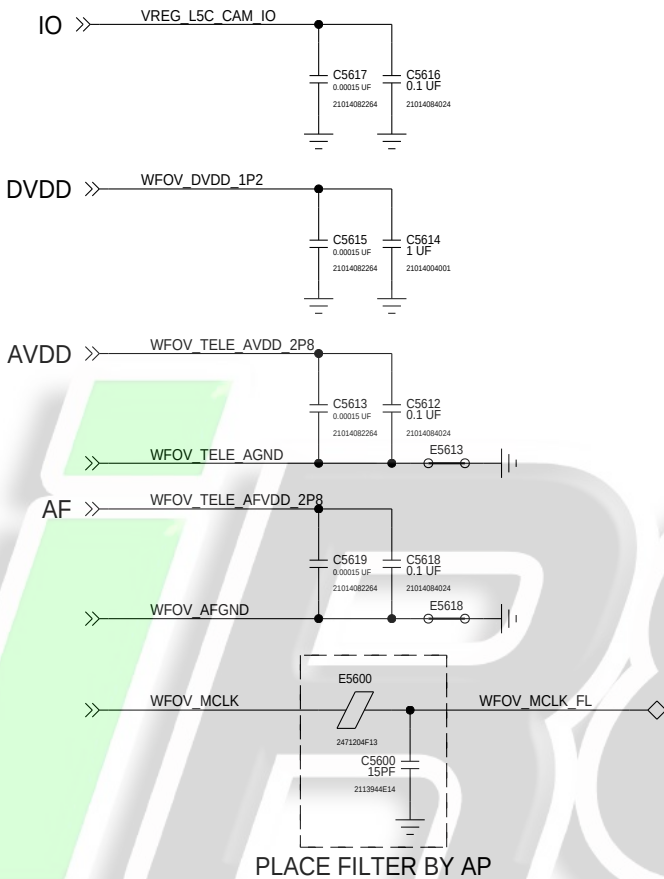
REAR MAIN 108MP CONNECTOR



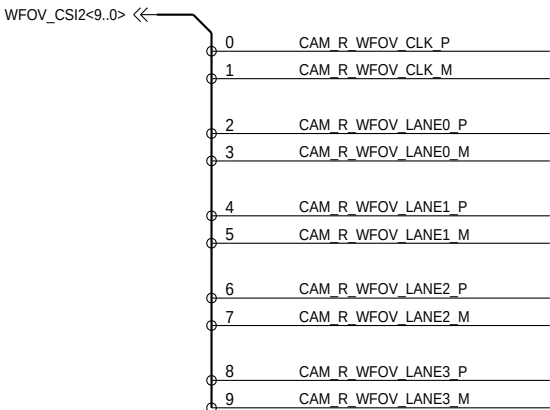
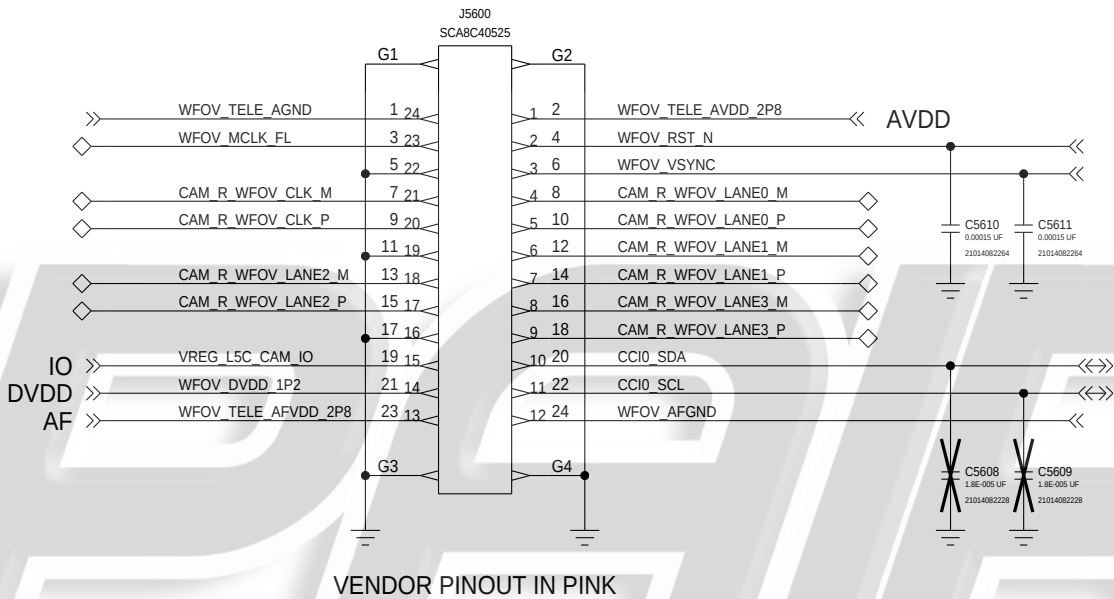
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:22 2020		imagers	
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		REV 03	
		PAGE 47 OF 97	

IMG: REAR WFOV CONNECTOR

REFDES: 5600-5699



REAR WFOV 16MP CONNECTOR

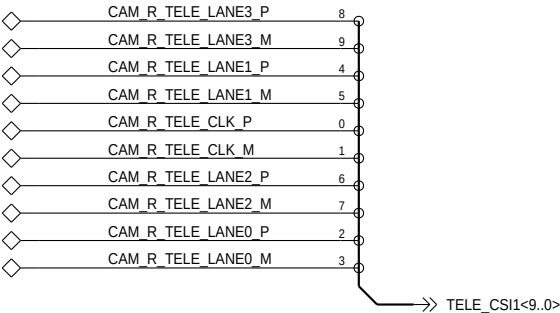
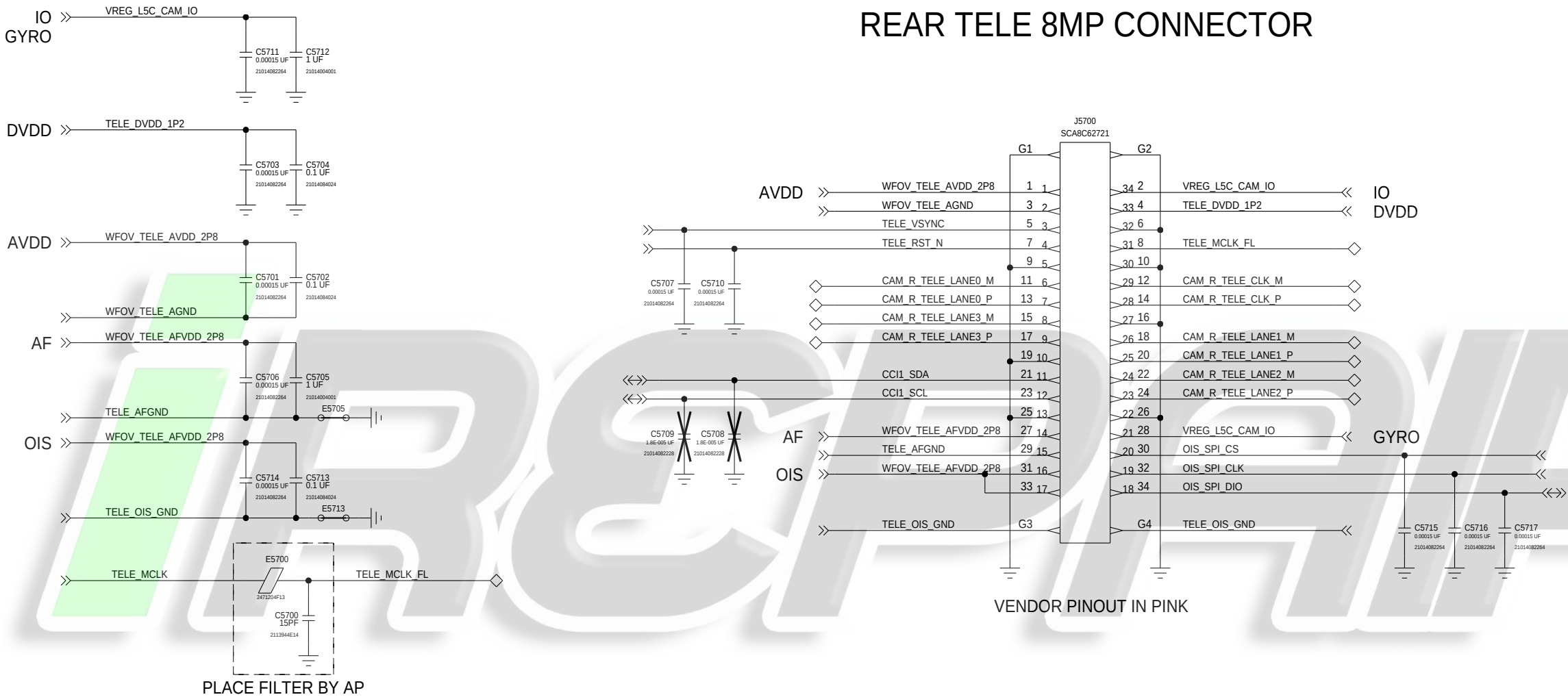


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:23 2020		imagers	
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		CAD FILE NAME:	
		PAGE 48 OF 97	

IMG: REAR TELE CONNECTOR

REFDES: 5700-5799

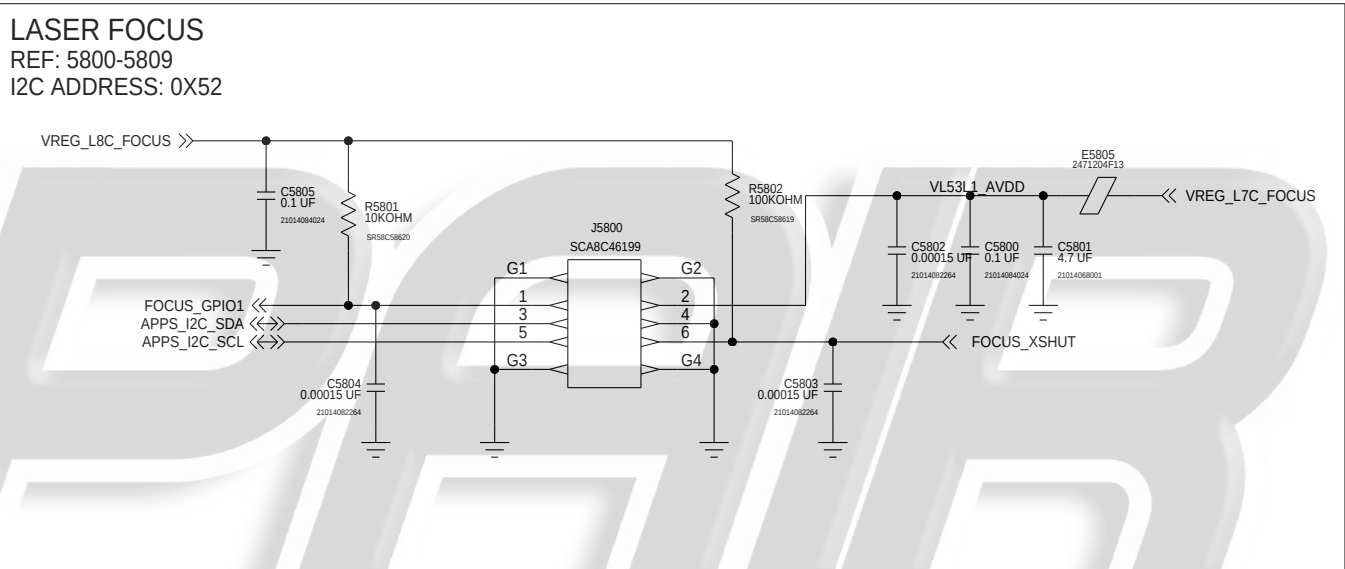
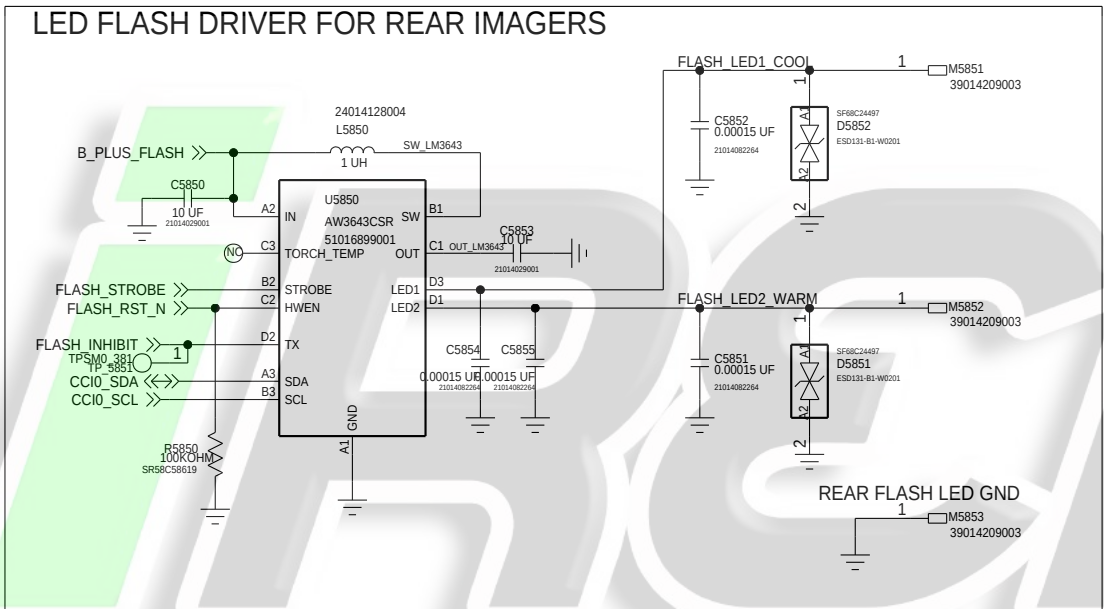
REAR TELE 8MP CONNECTOR



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: imagers	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:24 2020		PROJECT NAME B sr28c51351_b.cpm	
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IMG: REAR FLASH / LASER FOCUS

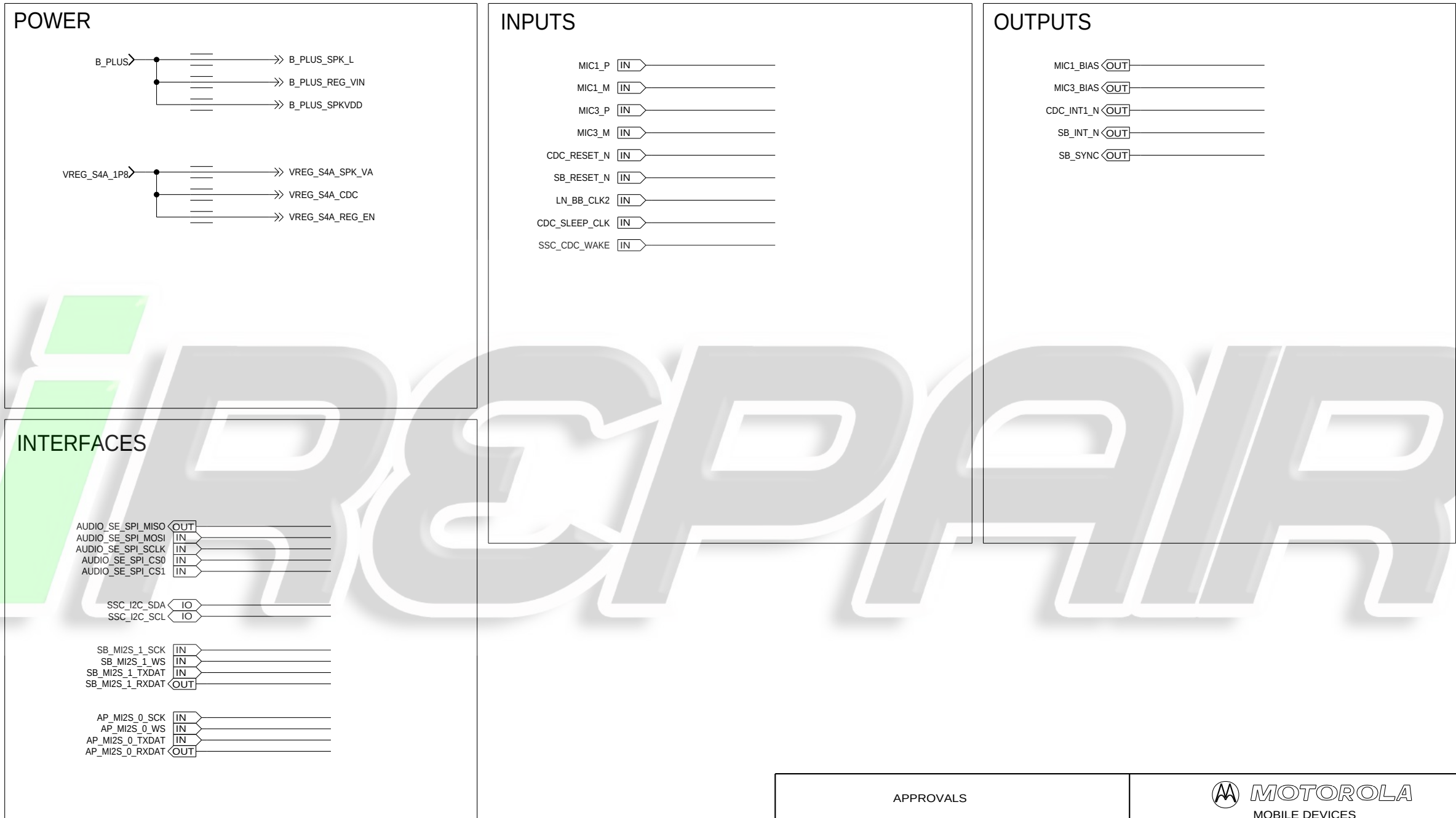
REFDES: 5800-5899



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:25 2020		imagers	
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		B sr28c51351_b.cpm	
		REV 03	
		CAD FILE NAME:	
		PAGE 50 OF 97	

AUDIO: PORTS

REF 4500-4999

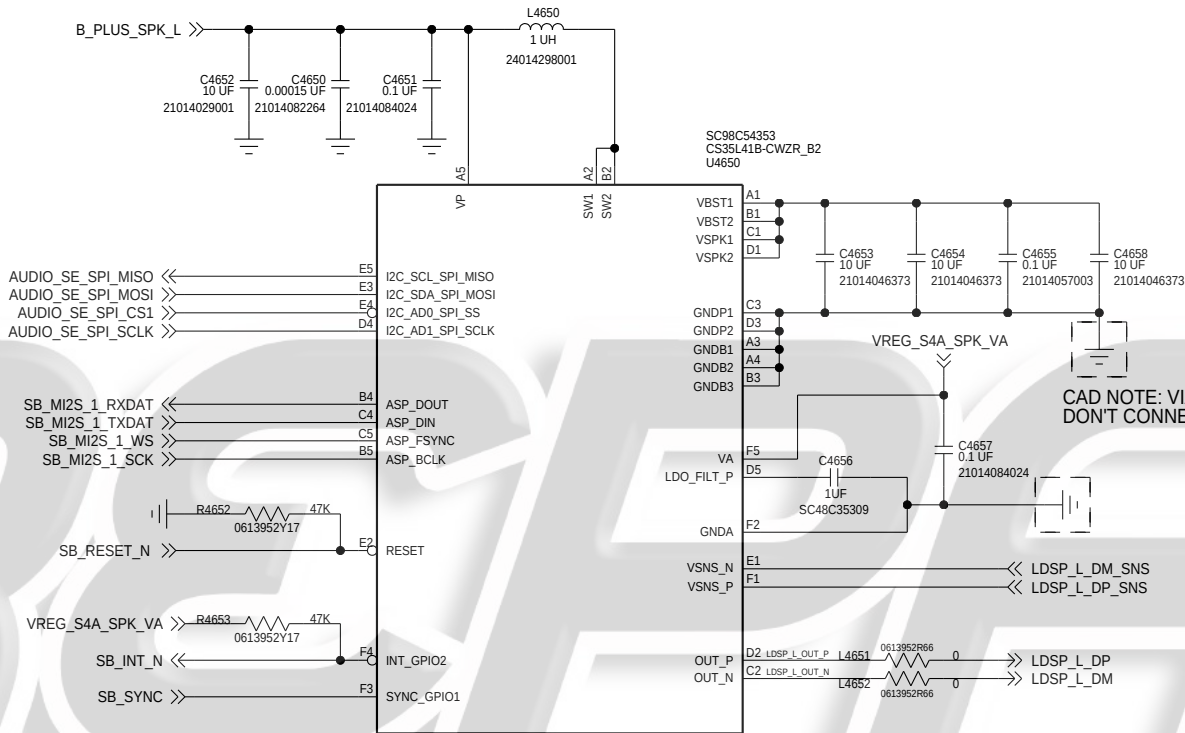


APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:14 2020			
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		CAD FILE NAME:	PAGE 51 OF 97

AUDIO: SMART BOOST TOP

REFDES: 4650-4699

*** BOOST IC I2C ADD. 0X41



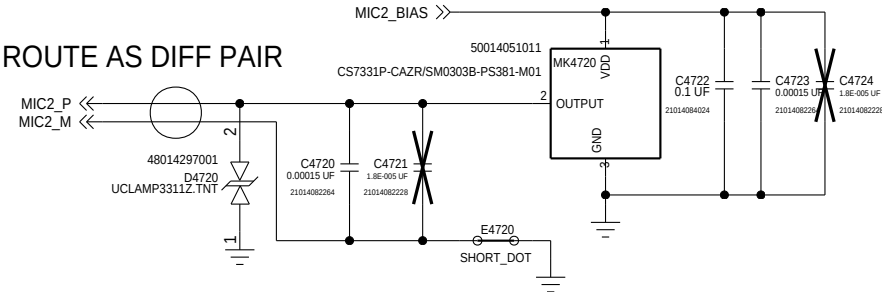
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ENGINEER: MOTOROLA		BLOCK TITLE: audio	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:17 2020			
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		CAD FILE NAME: PAGE 53 OF 97	

AUDIO: MICS/SPEAKERS

REFDES: 4700-4799

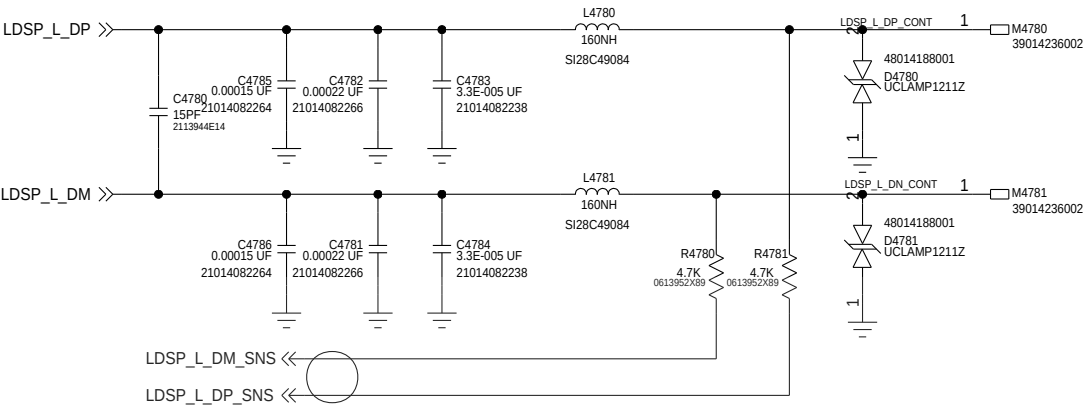
MIC 2

ROUTE AS DIFF PAIR



REFDES: 4720-4729

LOUDSPEAKER/EAR PIECE



REFDES: 4780-4789

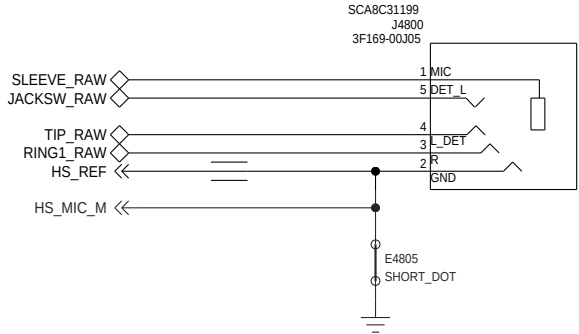
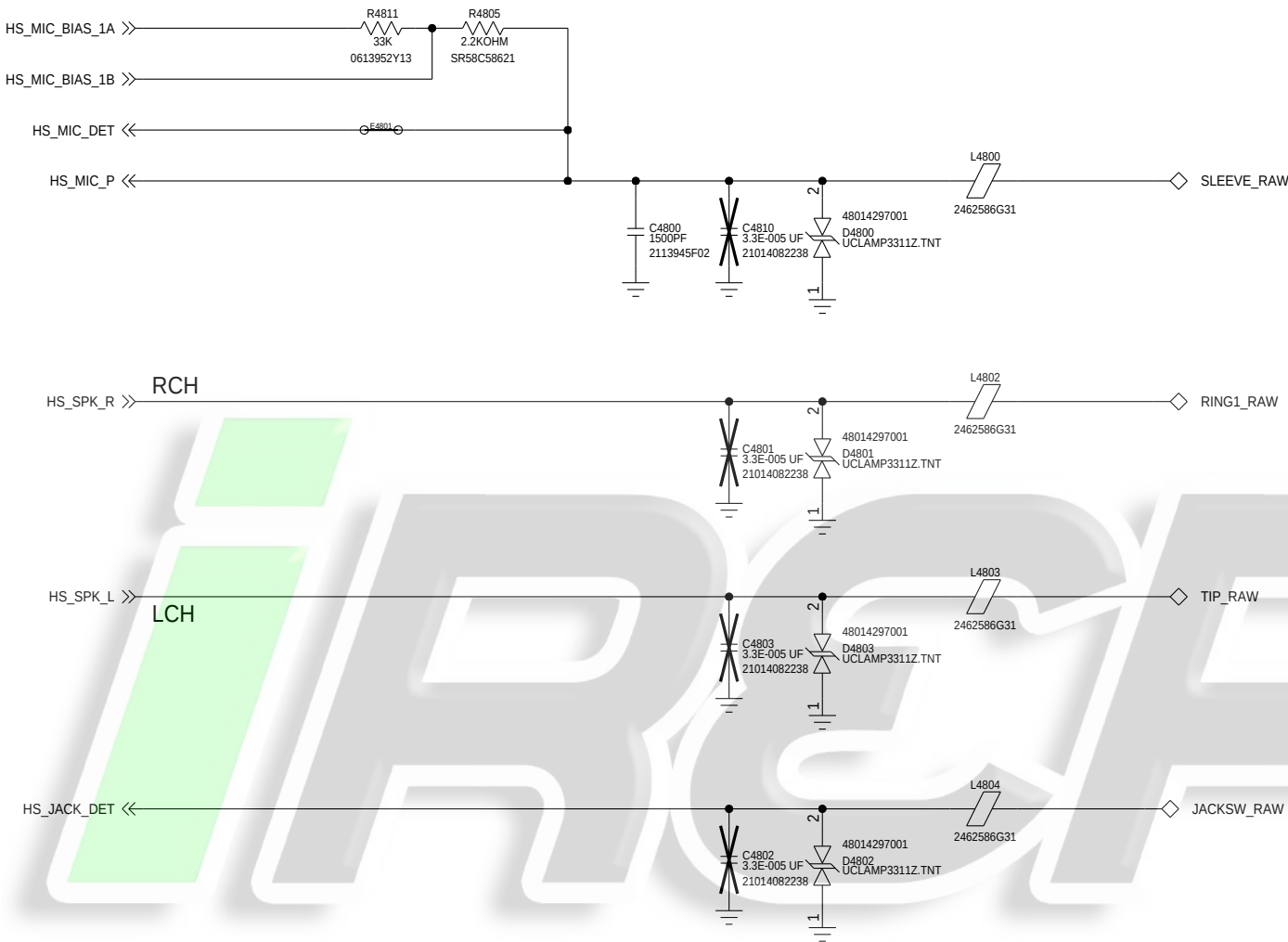
ROUTE AS DIFF PAIR

IRCPAIR

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: audio	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:18 2020			
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		CAD FILE NAME: PAGE 54 OF 97	

AUDIO: 3.5MM HEADSET
REFDES: 4800-4899

ANIBAL GARCIA VENEZUELA

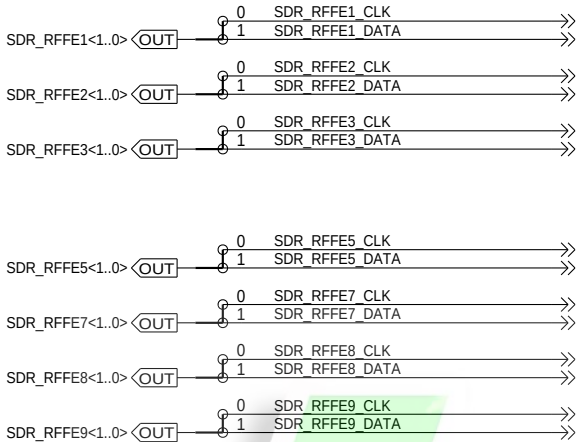


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE: audio	
LAST MODIFIED BY: MMI Tue Feb 18 16:06:18 2020			
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		CAD FILE NAME: PAGE 55 OF 97	

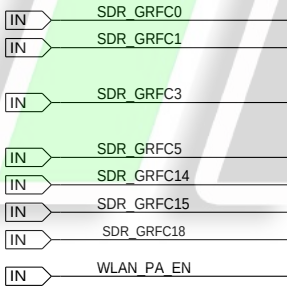
RFFE: PORTS

REFDES: 7000-8199

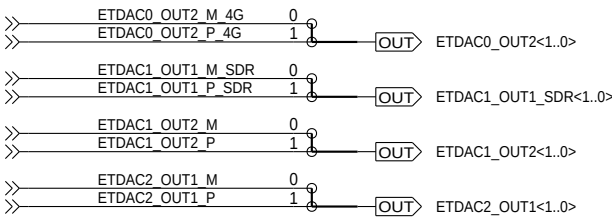
INTERFACES - RFFE MIPI



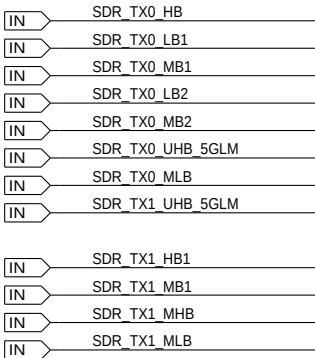
INTERFACES - GRFC



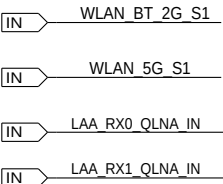
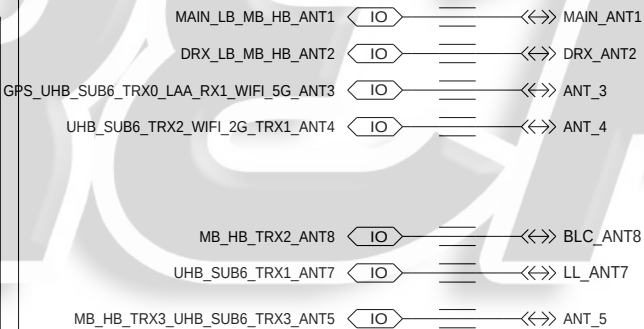
ET DAC



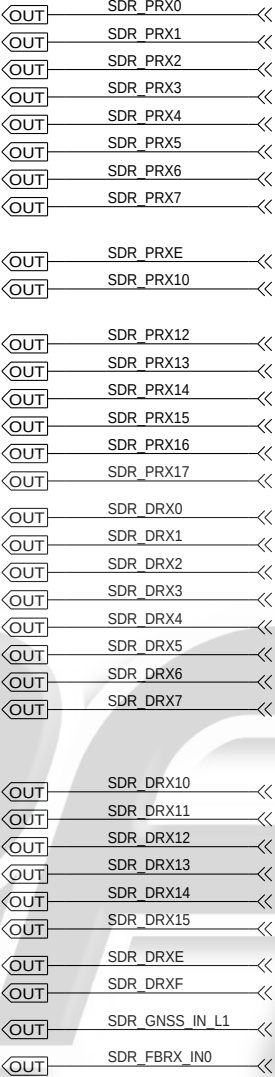
TRANSMITTER - RF



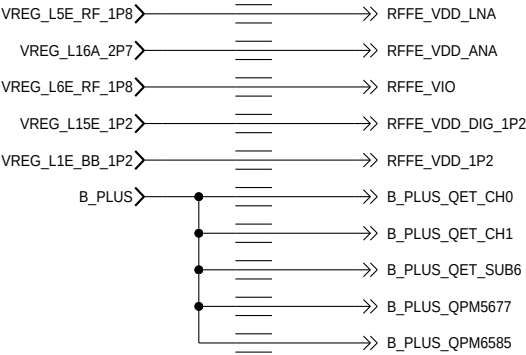
OUTPUTS - RF ANTENNA CONNECTORS



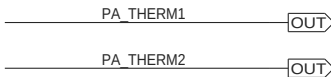
RECEIVER - RF



POWER



THERMISTORS



QUALCOMM RECOMMENDS PLACING BOTH THERMISTORS FOR FUTURE SW IMPLEMENTATION THEY ARE WORKING ON

APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
Tue Feb 18 16:05:51 2020

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BLOCK TITLE:

rffe

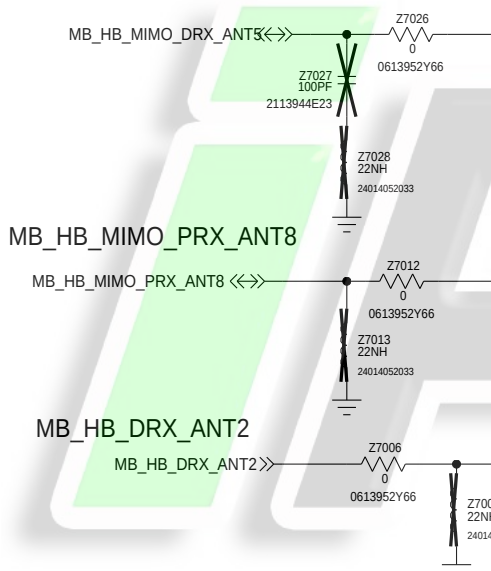
PROJECT NAME
B sr28c51351_b.cpm

CAD FILE NAME:

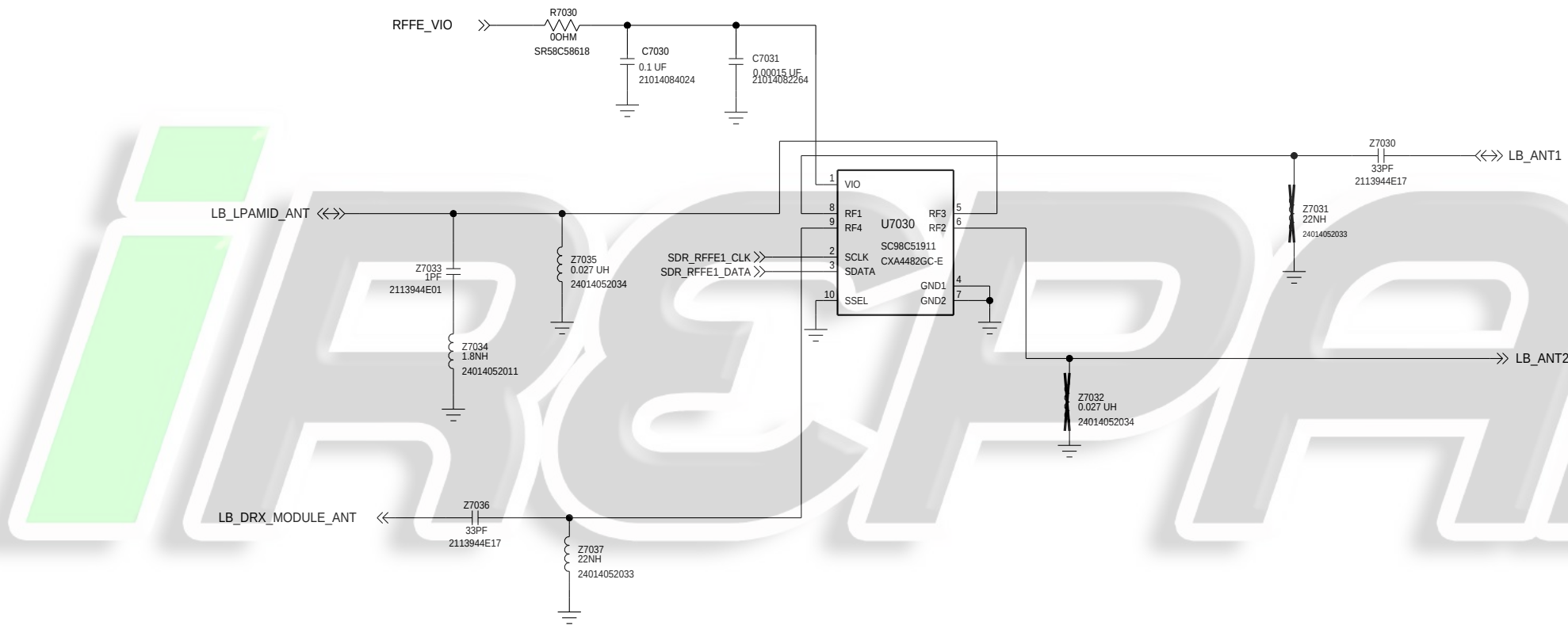
REV
03

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REFDES: 7000-7029

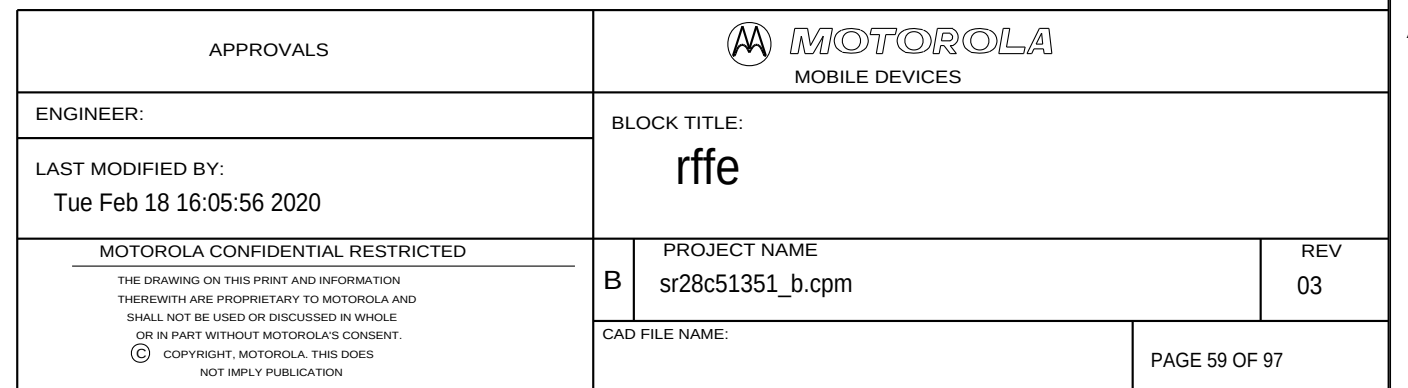

MOTOROLA
 MOBILE DEVICES

RFFE: LB SWITCH
REFDES: 7030-7039

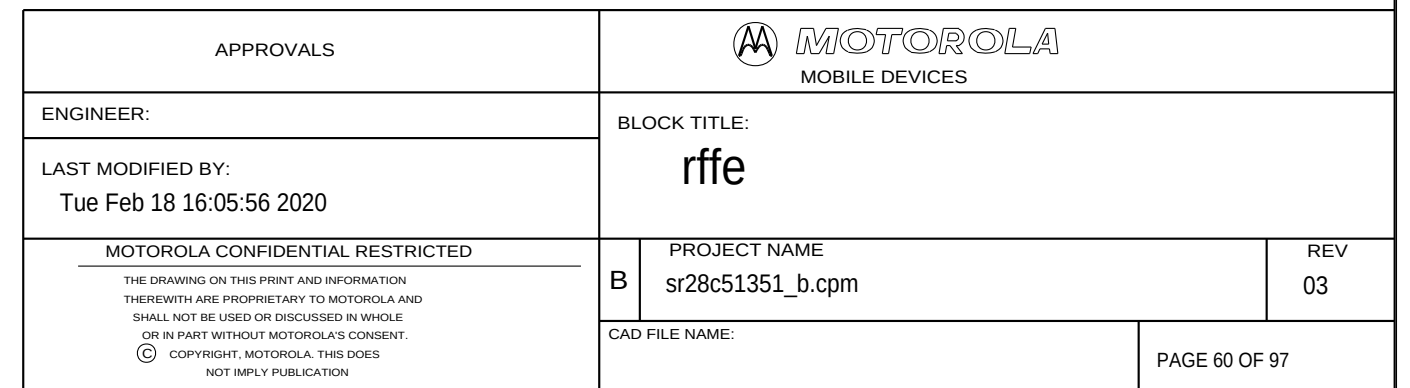


APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER:	BLOCK TITLE: rffe			
LAST MODIFIED BY: Tue Feb 18 16:05:55 2020				
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		B sr28c51351_b.cpm		03
		CAD FILE NAME:		PAGE 58 OF 97

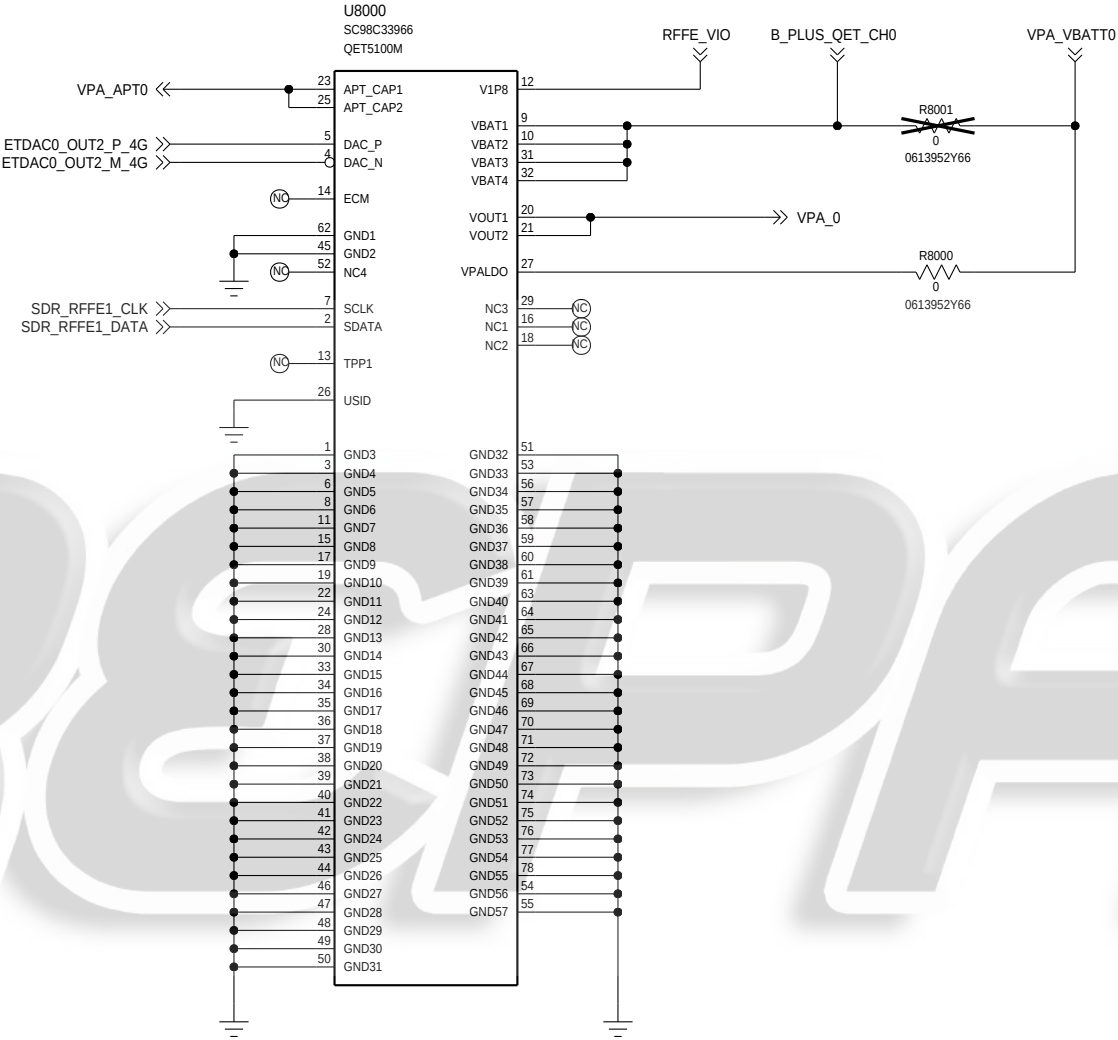
REFDES: 7050-7079



REFDES: 7080-7099



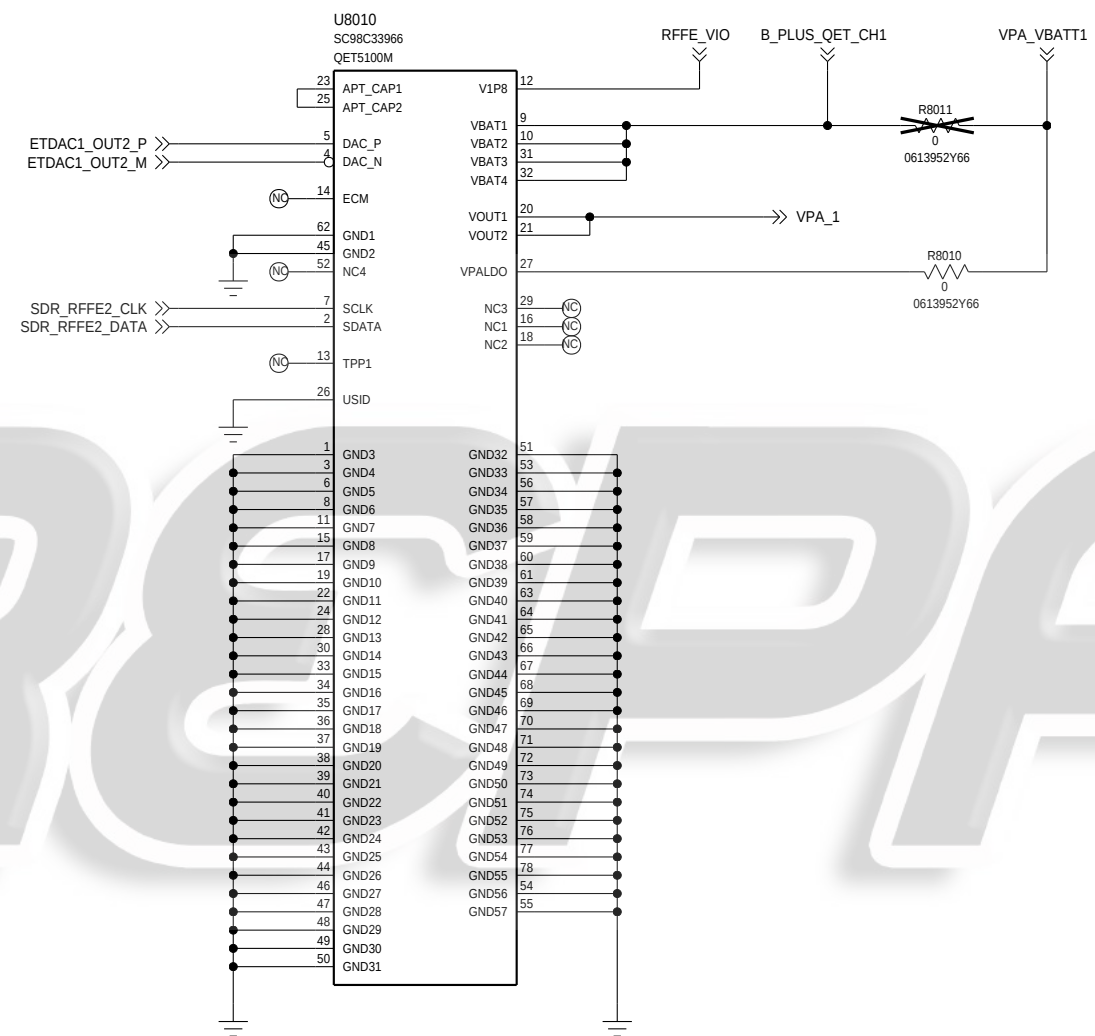
RFFE: QET5100 CH0
REF 8000-8099



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: MOTOROLA		BLOCK TITLE: rffe		
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:05:57 2020				
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		B sr28c51351_b.cpm		03
		CAD FILE NAME:		PAGE 61 OF 97

RFFE: QET5100 CH1

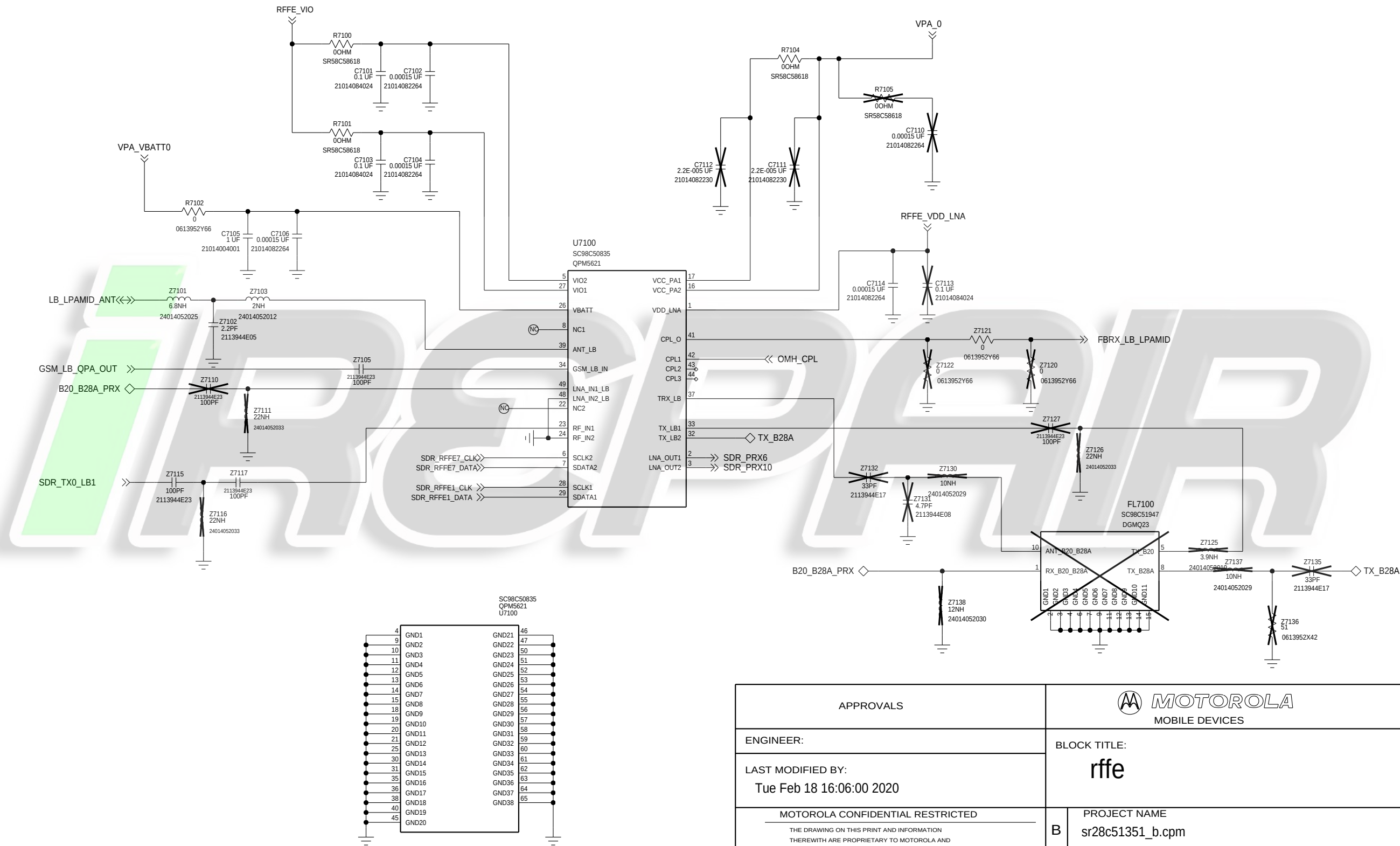
REFDES: 8100-8199



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER:		BLOCK TITLE: rffe		
LAST MODIFIED BY: Tue Feb 18 16:05:58 2020				
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		B sr28c51351_b.cpm		03
		CAD FILE NAME:		PAGE 62 OF 97

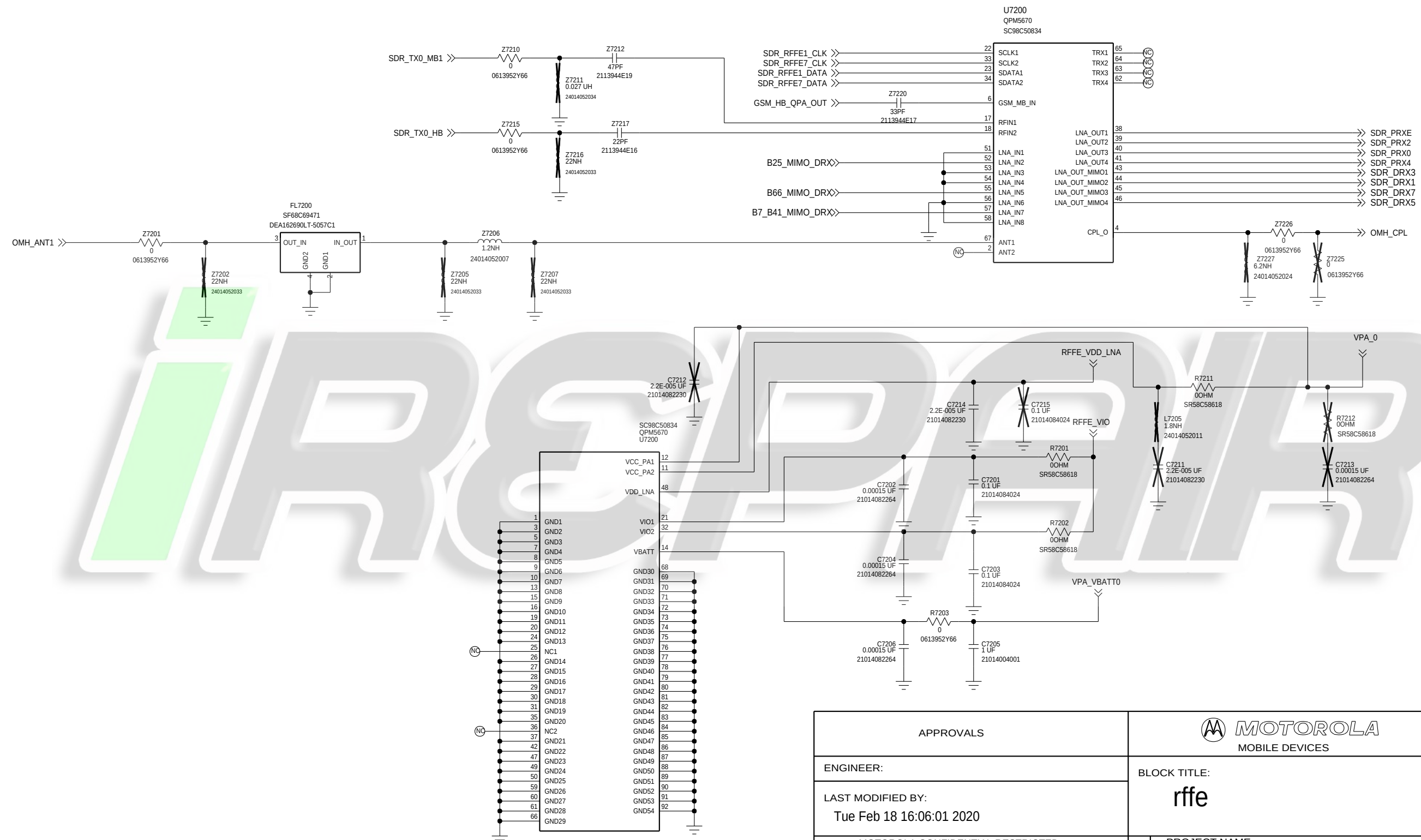
RFFE: LB LPAMID TX0

REFDES: 7100-7199



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY:		rffe	
Tue Feb 18 16:06:00 2020			
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		REV	
		03	
		PAGE 63 OF 97	

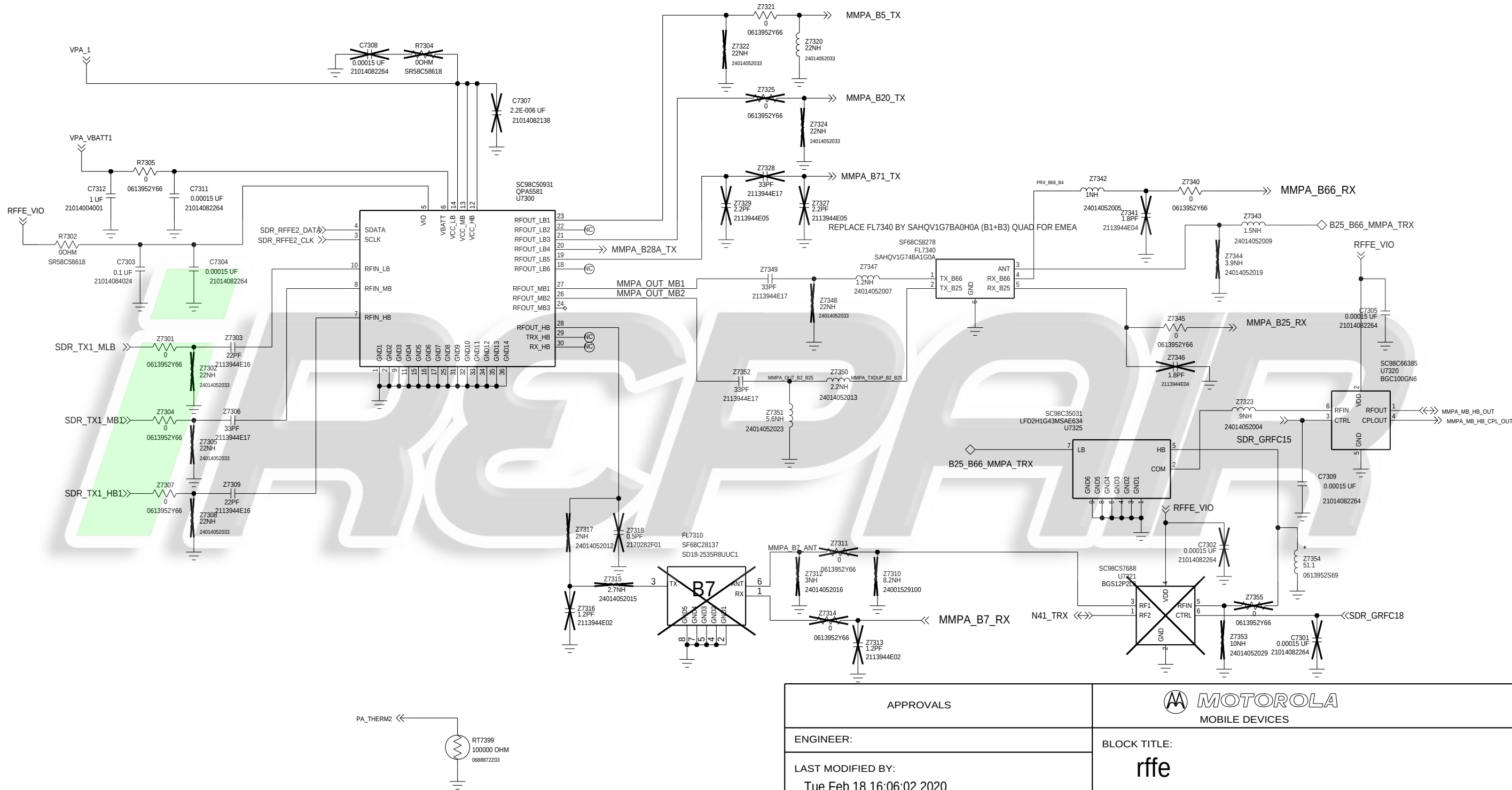
RFFE: MHB LPAMID TX0
REF 7200-7299



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER:		BLOCK TITLE: rffe		
LAST MODIFIED BY: Tue Feb 18 16:06:01 2020				
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		CAD FILE NAME:		PAGE 64 OF 97

RFFE: MMPA TX1

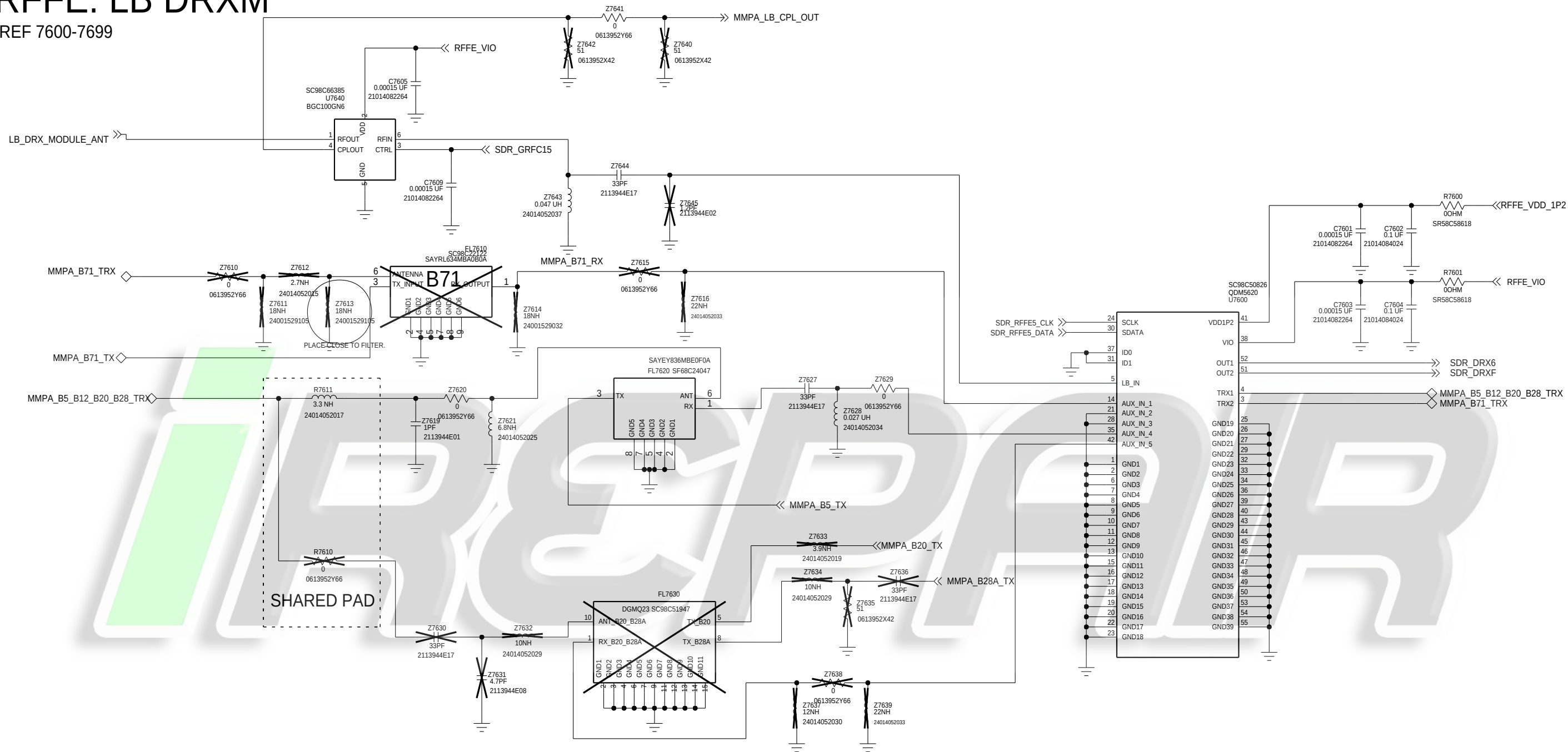
REFDES: 7300-7399



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: rffe	
LAST MODIFIED BY: Tue Feb 18 16:06:02 2020		PROJECT NAME B sr28c51351_b.cpm	
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		CAD FILE NAME: PAGE 65 OF 97	

RFFE: LB DRXM

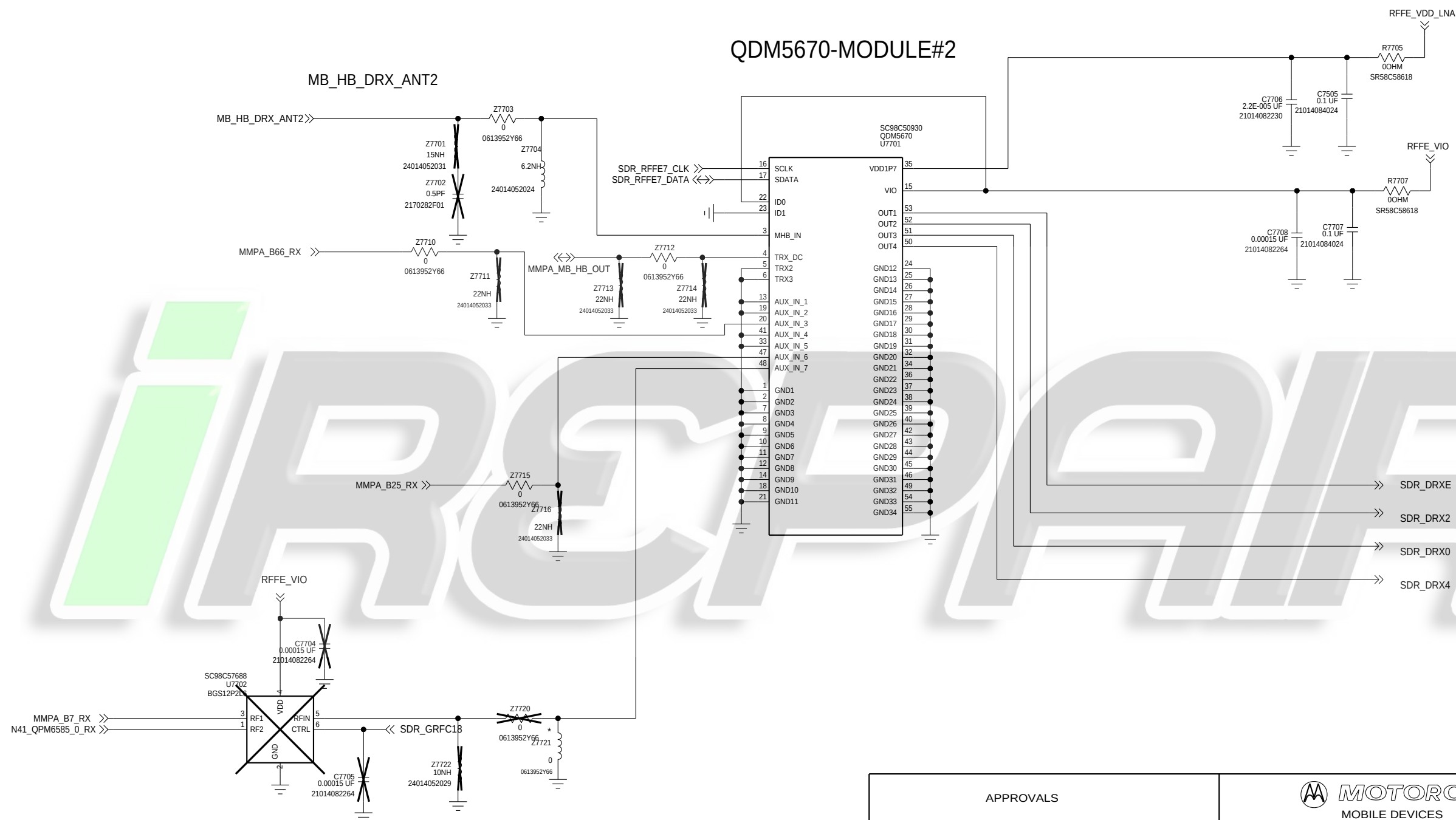
REF 7600-7699



APPROVALS		BLOCK TITLE:	
ENGINEER: MOTOROLA		rffe	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:03 2020		PROJECT NAME B 8020150851_b.cpm	
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		PAGE 66 OF 97	

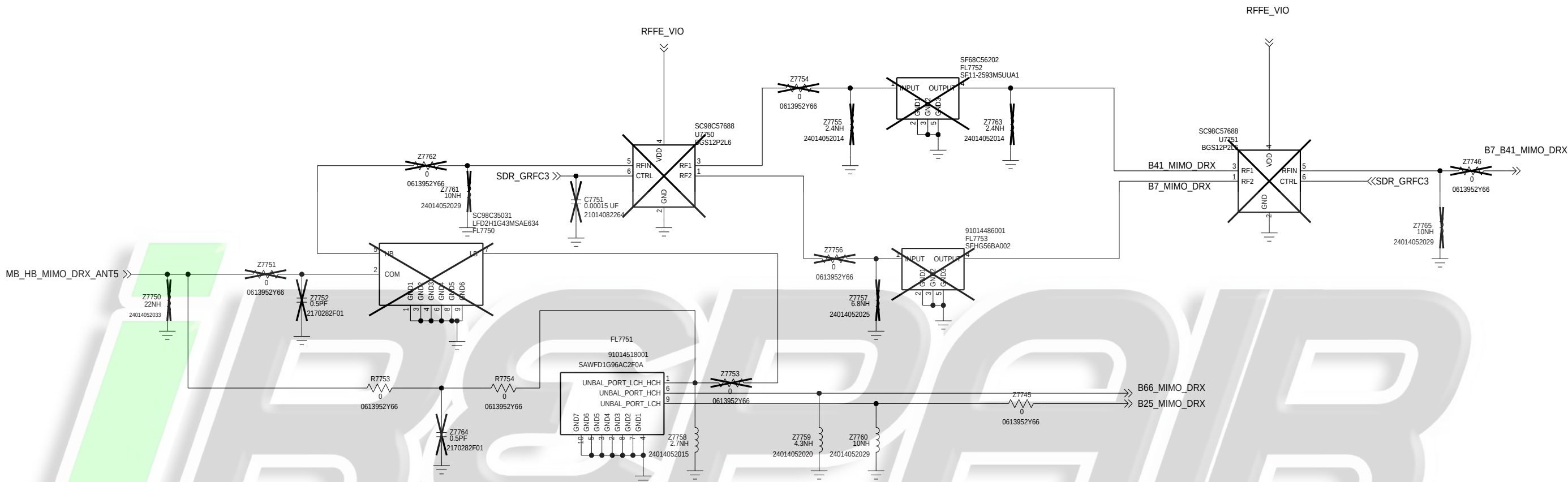
RFFE: MHB DRX

REFDES: 7700-7729



APPROVALS		BLOCK TITLE:	
ENGINEER:		rffe	
LAST MODIFIED BY:		Tue Feb 18 16:06:03 2020	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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CAD FILE NAME:		PAGE 67 OF 97	

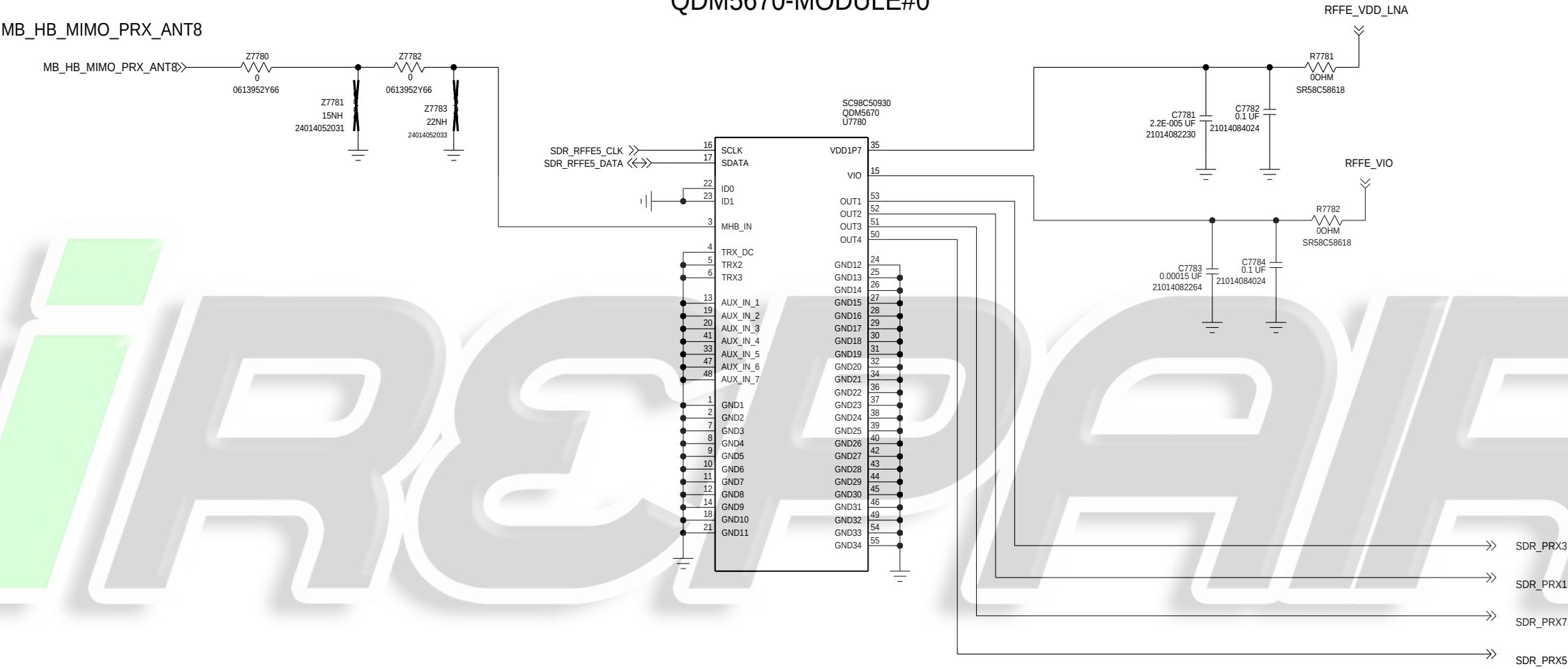
RFFE: MHB MIMO DRX
REFDES: 7750-7759



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: rffe	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:04 2020			
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		CAD FILE NAME: PAGE 68 OF 97	

RFFE: MHB MIMO PRX
REFDES: 7780-7789

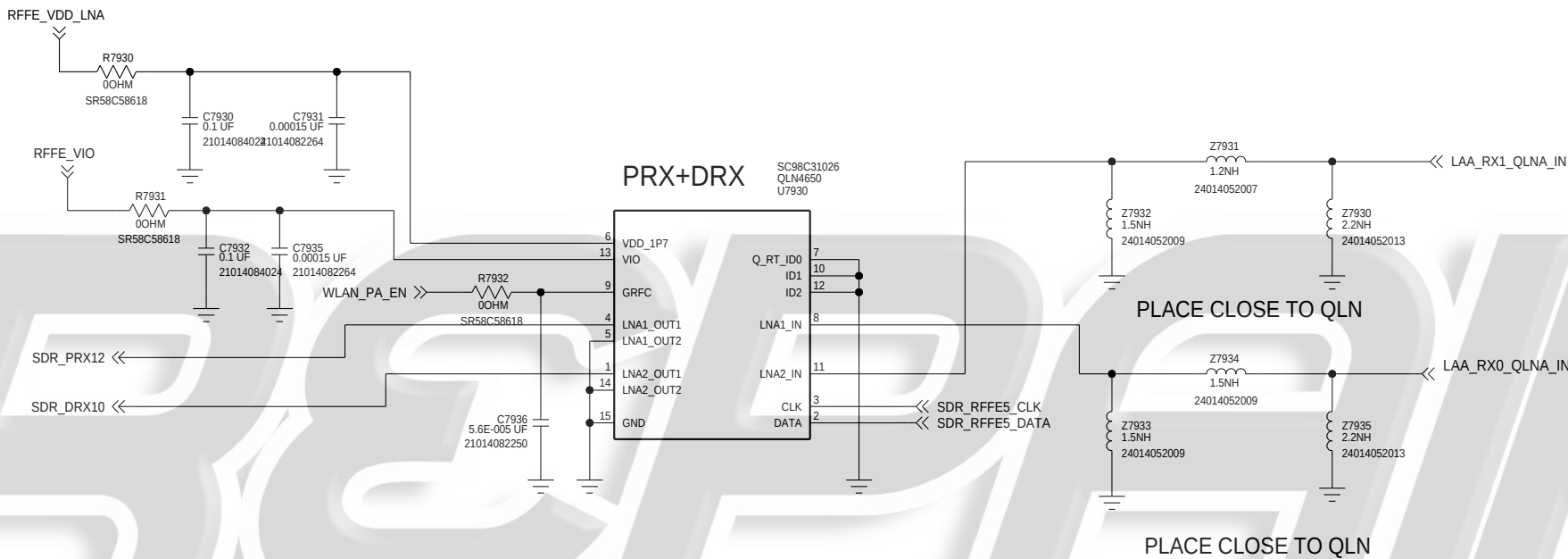
QDM5670-MODULE#0



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY:		rffe	
Tue Feb 18 16:06:05 2020			
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		REV 03	
		PAGE 69 OF 97	

RFFE: LAA LNA

REF 7900-7999



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: rffe	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:06 2020			
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		CAD FILE NAME:	PAGE 70 OF 97

REF 8500-8599

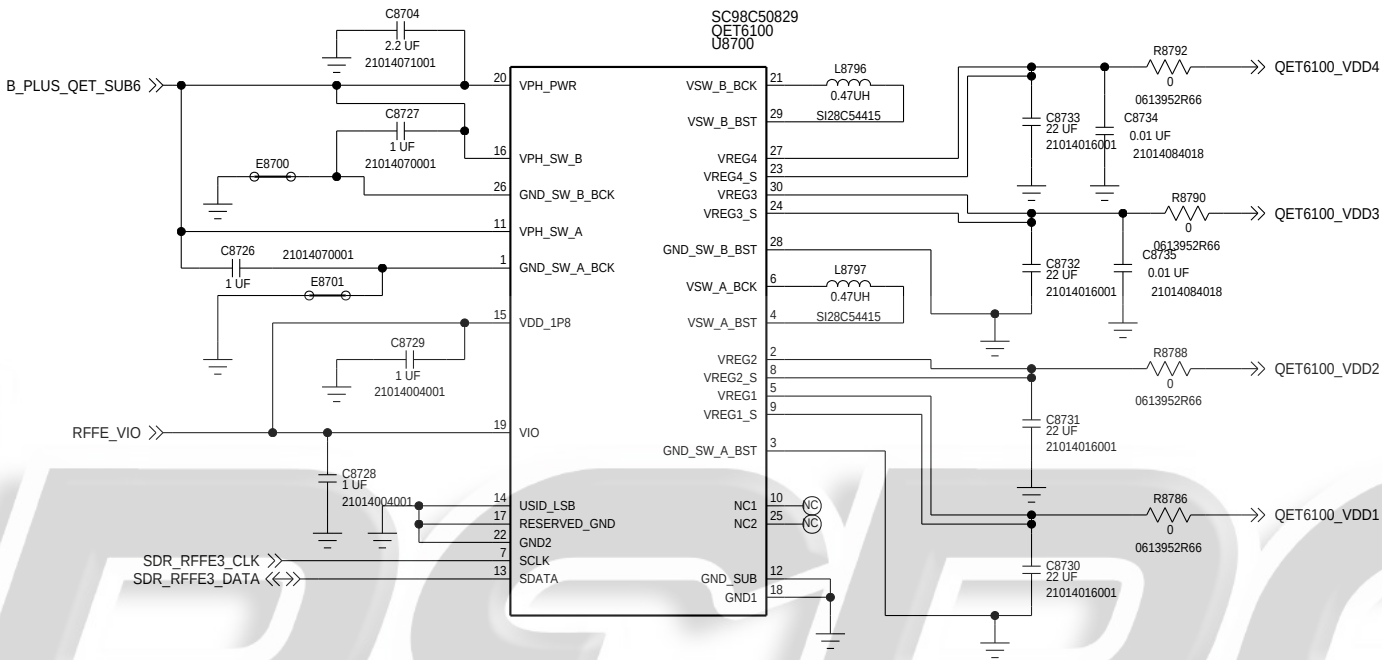


B13 NOTCH

APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER:	BLOCK TITLE: rffe			
LAST MODIFIED BY: Tue Feb 18 16:06:07 2020				
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		CAD FILE NAME:		PAGE 71 OF 97

RFFE: SUB6 QET6100

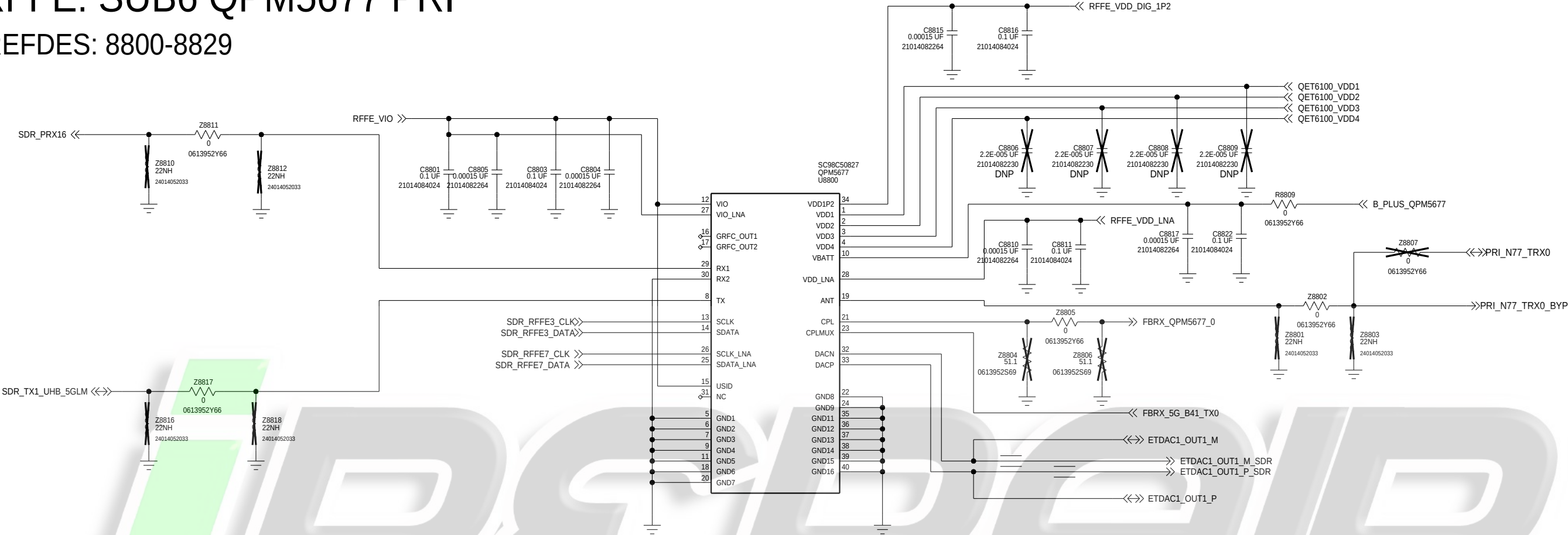
REFDES: 8700-8799



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: rffe	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:09 2020			
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		CAD FILE NAME: PAGE 72 OF 97	

RFFE: SUB6 QPM5677 PRI

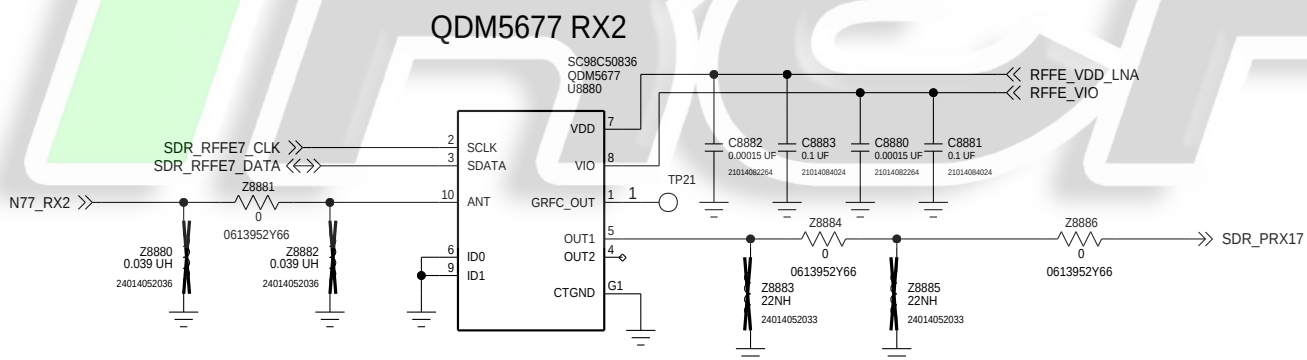
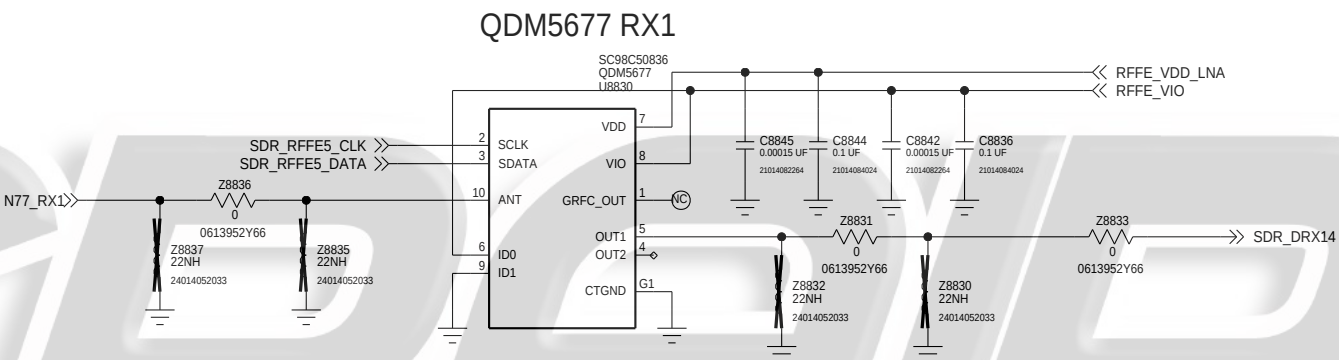
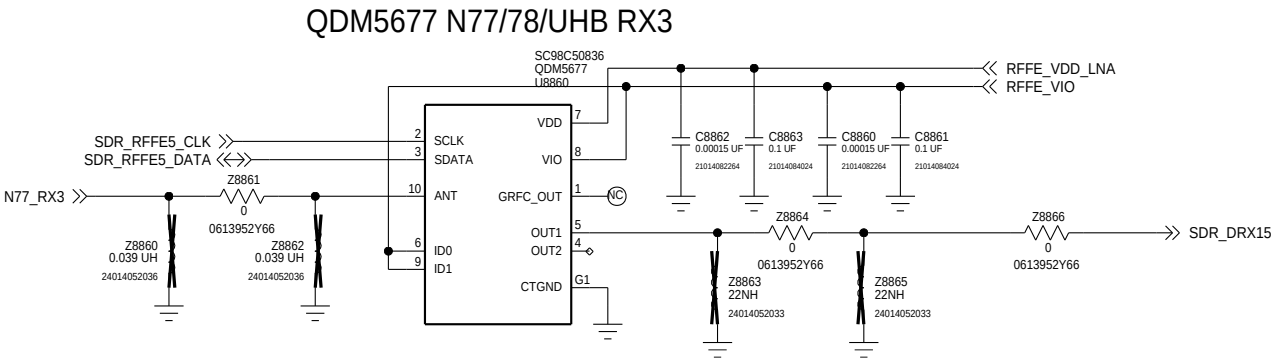
REFDES: 8800-8829



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: rffe	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:09 2020			
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CAD FILE NAME:			PAGE 73 OF 97

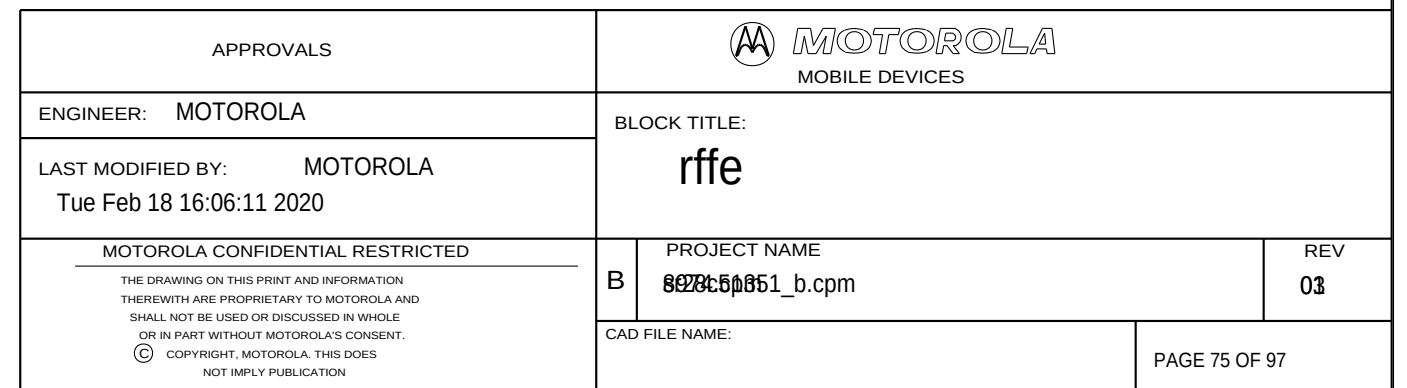
RFFE: SUB6 QDM5677 RX

REFDES: 8830-8899



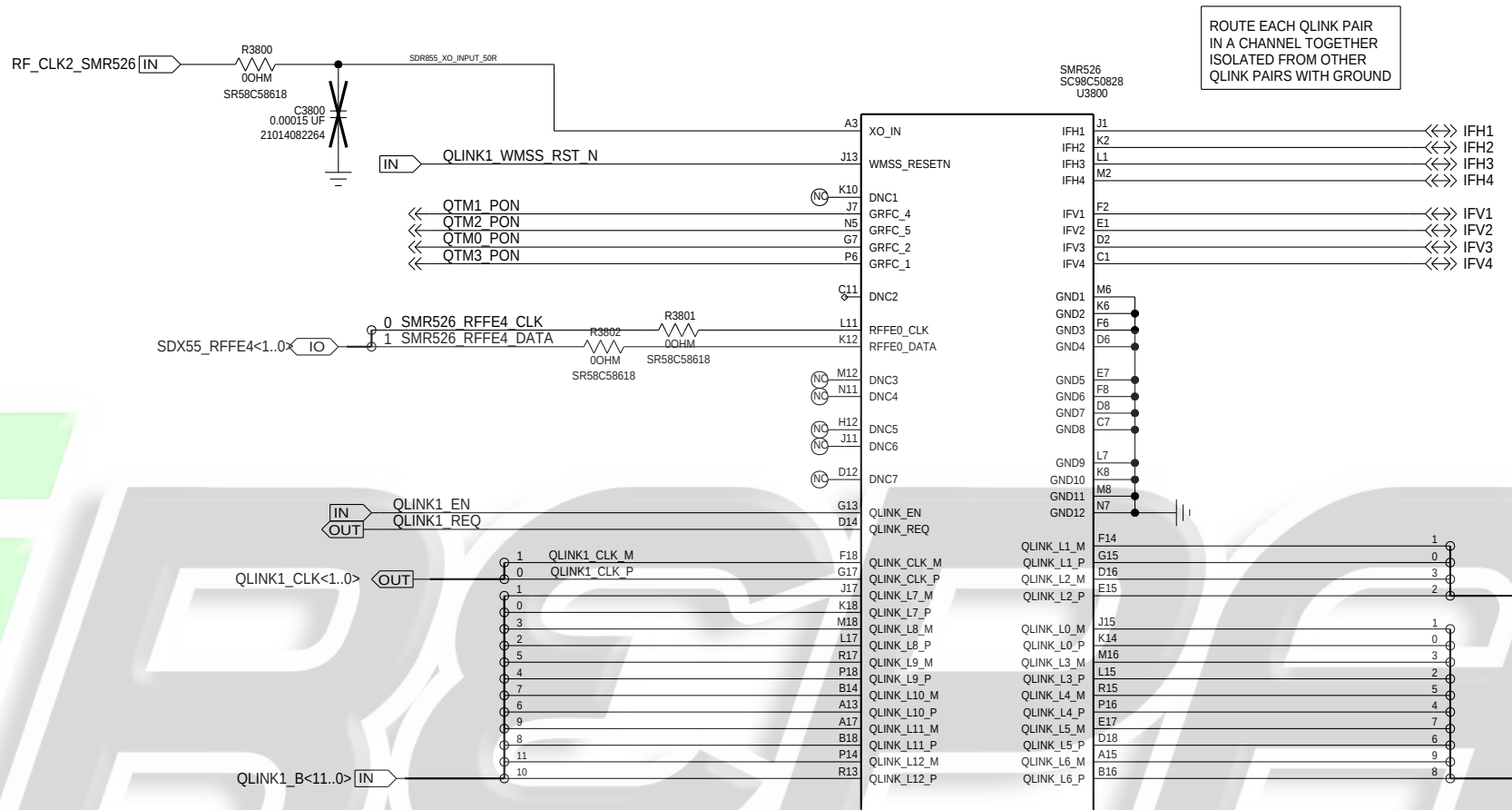
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: rffe	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:06:10 2020		PROJECT NAME B 802150851_b.cpm	
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		CAD FILE NAME: PAGE 74 OF 97	

REFDES: 8900-8929



SMR526: CTRL

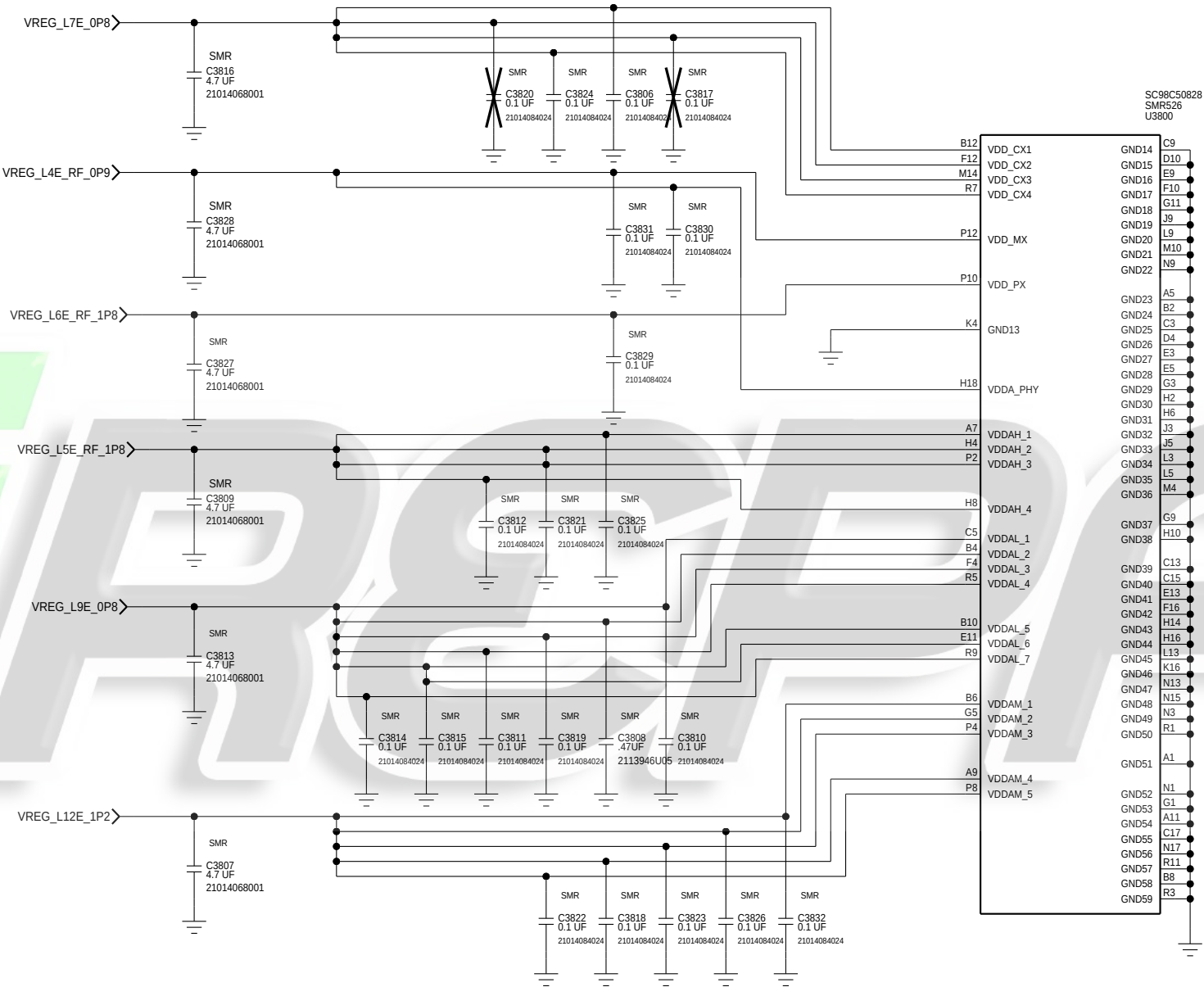
REFDES: 3800-3899



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: mmwave	
LAST MODIFIED BY: Tue Feb 18 16:05:48 2020			
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		CAD FILE NAME: PAGE 76 OF 97	

SMR526: PWR

REFDES: 3800-3899

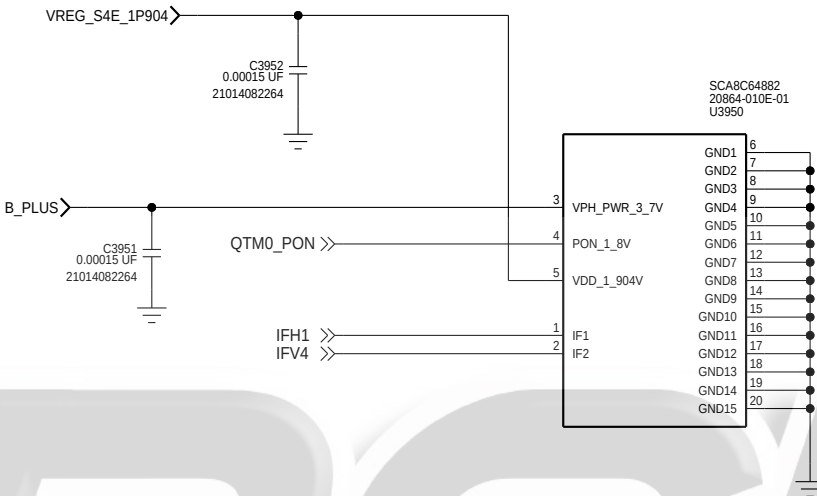


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Tue Feb 18 16:05:50 2020		mmwave	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B	PROJECT NAME sr28c51351_b.cpm
		REV 03	
		PAGE 77 OF 97	

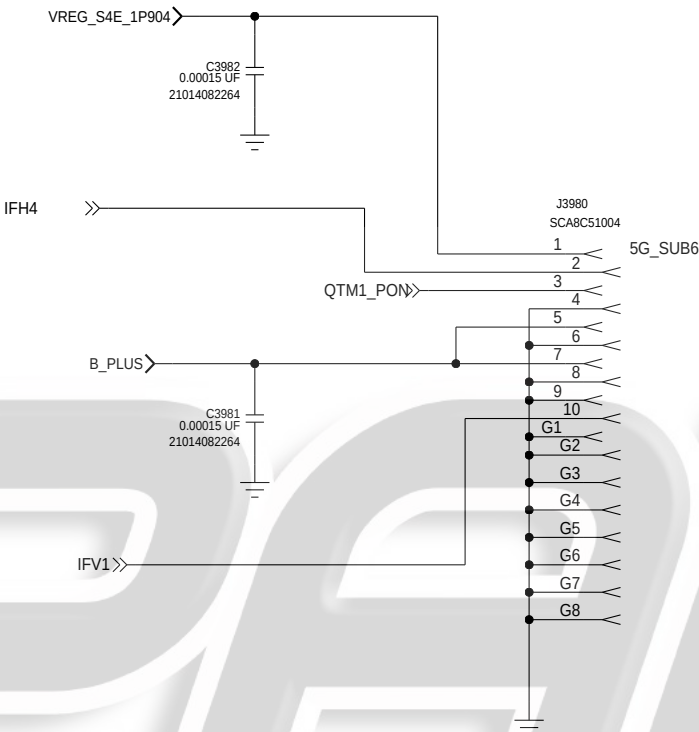
QTM525: CONNECTORS

REFDES: 3900-3999

QTM0 (BACK)



QTM1 (TOP)



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Tue Feb 18 16:05:50 2020		mmwave	
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		CAD FILE NAME: PAGE 78 OF 97	

D

C

B

A



C

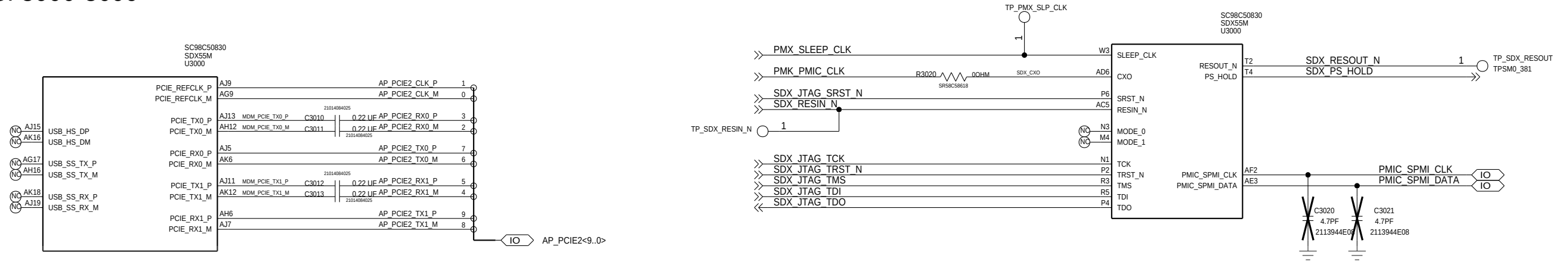
B

A

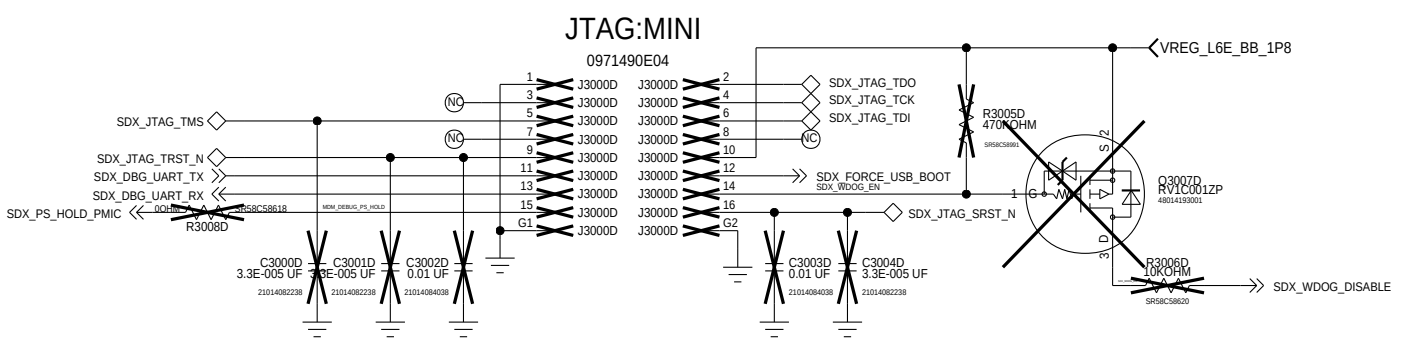
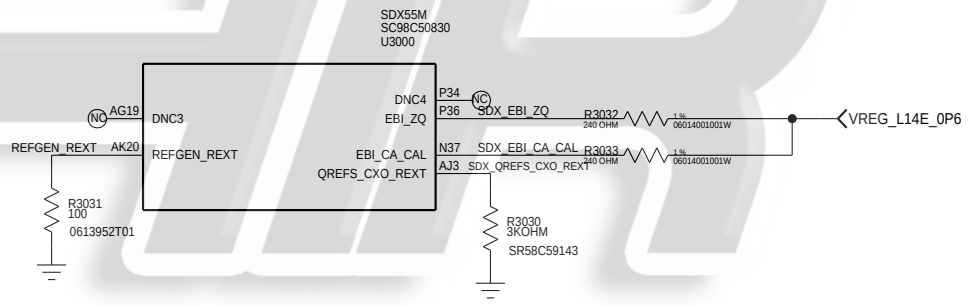
PAGE 79 OF 97

SDX55M: PCIE/CTRL/MEM/USB

REFDES: 3000-3099



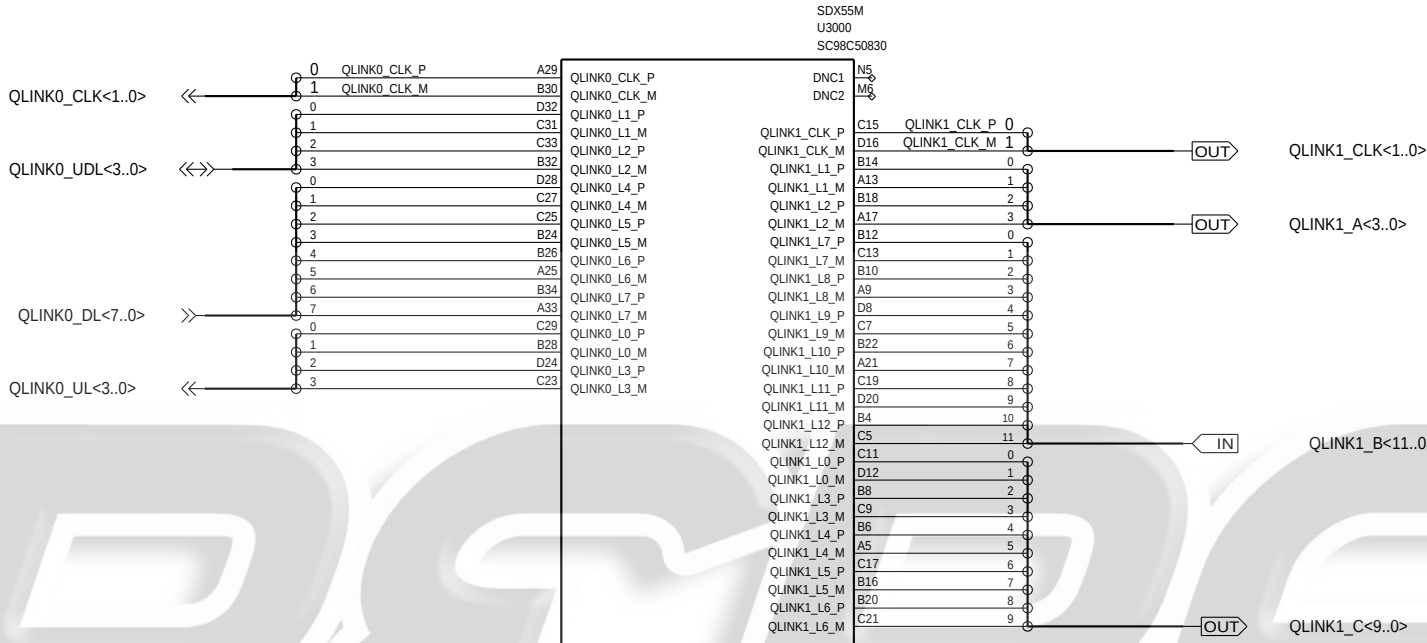
REPAIR



APPROVALS		BLOCK TITLE:	
ENGINEER:		modem	
LAST MODIFIED BY:		PROJECT NAME	
Tue Feb 18 16:05:39 2020		B sr28c51351_b.cpm	
MOTOROLA CONFIDENTIAL RESTRICTED		REV	
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CAD FILE NAME:		PAGE 80 OF 97	

SDX55M: QLINKS

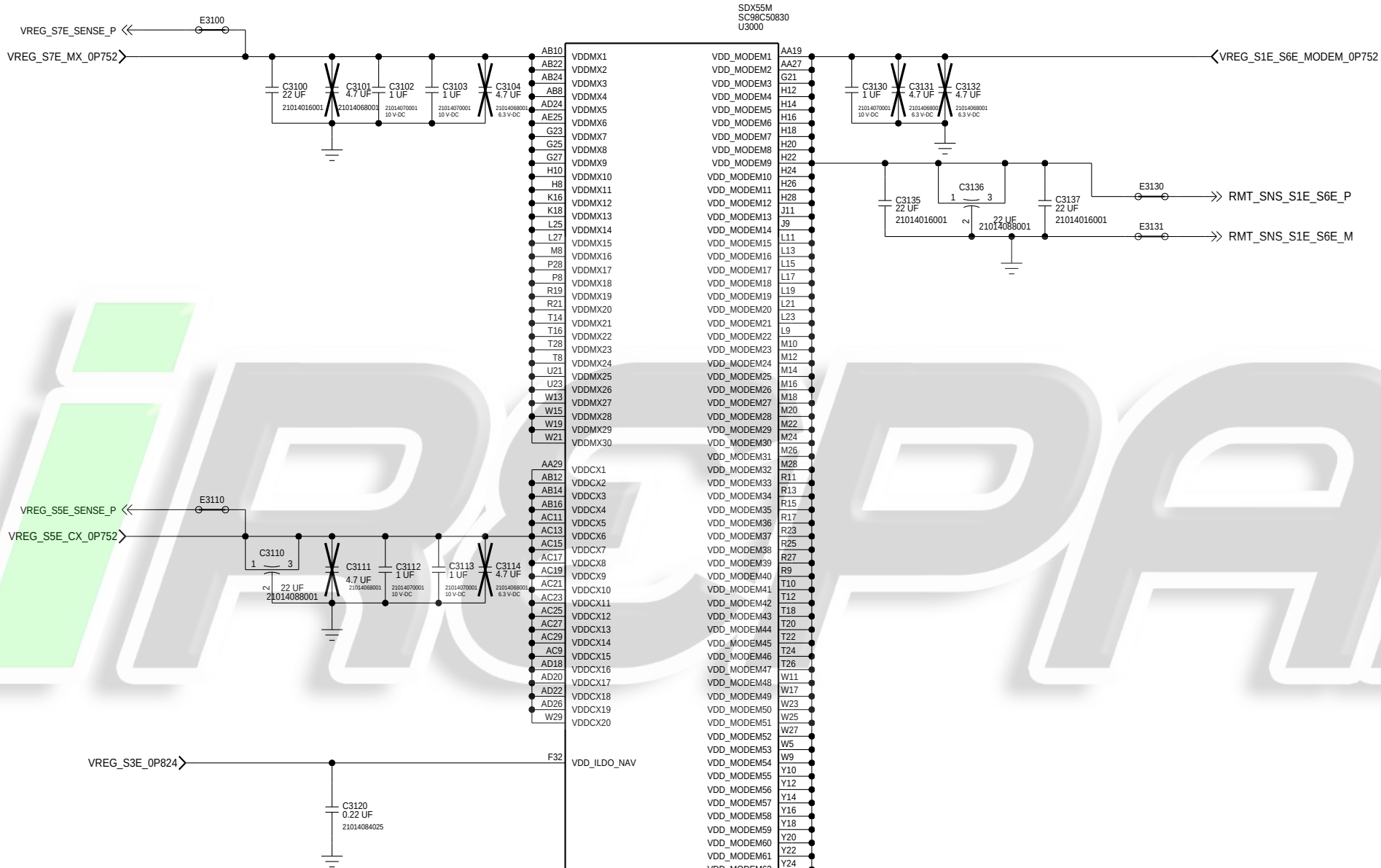
REFDES: 3000-3099



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: modem	
LAST MODIFIED BY: Tue Feb 18 16:05:39 2020			
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		CAD FILE NAME:	

SDX55M: PWR1

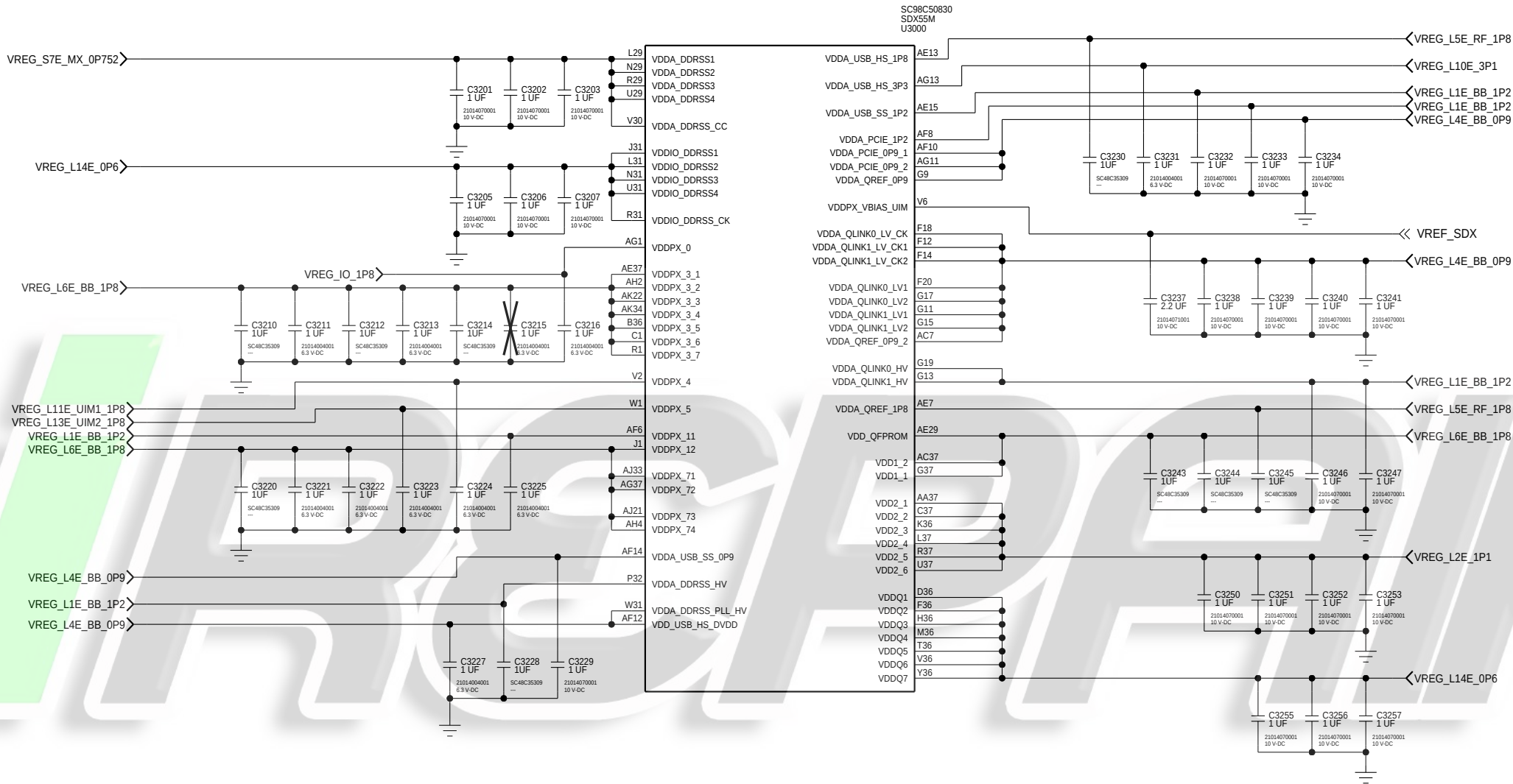
REFDES: 3100-3199



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Tue Feb 18 16:05:40 2020		modem	
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		CAD FILE NAME: PAGE 82 OF 97	

SDX55M: PWR2

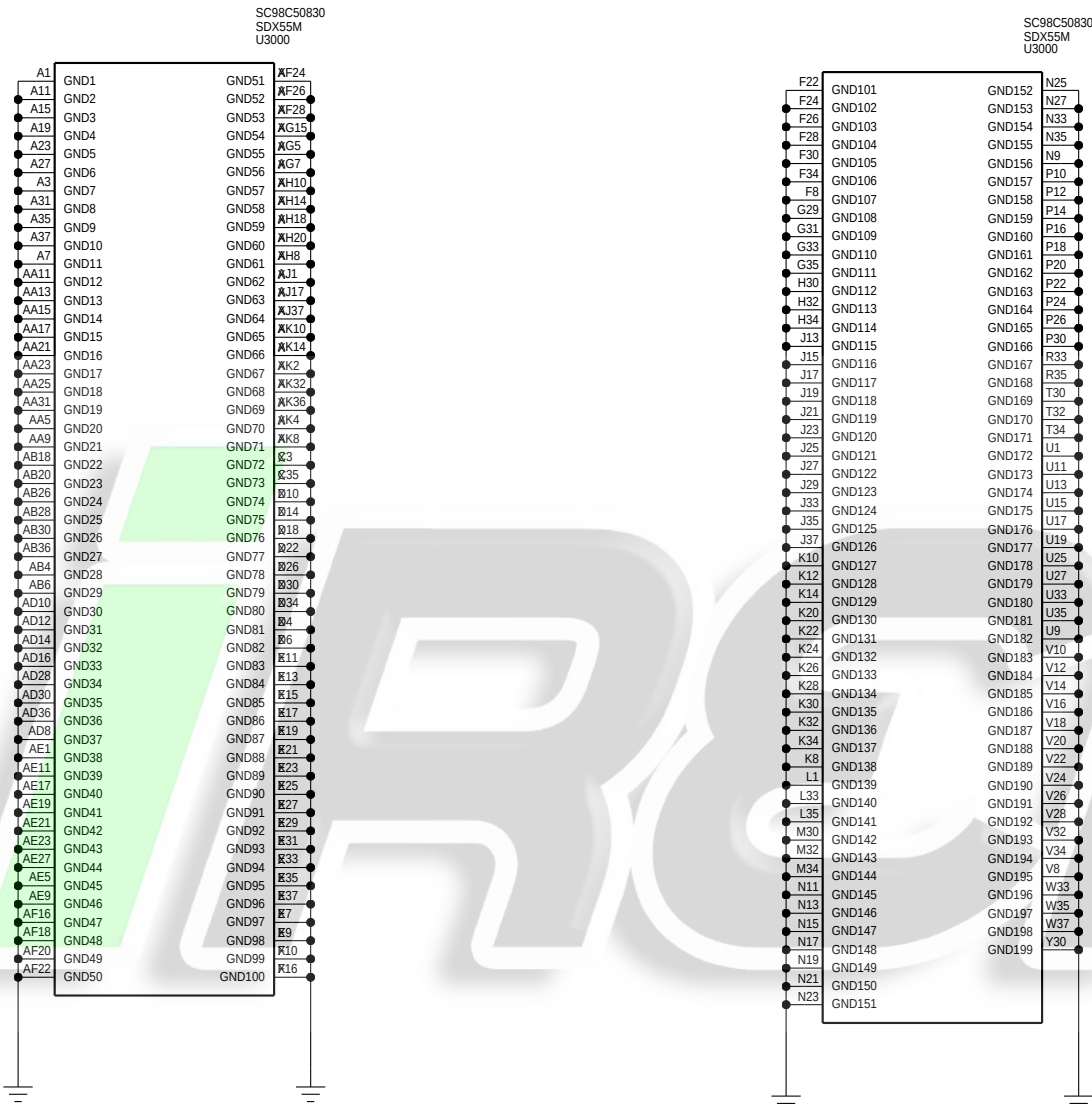
REFDES: 3200-3299



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: modem	
LAST MODIFIED BY: Tue Feb 18 16:05:41 2020			
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		CAD FILE NAME:	PAGE 83 OF 97

SDX55M: GND

REFDES: 3000-3099



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: modem	
LAST MODIFIED BY: Tue Feb 18 16:05:41 2020			
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		CAD FILE NAME:	PAGE 84 OF 97

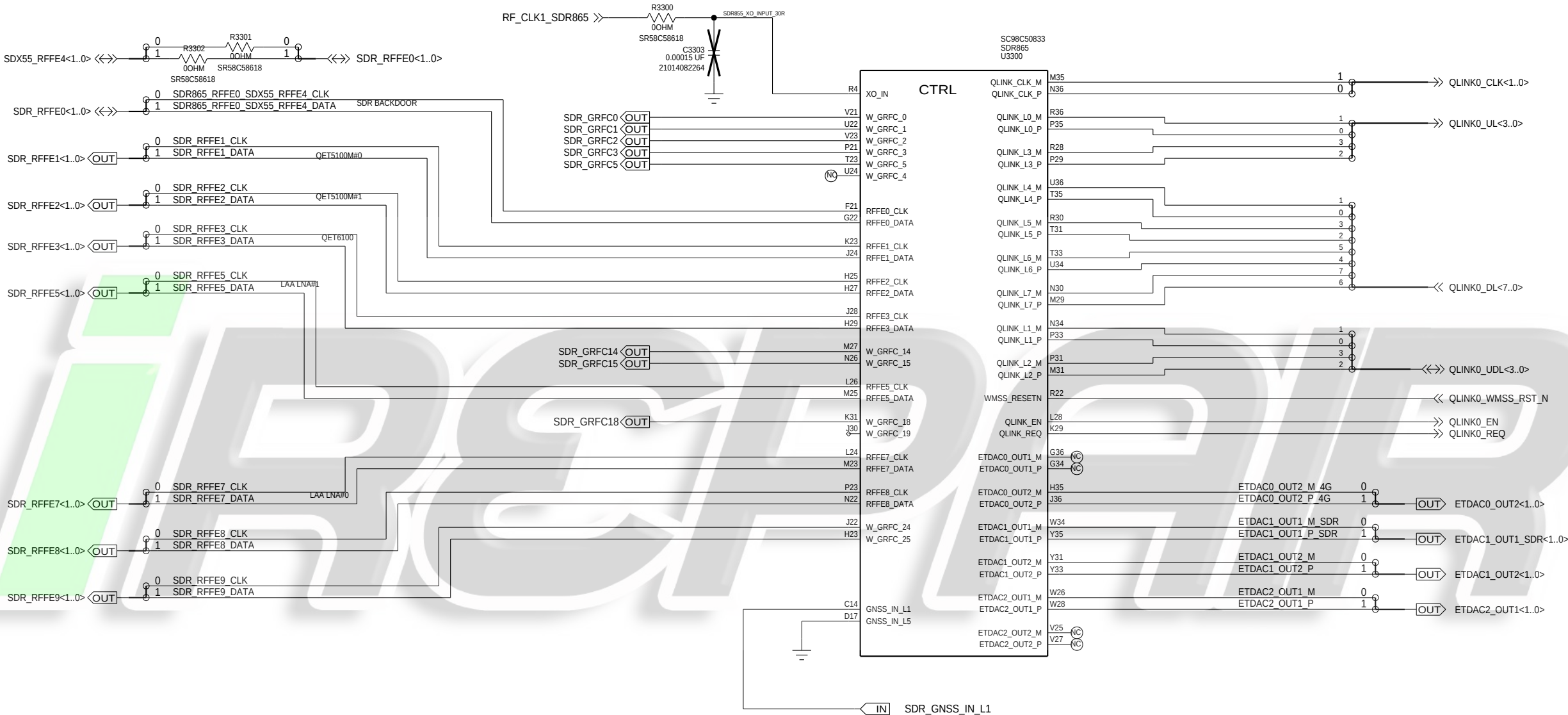
SDR865: CTRL / QLINK

REFDES: 3300-3399

NOTE: CONTROLLED IMPEDANCE FOR XO LINE MUST BE BETWEEN 30 OHM AND 50OHM
LOW IMPEDANCE IS PREFERRED.

THE SERIES PLACEHOLDER MUST BE A 0 OHM RESISTOR SINCE THE XO SIGNAL CARRIES A DC BIAS TO THE XO_IN INPUT.
DO NOT SHUNT THE CAPACITOR ON THE XO_IN TRACE

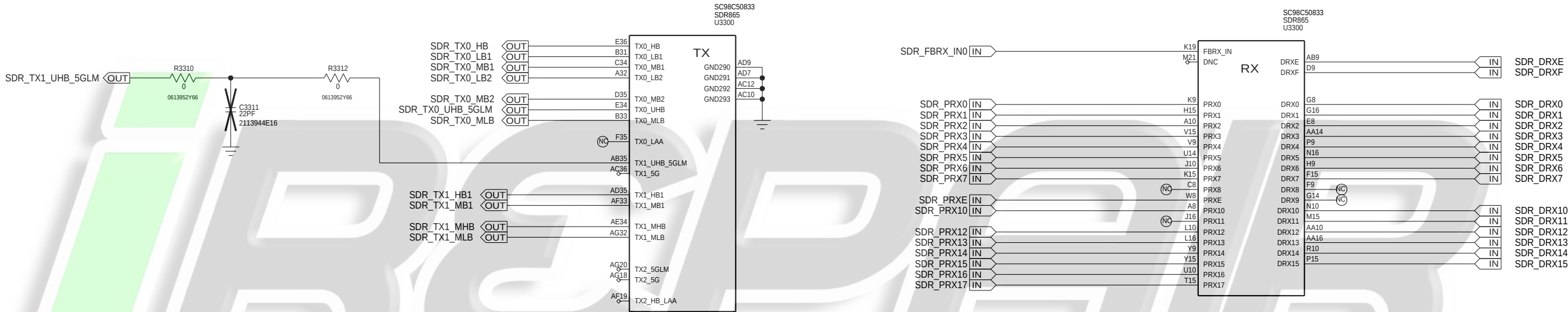
ROUTE EACH QLINK PAIR
IN A CHANNEL TOGETHER
ISOLATED FROM OTHER
QLINK PAIRS WITH GROUND



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:05:42 2020		modem	
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		REV 03	
		CAD FILE NAME:	
		PAGE 85 OF 97	

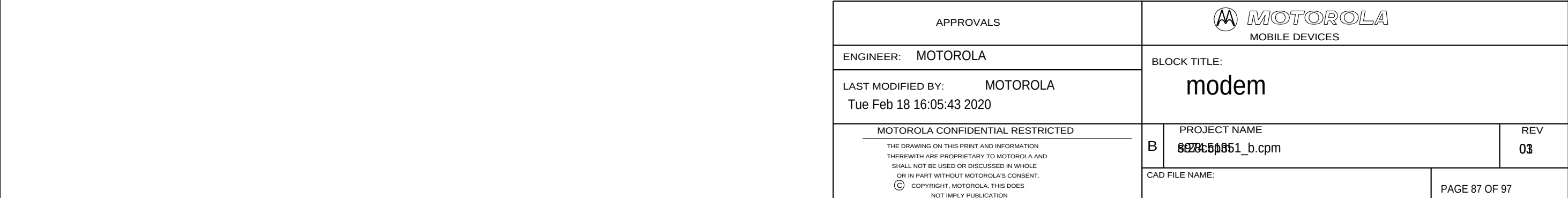
SDR865: TX & RX

REFDES: 3300-3399



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: modem	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:05:43 2020			
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		CAD FILE NAME:	

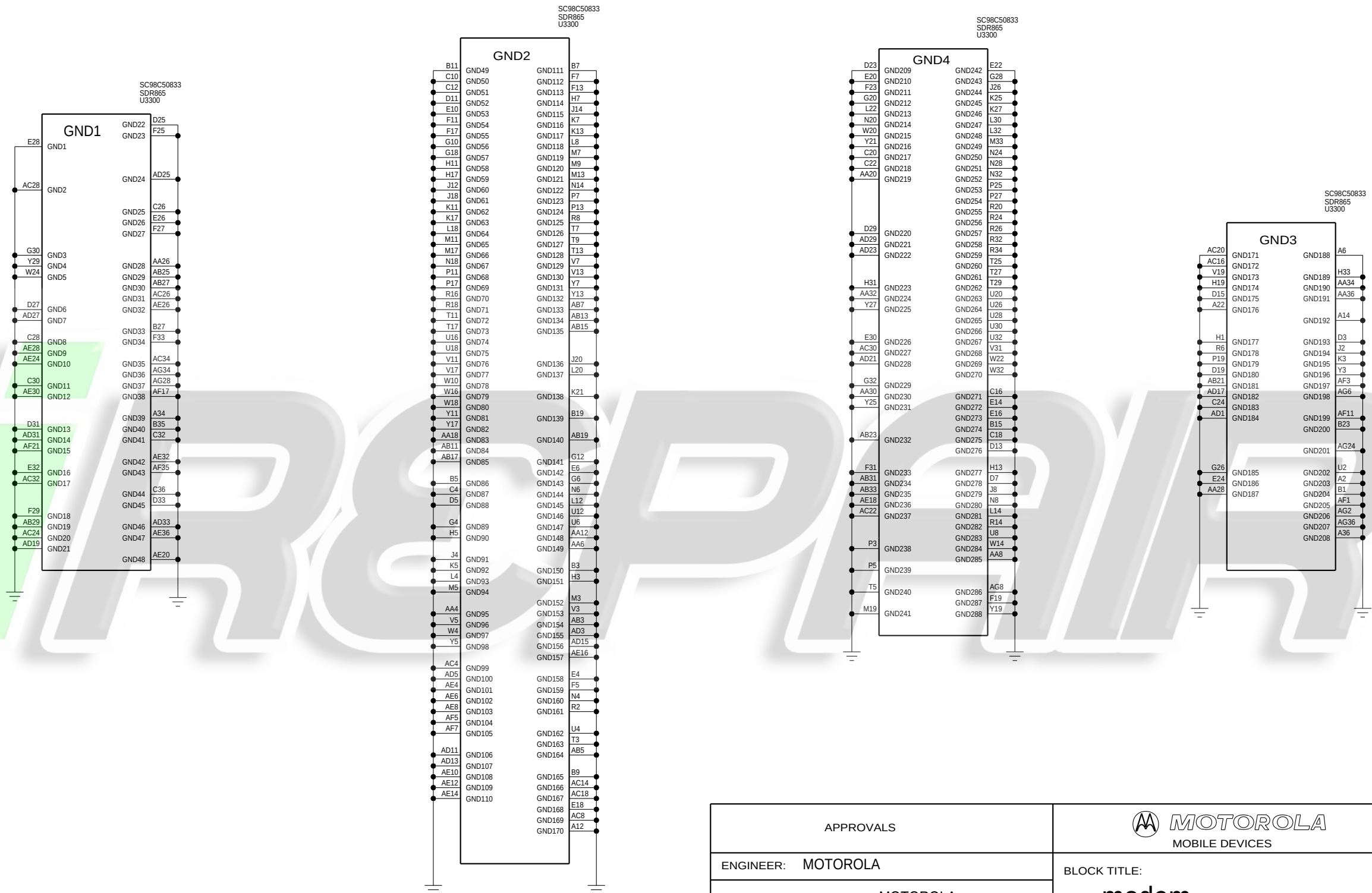
REFDES: 3400-3499



SDR865: GND

REFDES: 3300-3399

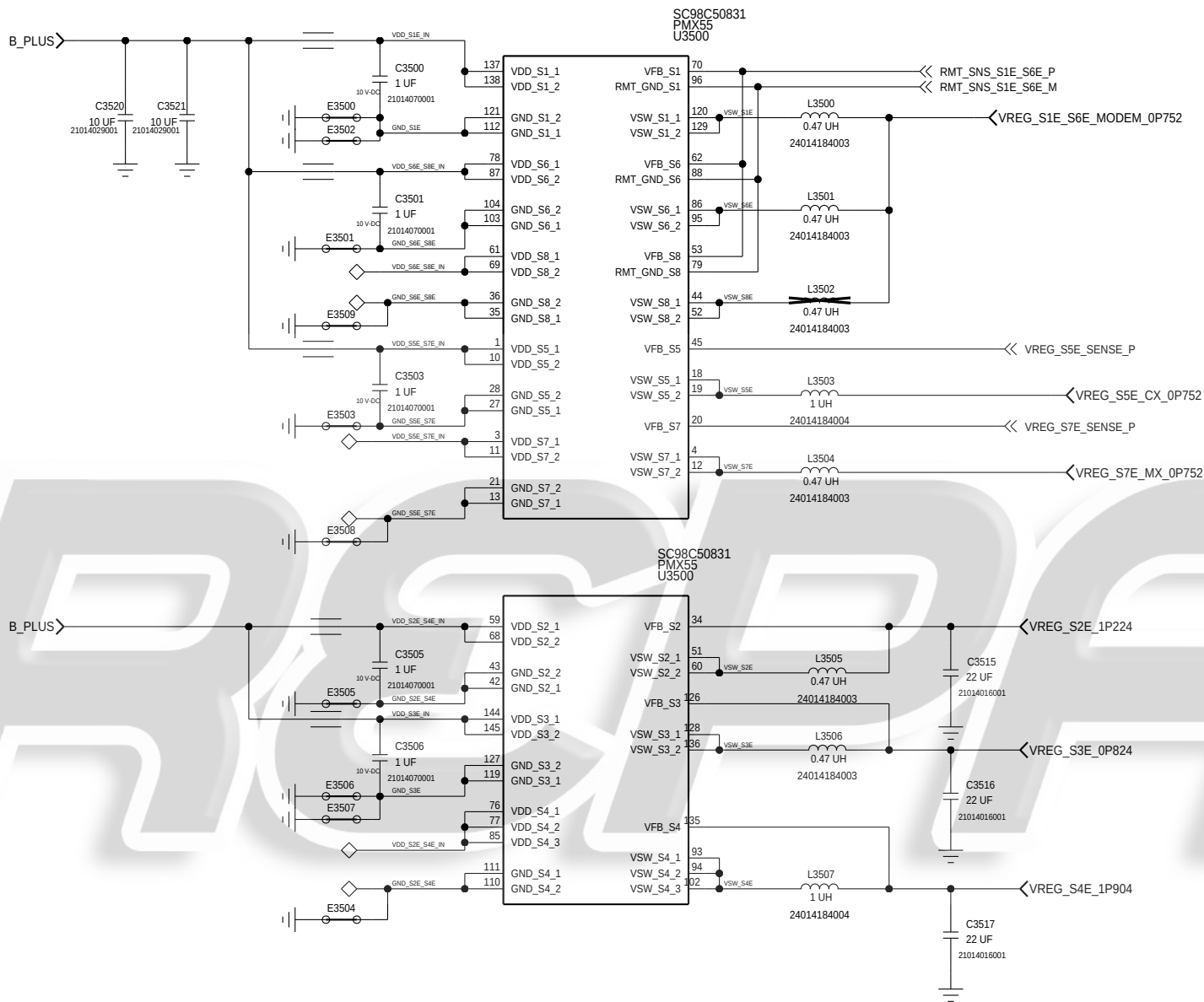
ANIBAL GARCIA VENEZUELA



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: modem	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:05:44 2020			
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		CAD FILE NAME: 0020150851_b.cpm	
		PAGE 88 OF 97	

PMX55: BUCKS 1/2

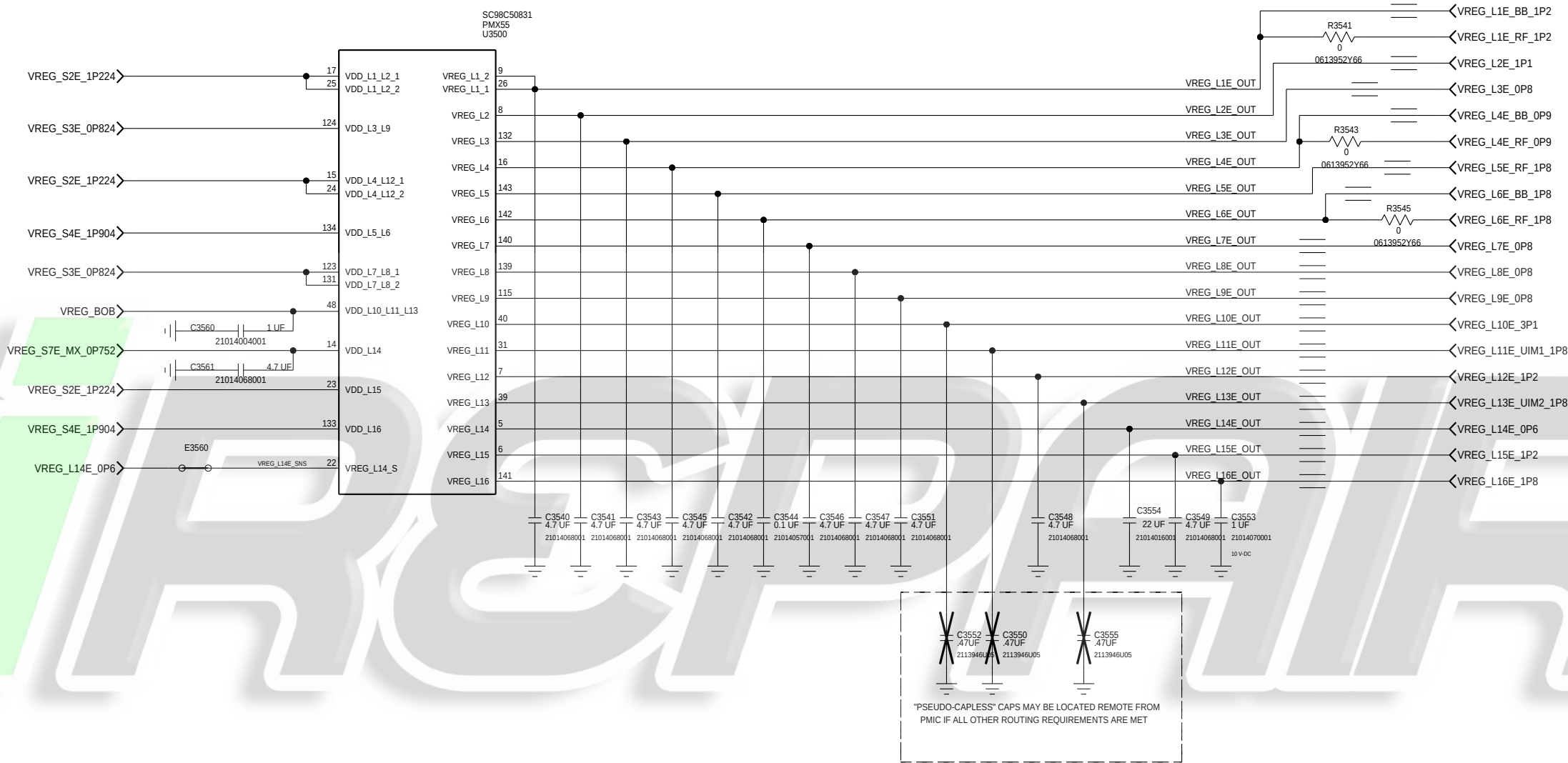
REFDES: 3500-3599



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Tue Feb 18 16:05:45 2020		modem	
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		CAD FILE NAME: PAGE 89 OF 97	

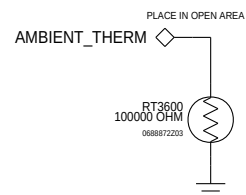
PMX55: LDOS

REFDES: 3500-3599



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: modem	
LAST MODIFIED BY: Tue Feb 18 16:05:46 2020			
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		CAD FILE NAME: PAGE 90 OF 97	

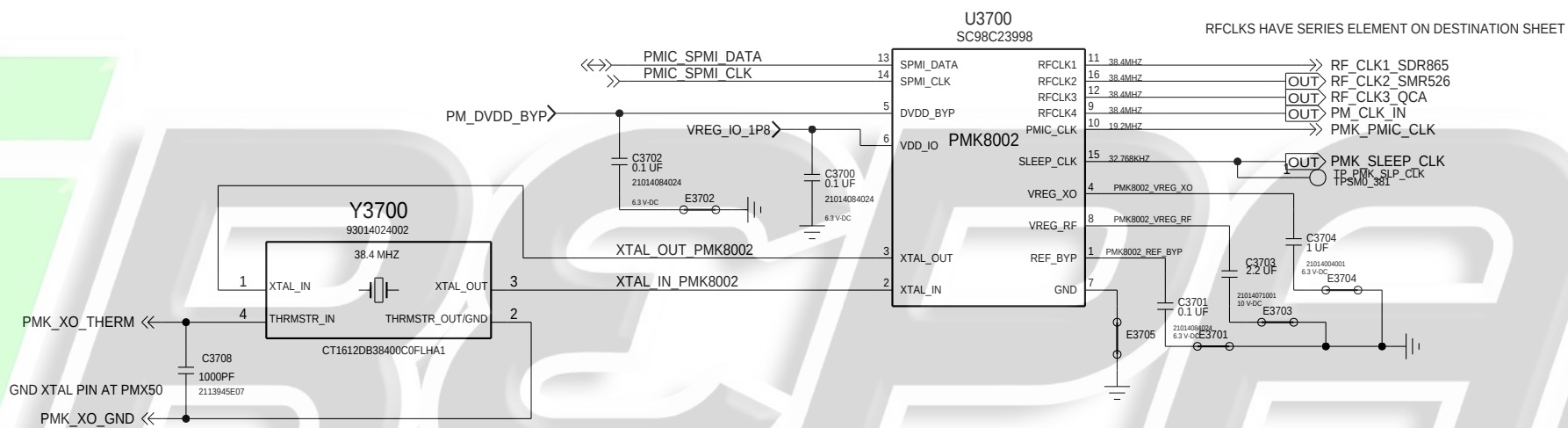
ANIBAL GARCIA VENEZUELA



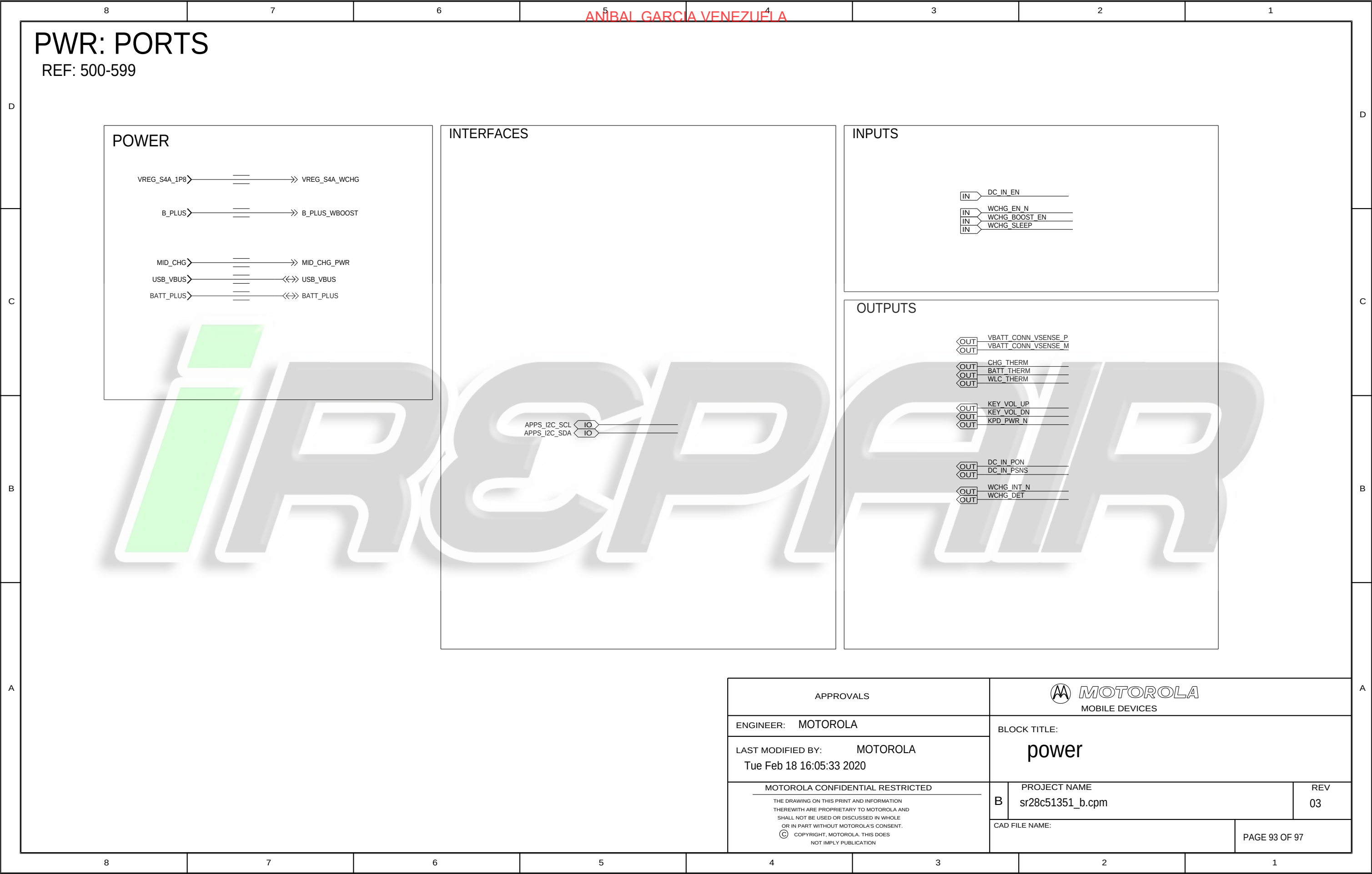
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: modem	
LAST MODIFIED BY: Tue Feb 18 16:05:46 2020			
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		CAD FILE NAME:	PAGE 91 OF 97

PMK8002: CLKS

REFDES: 3700-3799



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: modem	
LAST MODIFIED BY: Tue Feb 18 16:05:47 2020			
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		CAD FILE NAME: PAGE 92 OF 97	



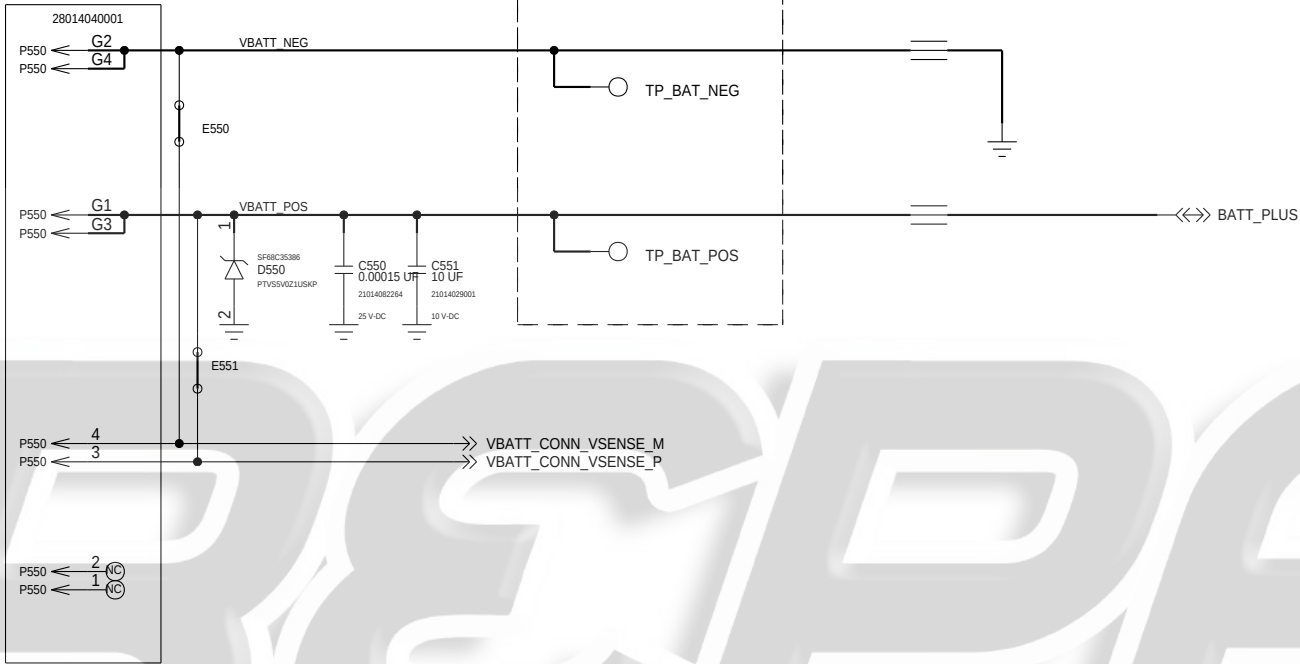
irAPR

PWR: BATT

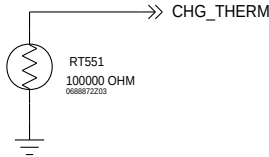
REF: 550-559

BATTERY CONN

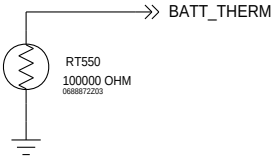
FACTORY TEST POINTS



AUX_THERM



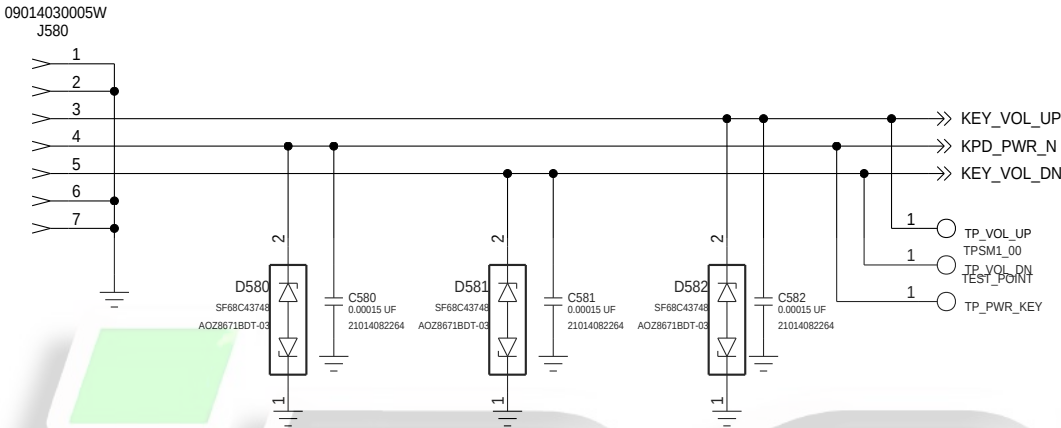
BATT_THERM



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:05:34 2020		power	
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		CAD FILE NAME: PAGE 94 OF 97	

PWR: SIDEKEYS

REF: 580-589

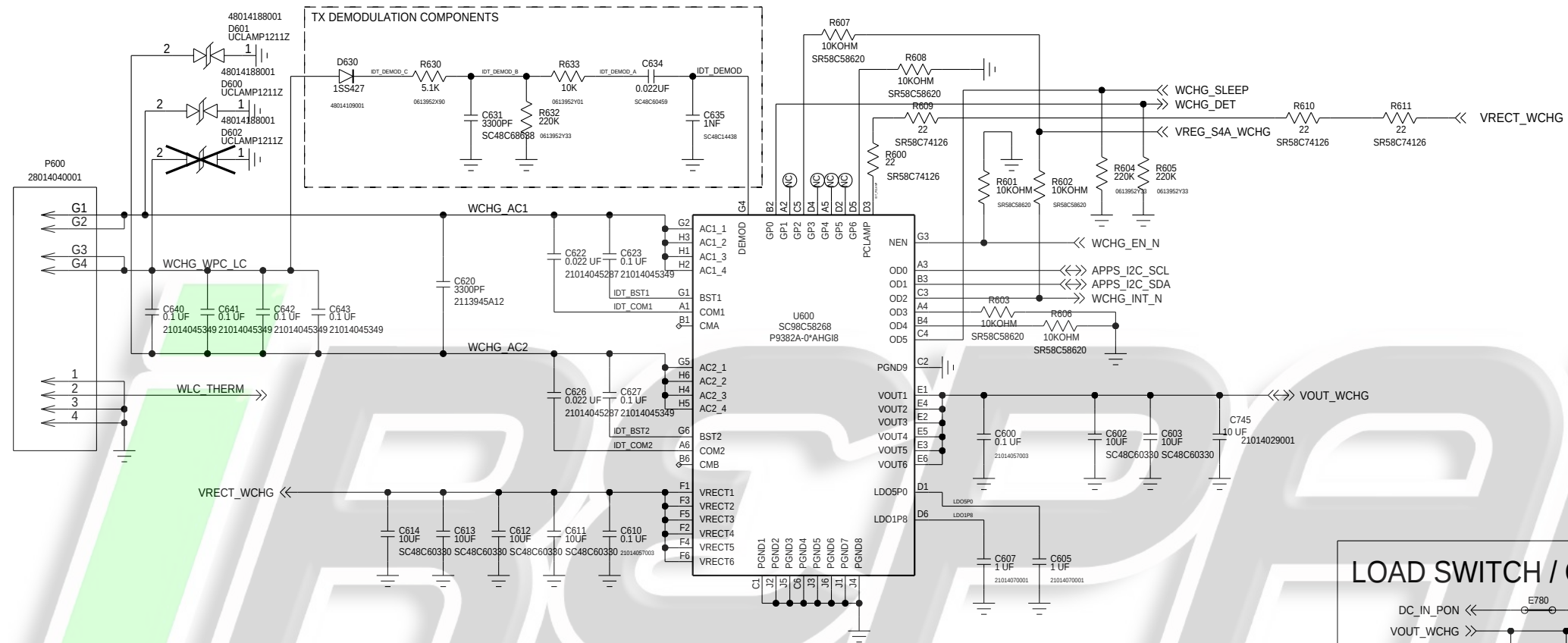


REPAIR

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: power	
LAST MODIFIED BY: MOTOROLA Tue Feb 18 16:05:34 2020			
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REFDES: 600-699

I2C ADDR: 0X3B



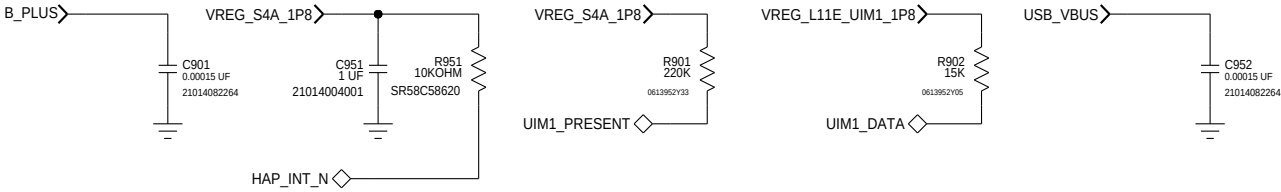
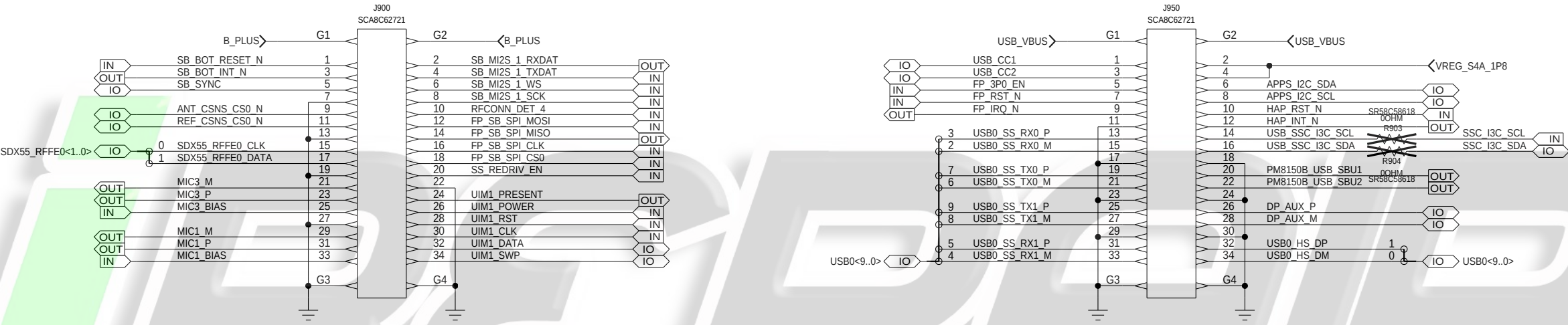
The schematic diagram illustrates a boost converter circuit. The input voltage **B_PLUS_WBOOST** is filtered by capacitors **C741** (10 uF) and **C742** (0.1 uF) before entering the **VIN** pin (D4) of the **SC98C67745 TPS61372 U740** IC. A 1 uH inductor **L740** is placed between the input filter and the IC. The **MODE** pin (A4) is connected to ground. The **EN** pin (B4) is connected to the **WCHG_BOOST_EN** signal. The **NC** pin (A3) is connected to ground through a network of resistors (**R742**, **R743**, **R744**) and capacitors (**C743**, **C744**). The output is taken from the **VOUT1** pin (D1) through a network of resistors (**R740**, **R741**) and capacitors (**C740**, **C741**) to ground. The output is labeled **VOUT_WCHG**.

[illegible]

APPROVALS		 MOTOROLA MOBILE DEVICES	
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BRIDGE: CONNECTORS

REFDES: 900-999



APPROVALS		BLOCK TITLE:	
ENGINEER: MOTOROLA		bridge_conn	
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